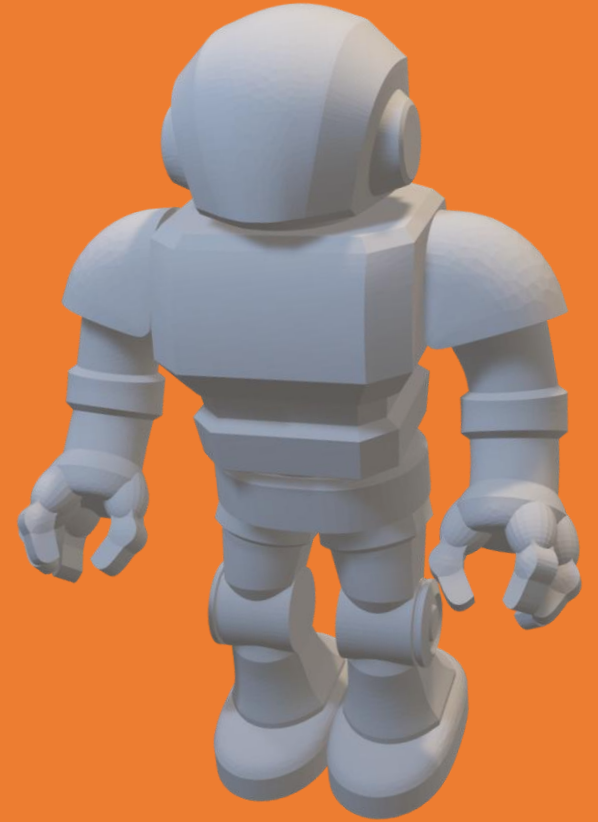


Udemy Course on Amazon Bedrock & Generative AI Beginner to Advanced

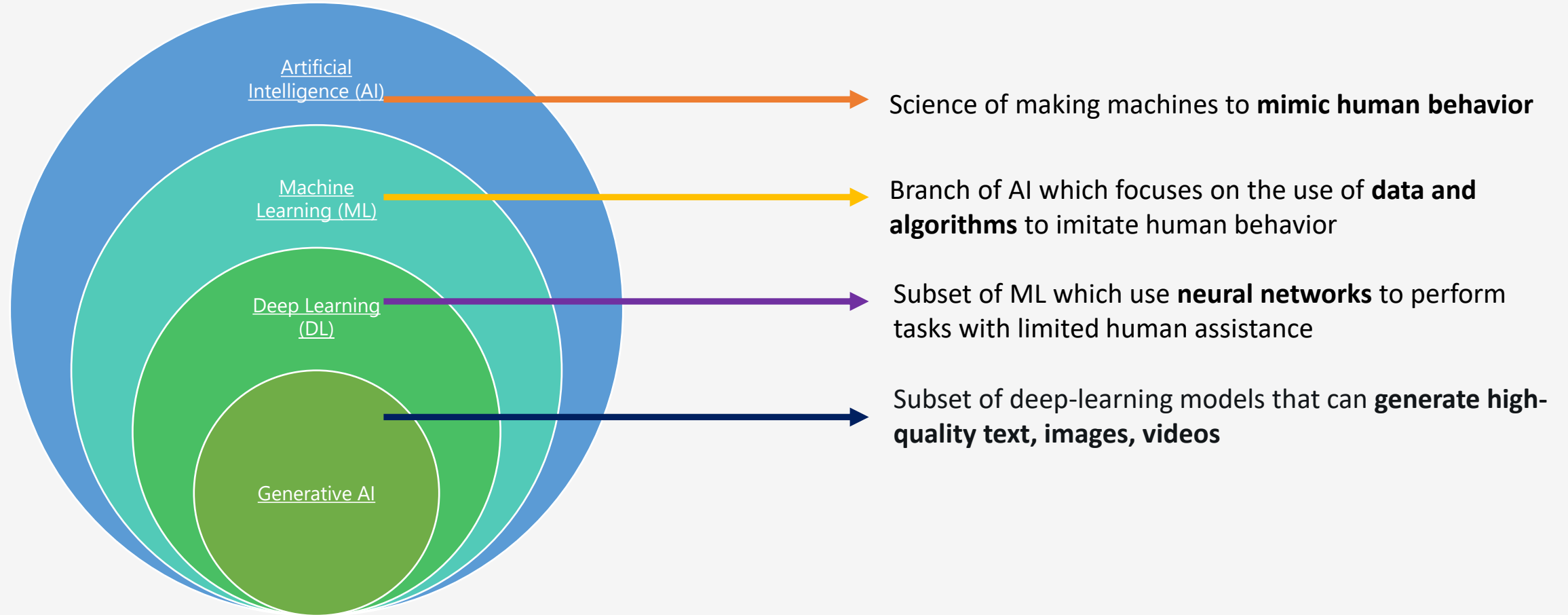


Image Generated using the use case built in this course

*Section on :
Evolution of Generative AI*



What is Generative Artificial Intelligence – Context ?



Artificial Intelligence - Overview



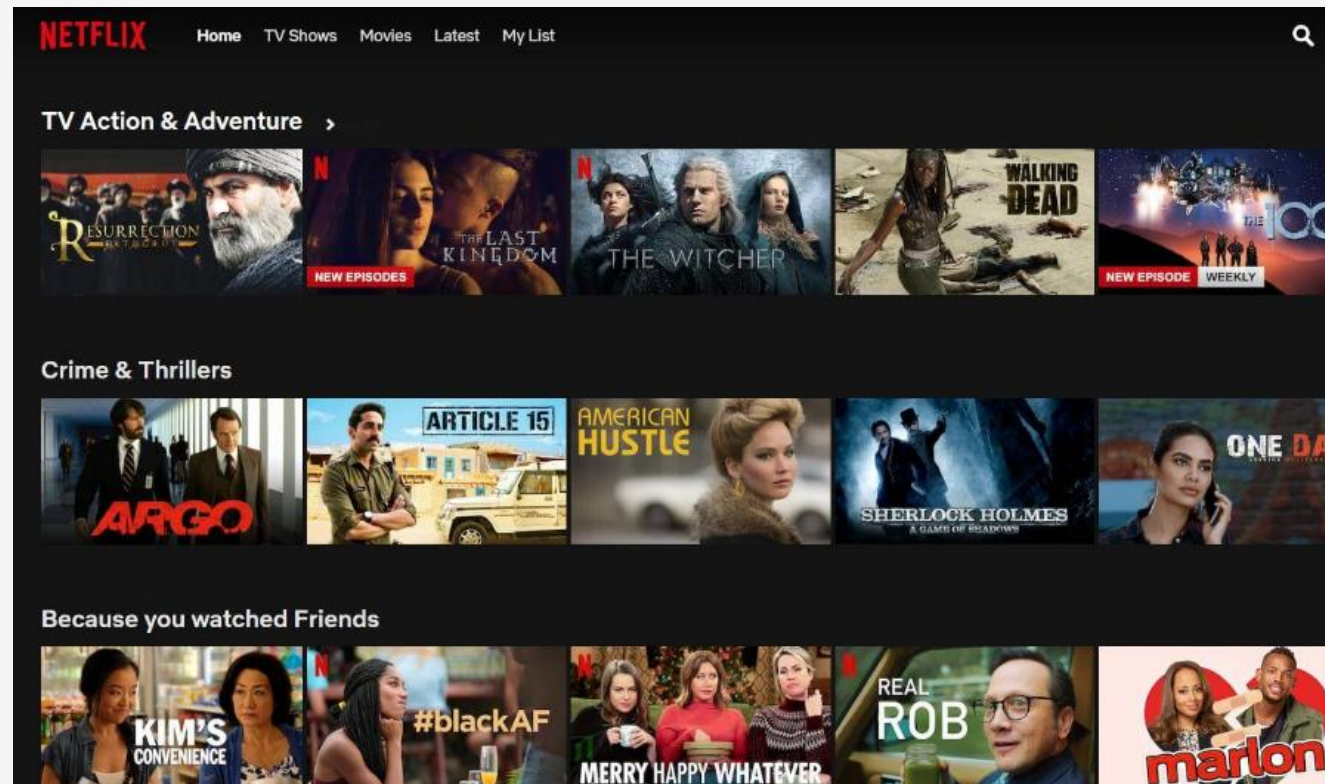
Artificial Intelligence is the science and engineering of **making** intelligent machines, especially **intelligent computer programs** that **simulate human intelligence and decision making**.

Artificial Intelligence - Examples



Self-driving cars use artificial intelligence to make real-time decisions based on sensor data.

Artificial Intelligence - Examples

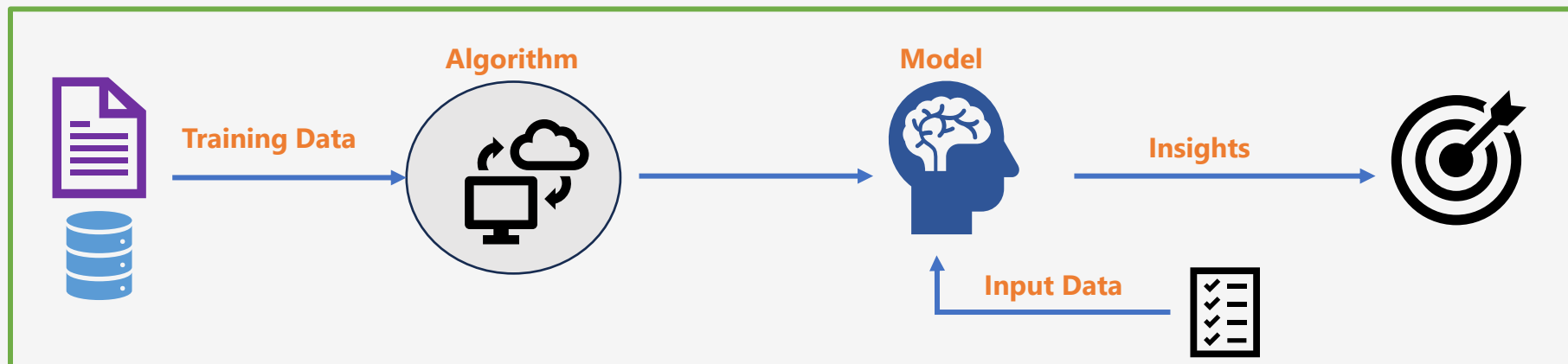


Recommendation engines such as ones from **Netflix and Amazon** use AI to provide custom insights

Machine Learning - Overview



- Machine Learning is a **subset of AI**
- In ML the **algorithm** is trained using the **historical data** to make **predictions on new data**.

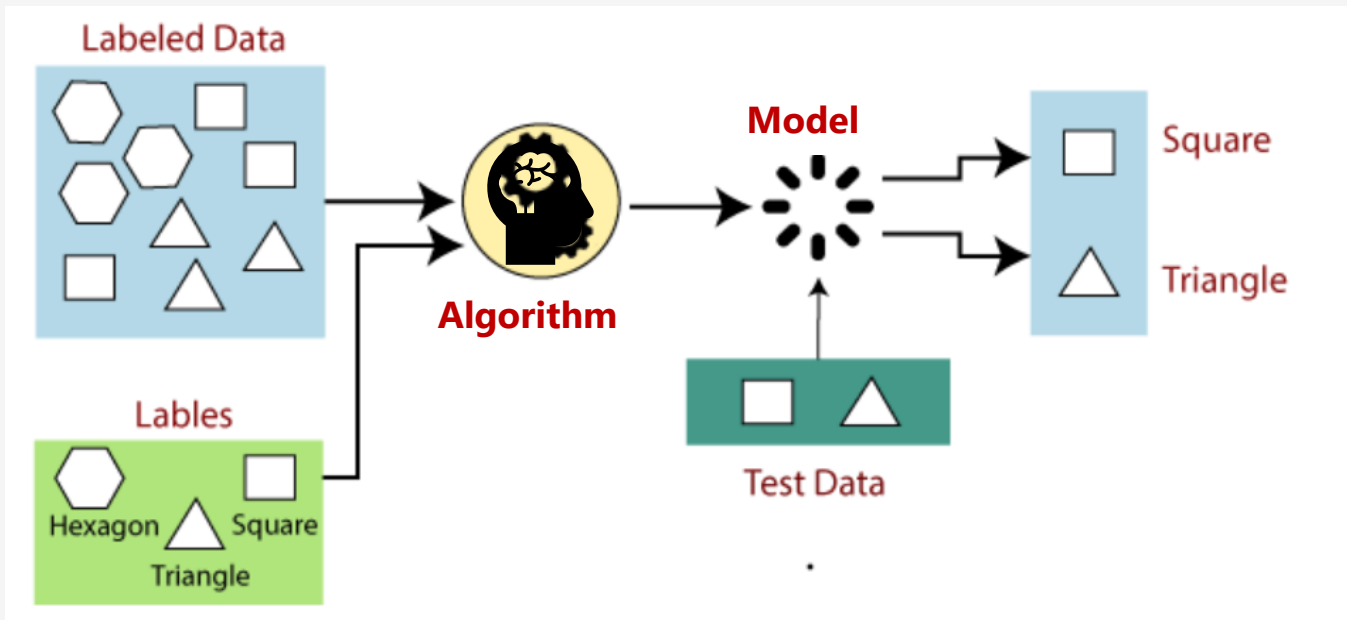


Machine Learning – Broad Categorization



Machine Learning – Supervised Learning

- In supervised learning, **algorithms are trained using labelled dataset.**
- Once the **training is completed**, the **model is tested using test data** and then it predicts the output.



Use Case (Classification) – Gmail Spam Filter



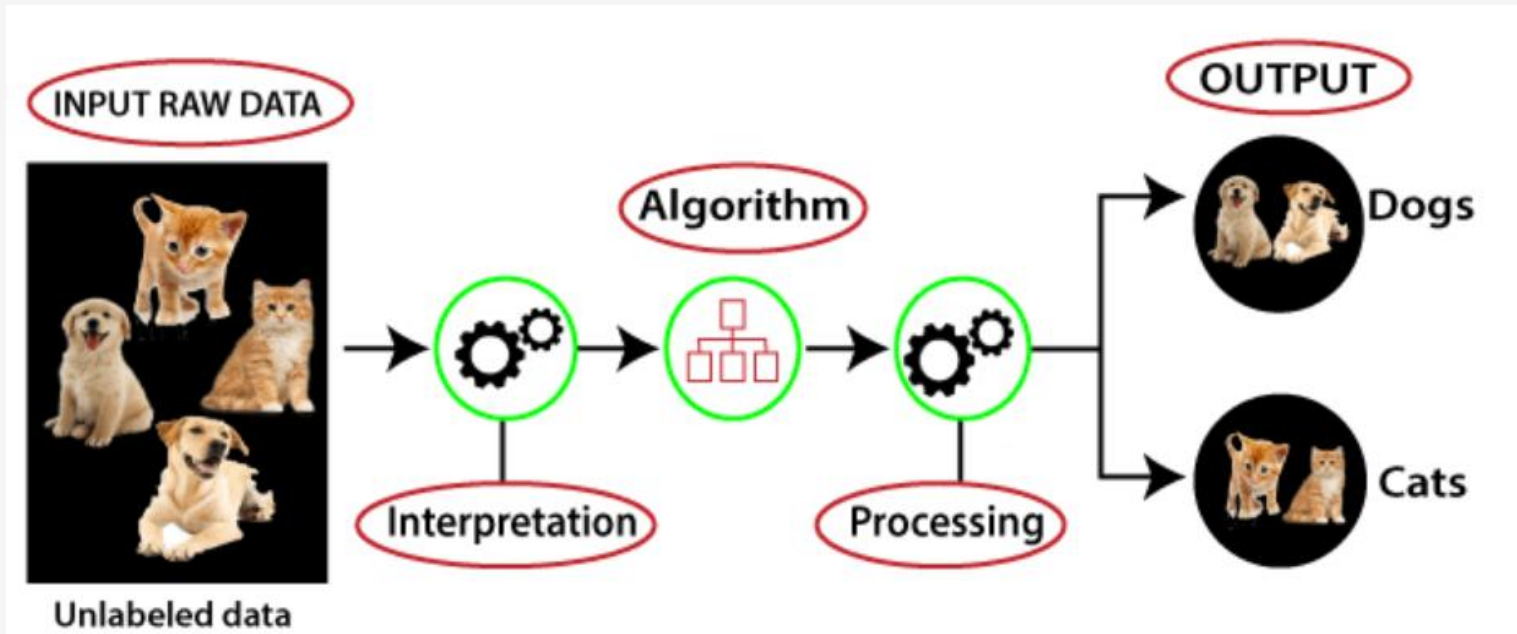
Advertisement	Sales
\$90	\$1000
\$120	\$1300
\$150	\$1800
\$100	\$1200
\$130	\$1380
\$200	??

Use Case (Regression) – Forecasting

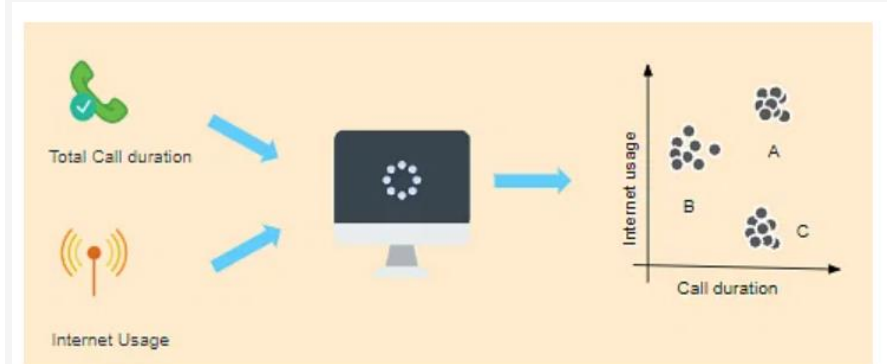
Common Algorithms - Classification and Regression

Machine Learning – Unsupervised Learning

- Unsupervised learning is a type of machine learning in which **algorithm are trained using unlabeled dataset**
- Based on the dataset, model finds **hidden patterns and insights** based on the **similarities of the data**



Use Case (Clustering) – Reduce Churn Rate



Use Case (Association) – Market Basket Analysis

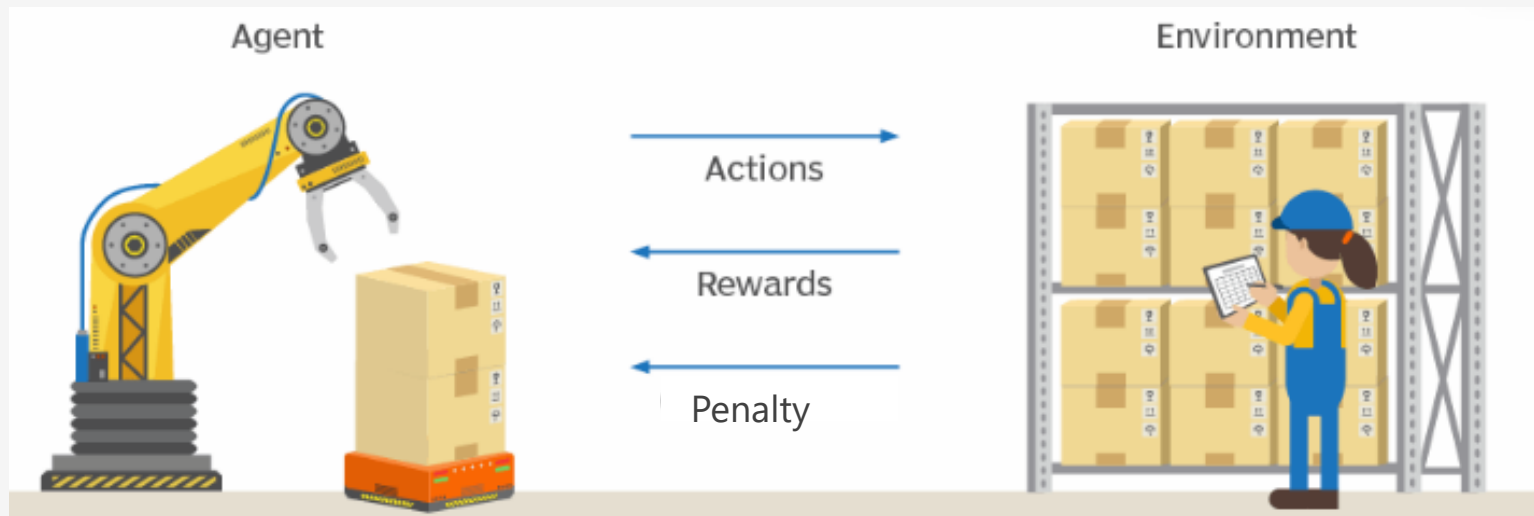


Common Algorithms - **Clustering** and **Association**

Machine Learning – Reinforced Learning



- Reinforcement learning (RL) is a ML technique that focuses on **training an algorithm following the trial and error approach**.
- No pre-defined dataset
- The **algorithm (agent) evaluates a current situation** (state), **takes an action**, and **receives feedback** (reward and feedback) from the environment after each act.

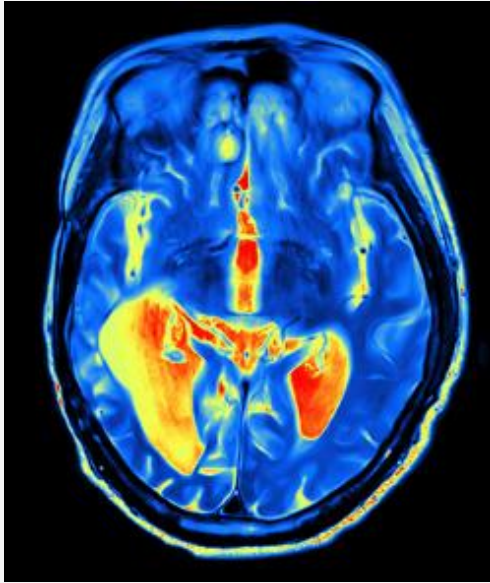


Real World Use Case

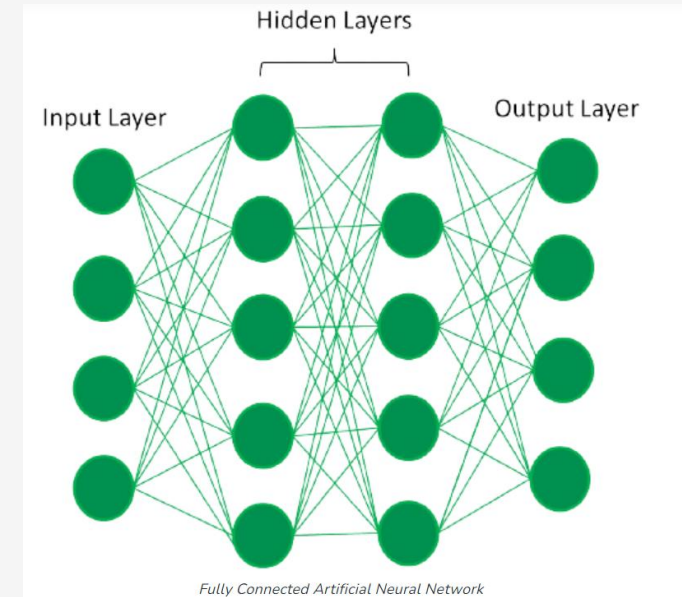
- Training Industrial Robots
- Training Self Driving Cars

Machine Learning – Deep Learning and Artificial Neural Networks

Neural Pathways in Human Brain



Artificial Neural Network



- **Artificial Neural networks** are designed to **mimic the structure of the human brain**
- Deep Learning is a subset of Machine Learning that **focuses on building artificial neural networks that can learn from data.**
- Neural network with **multiple hidden layers and multiple nodes** in each hidden layer is known as a **deep learning (>3)**

Machine Learning – Deep Learning with Artificial Neural Networks

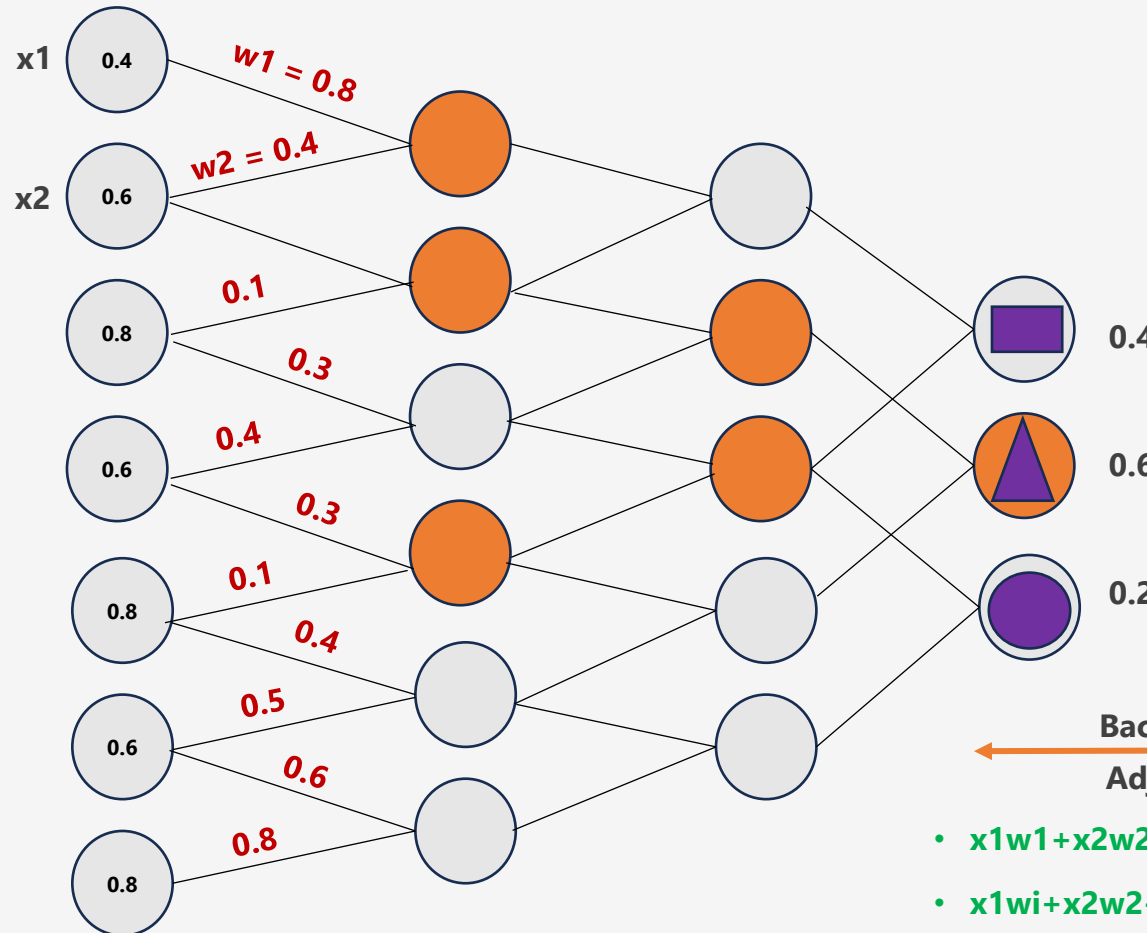


Input Layer

Hidden Layers

Output Layer

Key terms



Output	Error
1	+0.6
0	-0.6
0	-0.2

Backward Propagation
Adjust Weights

- $x1w1 + x2w2 + \dots > \text{threshold value}$
- $x1w1 + x2w2 - \text{threshold} > 0$
- $x1w1 + x2w2 + \text{Bias} \gg \text{Activation Function}$

- Neurons (Binary Value)
- Weights
- Bias/Threshold
- Activation Function
- Feedforward network
- Actual Output
- Error
- Forward Propagation
- Backward Propagation

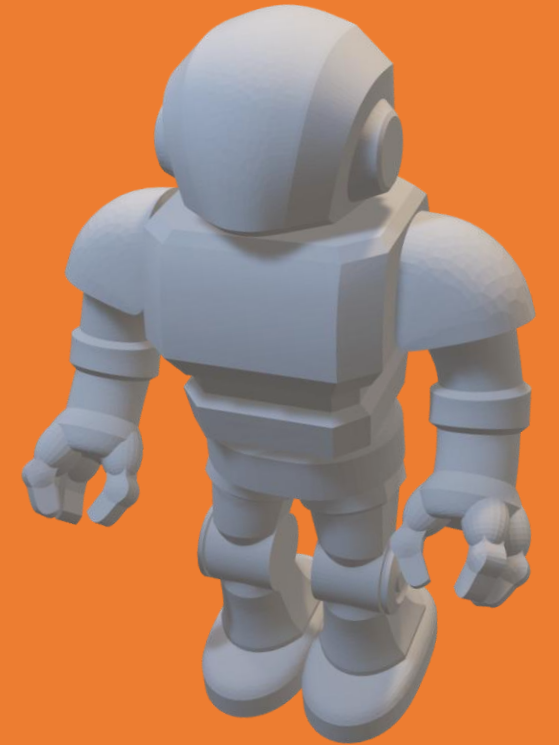
Generative Artificial Intelligence(AI)



Generative AI

- Generative AI is a **subset of Deep Learning** which focuses on **creating models** capable of **generating new content** :
 - Chat and Text
 - Images and Videos
 - Code Generation

*Section on :
Generative AI Fundamentals*



What is Generative AI ?

dynamoDB CodeCommit Elastic Container Service ElastiCache Route 53 RDS EC2 Cloud9 VPC Cognito

AI21
labs

AI21 Labs

Jurassic-2 Mid

Start new chat

Using on-demand capacity. [Switch Provisioned Throughput](#)

what is generative AI

Default



Generative AI is a branch of artificial intelligence (AI) that focuses on creating models or algorithms that can simulate or generate new data based on existing data. The purpose of generative AI is to mimic the process of natural generation, such as the way living organisms produce offspring with traits similar to their parents.

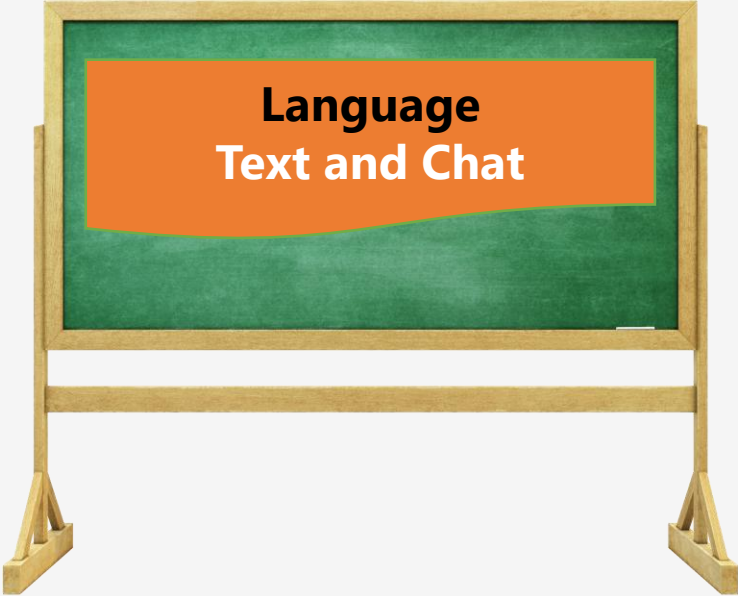
Generative models can be trained on various data types, including images, sounds, text, and even complex data structures such as DNA sequences. These models use techniques such as deep learning and reinforcement learning to generate new examples that are similar to or identical to the training examples.

Generative AI has a wide range of applications, including image generation, natural language processing, drug discovery, and robotics. It is also used in various fields such as art, music, and gaming to create more realistic and engaging content.

Generative AI refers to

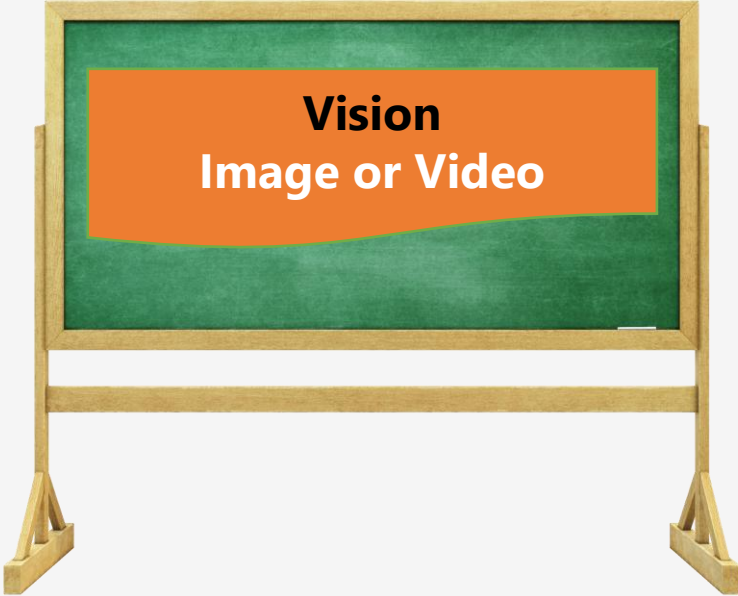
- Branch of AI that use deep-learning **models**
- Generate high-quality text, images, code based on the data they were trained on
- Popular examples of generative AI include
 - ChatGPT
 - Bard

Generative AI – Use Cases



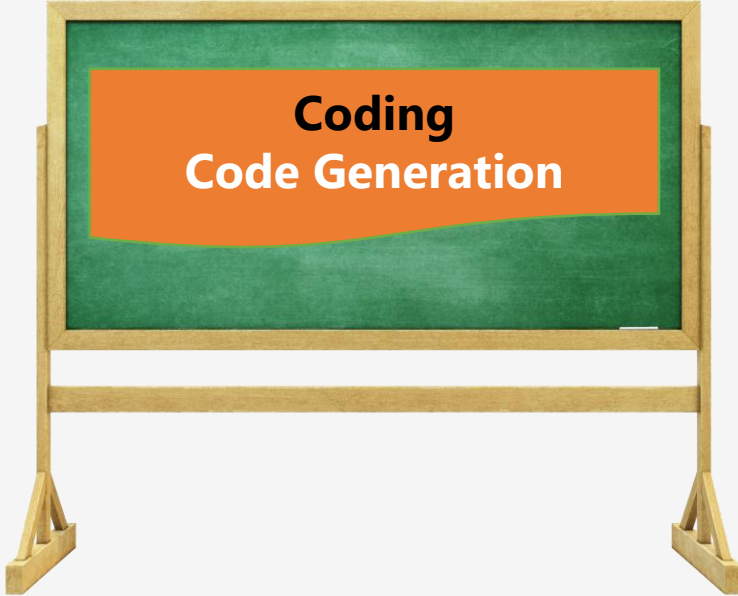
Language Text and Chat

- ✓ Text Summarization - **Healthcare**
- ✓ Content Generation – **HR & Legal**
- ✓ Chatbot/ Q&A – **Customer Service**
- ✓ Models : **Jurassic – 2 from AI21 Labs**



Vision Image or Video

- ✓ Image Generation - **3D-Design for construction industry**
- ✓ Video Generation - **Content for marketing campaigning**
- ✓ Models : **Stable Diffusion from Stability AI**



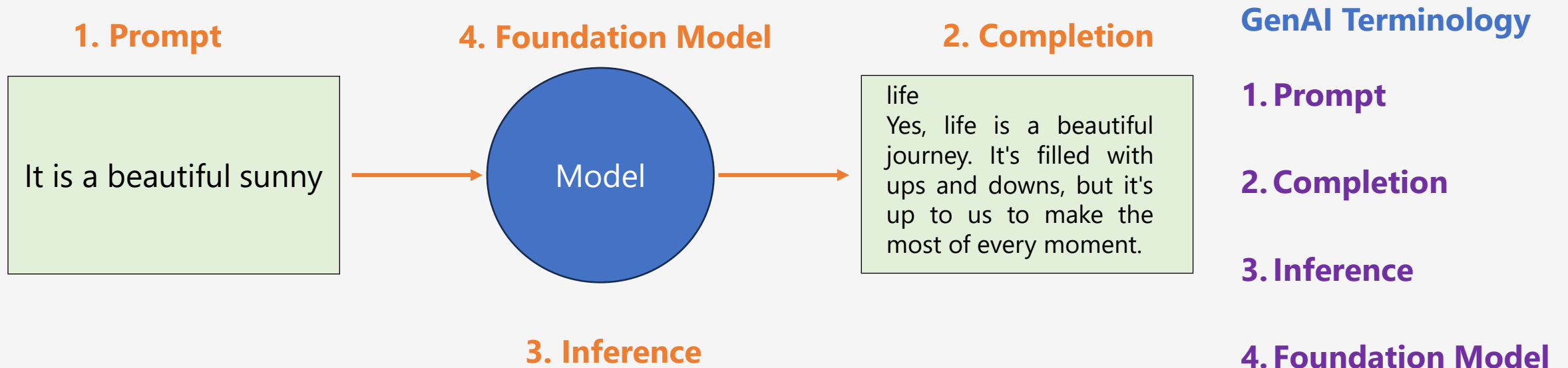
Coding Code Generation

- ✓ Code Generation
- ✓ 15+ Program Lang
- ✓ Multiple IDE's
- ✓ AWS Code Whisperer

Console Walkthrough Amazon Bedrock

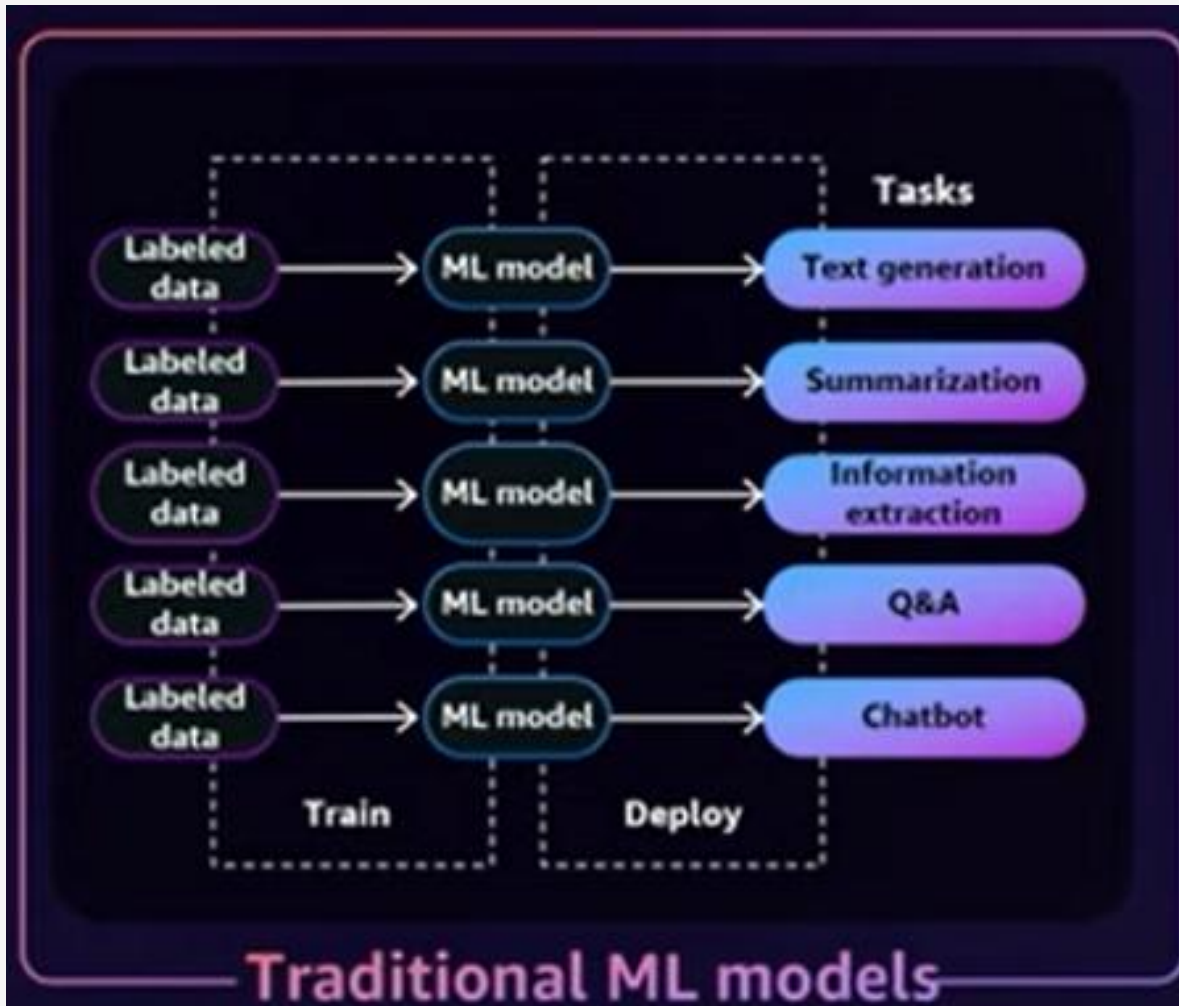
- Bedrock offers a choice of foundation models from leading AI companies.
- Provides access to variety of FMs from **Amazon, AI21 Labs, Anthropic, Cohere, Meta, and Stability AI**

How Generative AI works - Basic Concepts and Terminology - 1



1. Prompt – Input provided to Model
2. Completion – Output of Model
3. Inference : Act of using model to generate text is called Inference

Challenges in Traditional Machine Learning Models

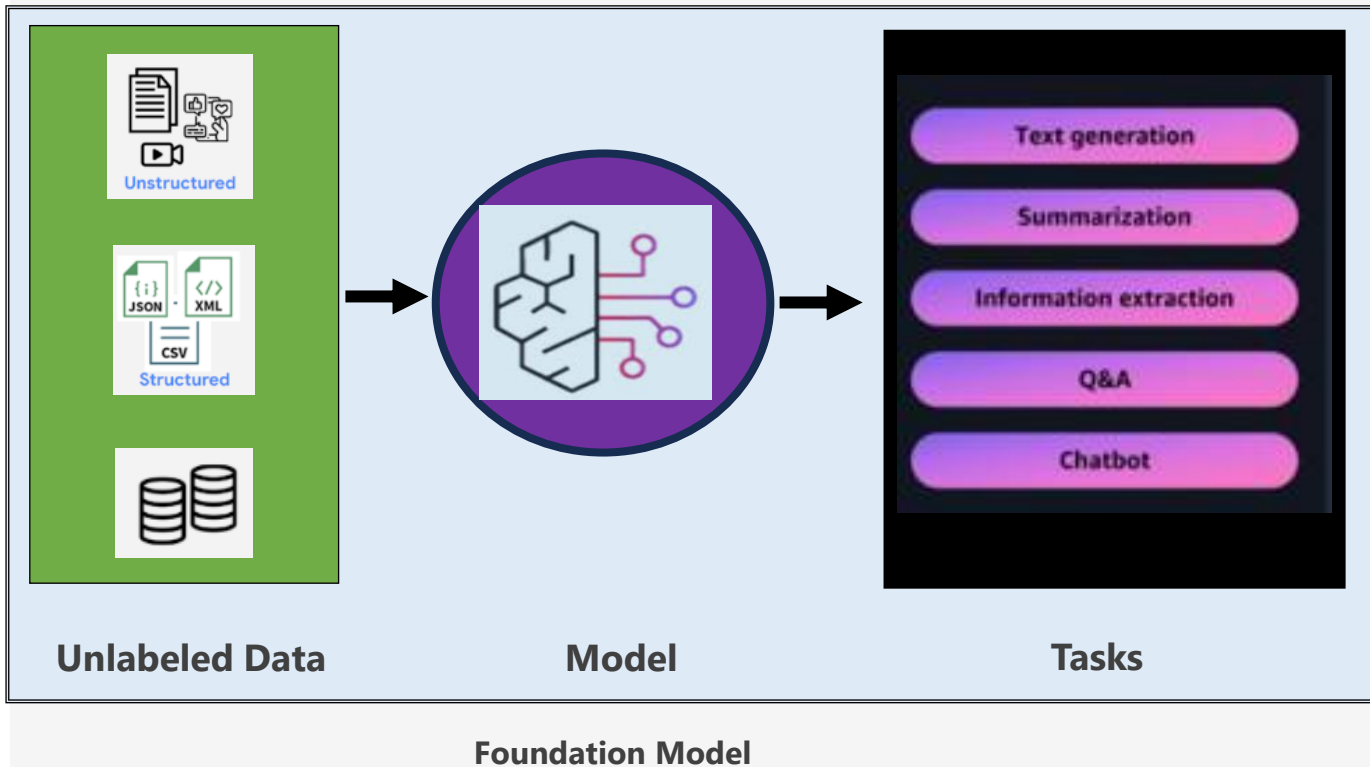


Source : AWS

Challenges in Traditional ML Models

- Each Model focused on specific Task such as
 - Text Generation
 - Summarization
 - Q&A
- Needs labeled data which requires human intervention and can be cost prohibitive

Key Characteristics of Foundational Models



Key Characteristics

- **Deep neural networks** techniques
- **Unlabeled data** - Trained on unlabeled data and Self-supervised learning such as internet data - Wikipedia
- **Generalized** – One FM for multiple tasks
- **Pre-trained** – FM's are pre-trained models
- **Large** – Exposed to vast amounts of data, enabling it to learn patterns & features from data.
- **Expensive** - Developing **FM's** is an **expensive process** that involves **large amounts of computational resources** and **expertise**.

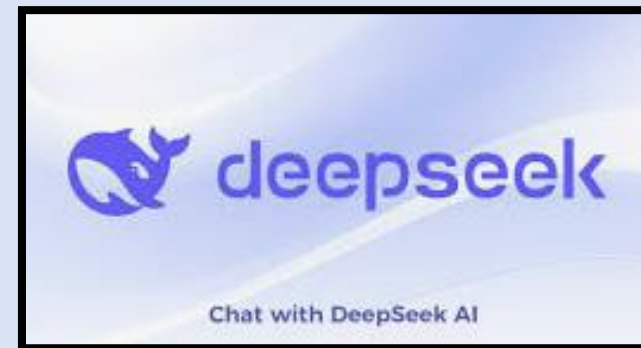
Organization's can utilize **existing FM's as the base for building** task-specific models and **Adapt** them

Popular Foundation Models – Proprietary and Open-Source

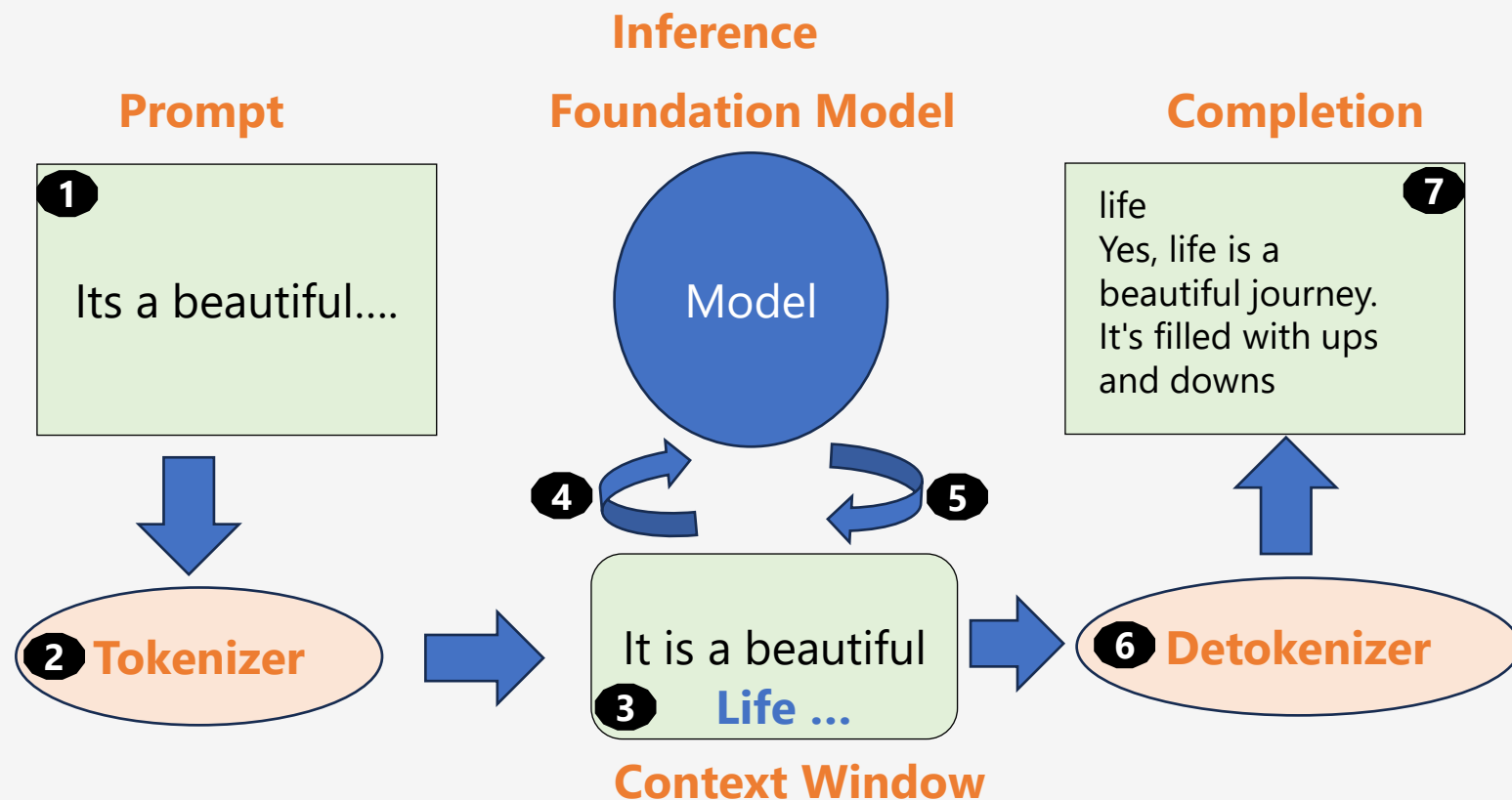
Proprietary Models



Open-Source Models

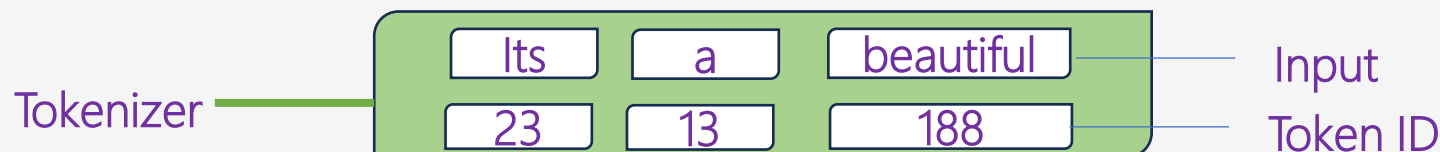


How Generative AI works - Basic Concepts and Terminology - 2



GenAI Terminology

- **Prompt**
- **Tokenizer**
- **Context Window**
- **Foundation Model**
- **Max Token/Stop Sequence**
- **Completion**
- **Inference**
- **Context Window** - number of tokens the model can take as input at once



Foundation Model – Tokens, Parameters and Temperature



Tokens



Parameters



Temperature

Model	Context length	Number of English pages (1 page = 500 words)
GPT 3.5	4,096	6
GPT 4	8,192	12
Llama 1	2,048	3
Llama 2	4,096	6

Tokens - **One token** generally corresponds to **~4 characters of text** for common English text

Parameters - can be considered count of **connections & weights** between nodes in a neural network.

Temperature - This parameter controls the **creative ability** of your model.

Popular Foundation Models (Parameters perspective)

**Access to private APIs
and open source models
drive progress for
generative ML**

Private



Open source



GPT 4

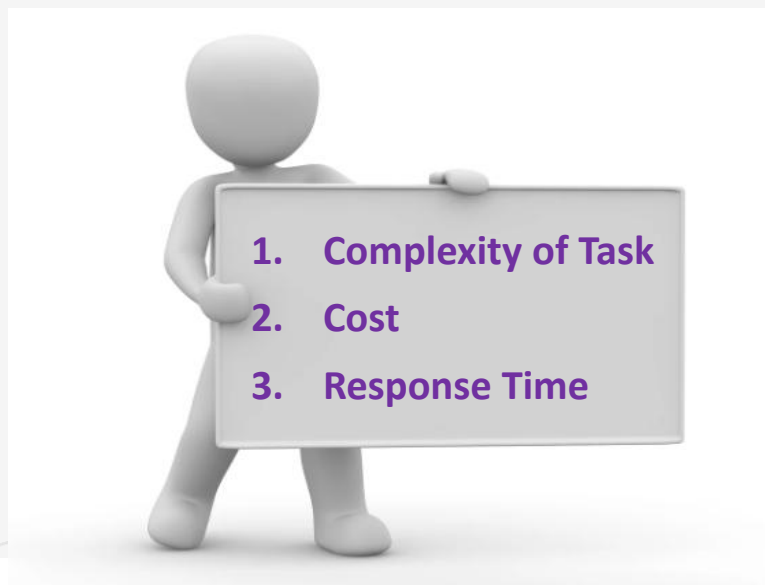
- ~1.8 trillion parameters
- 120 layers

Complexity of Foundation Model for your application (Parameter perspective)

Complexity of Foundation Models

- Large Language Models (LLM) - 8B to 100's of Billions of parameters
- Small Language Model (SLM) – Less than 8B parameters

Choice of Foundation Model for a Use Case



> 120 Layers

What is model size of GPT 4.5

As of now, OpenAI has not officially disclosed the exact model size of GPT-4.5. While GPT-4 (the predecessor to GPT-4.5) is known to have trillions of parameters, the specific number for GPT-4.5 hasn't been shared publicly.

It's important to note that the size of models like GPT-4.5 could be a step up from GPT-4 in terms of parameters or other architectural improvements, but OpenAI typically focuses on performance and capability enhancements rather than specific model size metrics.

More number of Foundation Model Parameters ==

Complex Response + Higher response time + Costlier

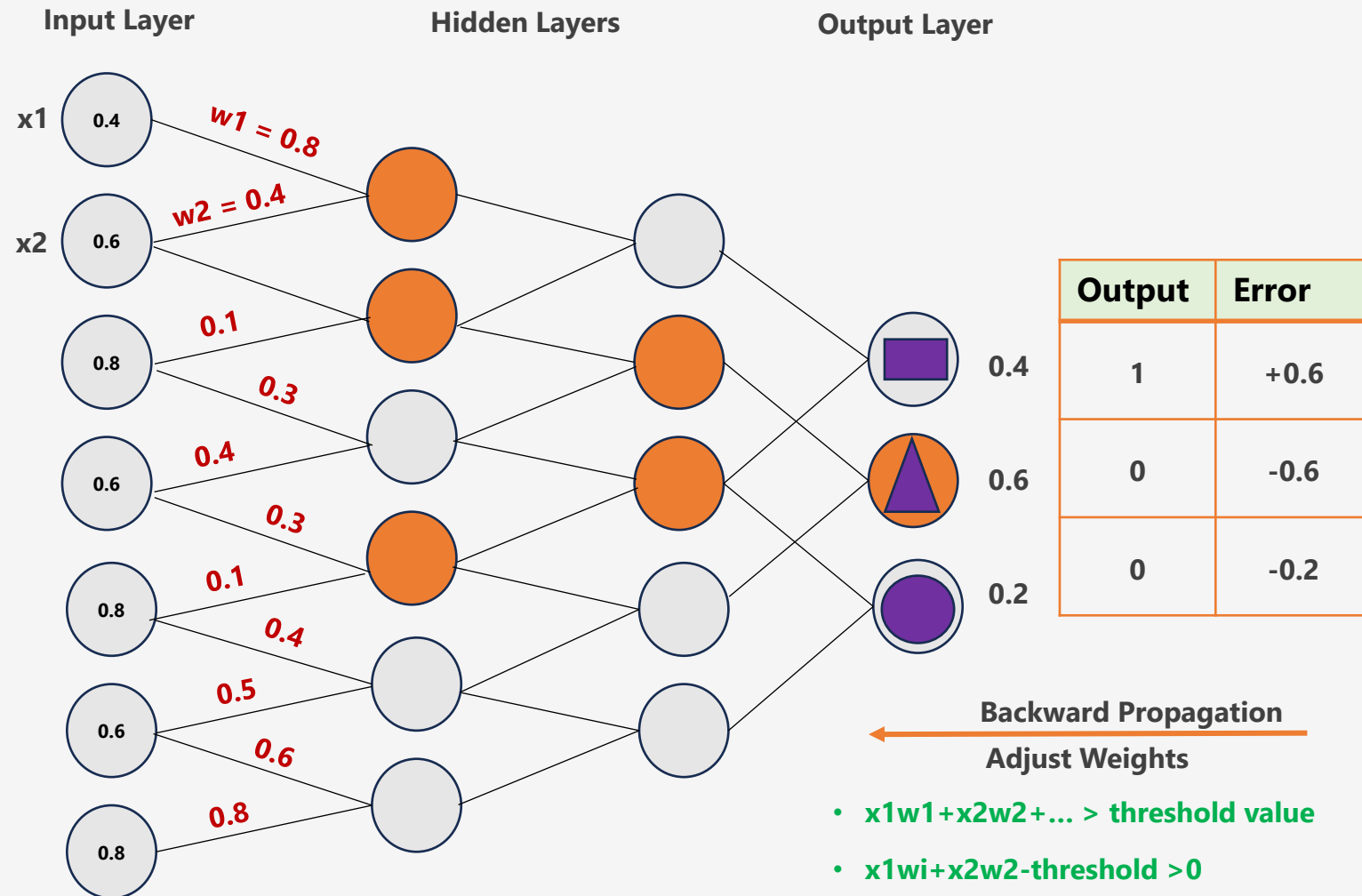
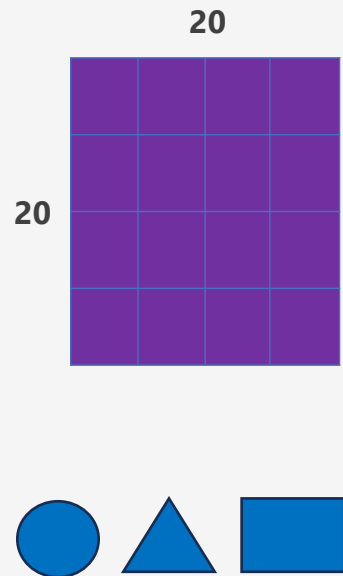


Lesser number of Foundation Model Parameters ==

Low complexity Response + Lower response time + Cheaper



Machine Learning – Deep Learning with Artificial Neural Networks



Key terms

- Neurons (Binary Value)
- Weights
- Bias/Threshold
- Activation Function
- Feedforward network
- Actual Output
- Error
- Forward Propagation
- Backward Propagation

- $x_1w_1 + x_2w_2 + \dots > \text{threshold value}$
- $x_1w_i + x_2w_2 - \text{threshold} > 0$
- $x_1w_1 + x_2w_2 + \text{Bias} \gg \text{Activation Function}$

Foundation Models vs LLM's vs Image Generators vs Multi-Modal



Foundation Models

VS

Large Language Models (LLM's)

VS

Text to Image Model (Stable Diffusion)

VS

Multi-Modal

Foundation Models - Categorization

Foundation Models – Broad Categorization of Models

- **Focus Area : Text Generation**
- **Large Language Models (LLM's)**
- Example : GPT 4, Llama 3

- **Focus Area : Image Generation**
- **Image Generation or Diffusion Model**
- Example : Stable Diffusion

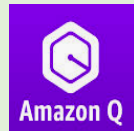
- **Multi Modal**
- **Focus Area : Text & Vision Model**
- Example : Claude3 – Multi-Modal

Foundation Model

- Large deep learning neural networks +
Trained on massive datasets
- Generate Text, Images, Videos or both
- **Large Language Models (LLM's)**
- **Image Generation Models – Text to Image or Image to Image**
- **Multi-Modal**

AWS Generative AI Service Offerings

Apps. that leverage Foundation Models



4

Amazon Q Developer || Amazon Q Business || Amazon Q in Quick Sight ||
Amazon Q in Supply Chain || Amazon Q in Connect

Model as a Service



3

Amazon Bedrock

Bedrock Knowledge Bases || Agents || Guardrails

Infra. for Training & Inferencing



2

Amazon SageMaker AI

Foundation Models



1

Amazon Tranium and Inferentia

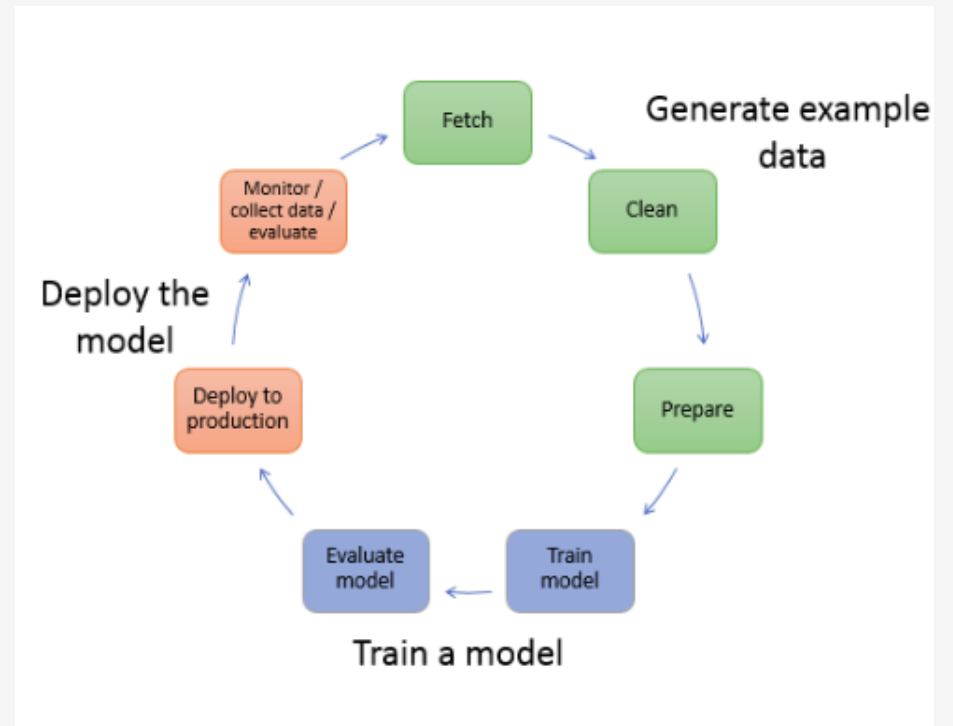
1. Train and Deploy own Models on Amazon EC2 Trn1n and EC2 Inf2

Compute Infrastructure to TRAIN and DEPLOY your AI Models



2. Amazon SageMaker AI (formerly Amazon SageMaker)

- Pre-generative AI era AWS Service
- Traditional AI/ML service to help you build, train and deploy Machine Learning Models.
- Example – Gmail Spam Filter, Forecasting
- GenAI capabilities with SageMaker Jumpstart – Train & Deploy Foundation Models
- Unified platform for data, analytics, and AI.



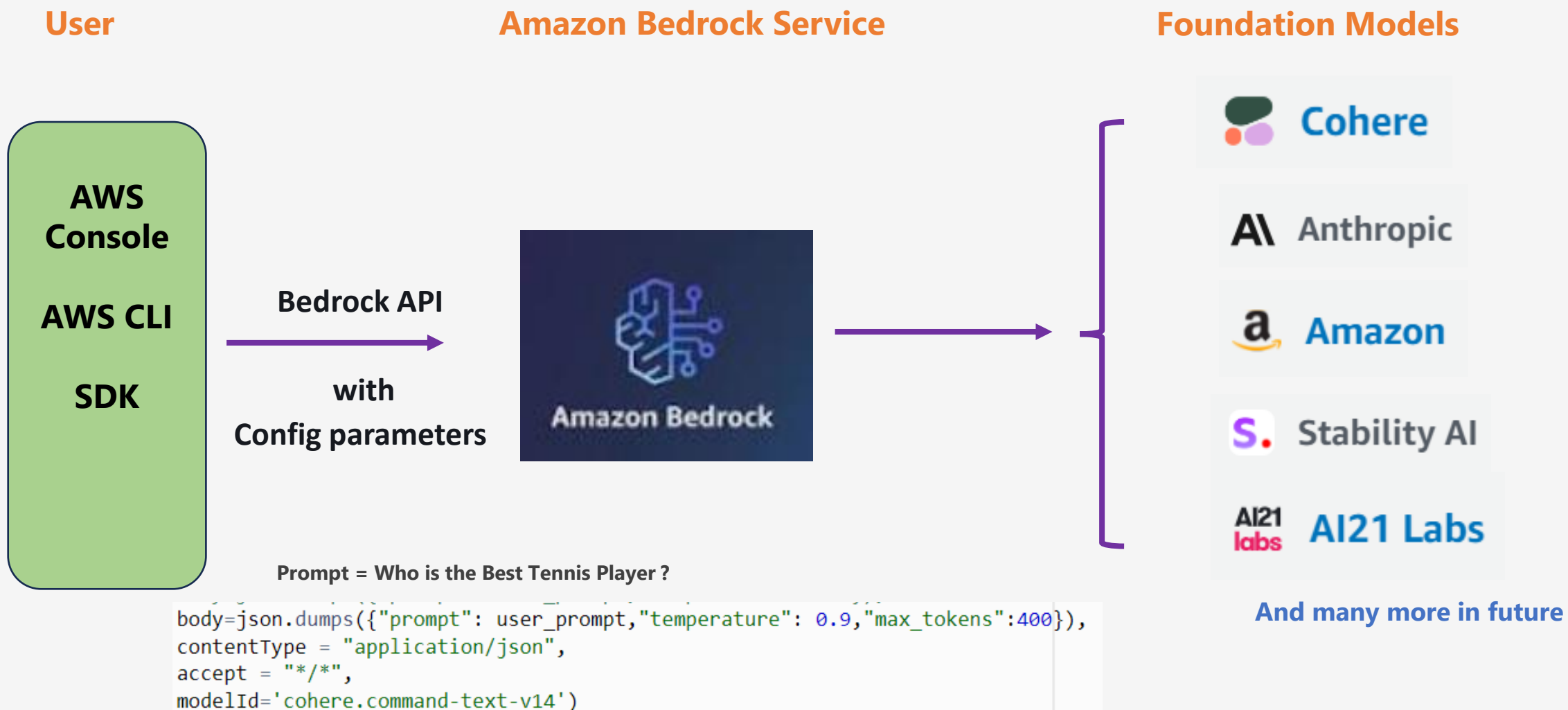
2. Foundational Models available on SageMaker JumpStart



The screenshot shows the Amazon SageMaker AI console interface. The left sidebar contains navigation links for Applications and IDEs, Admin configurations, JumpStart, Governance, HyperPod Clusters, Ground Truth, Processing, Training, Training Plans, Inference, Augmented AI, and AWS Marketplace. The main content area displays a grid of foundational models available on SageMaker JumpStart. The models are organized into a 3x3 grid, with the last cell containing a partial view of a fourth model. Each model card includes a search bar, a title, a description, and a 'View model' button.

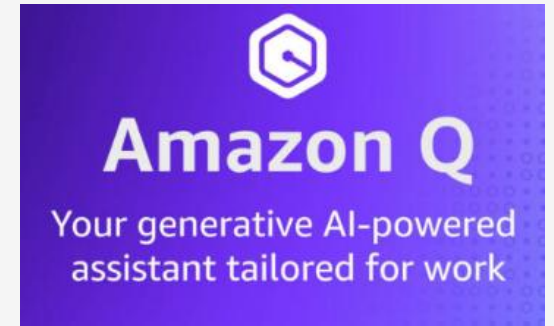
Model Name	Provider	Version	Description
Stable Diffusion XL 1.0	stability.ai	Ver 20230726	PROFESSIONAL: COMPARED TO PREVIOUS VERSIONS, SDXL 1.0 GENERATES MORE VIBRANT AND ACCURATE... The official foundation model for image generation from Stability. Deploy this optimized instance and serve generative AI within minutes.
Meta Llama 2 7B Chat	Meta	Ver 1.0.0	CHAT OPTIMIZED, TEXT GENERATION, LLAMA 2 7B dialogue use case optimized variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range...
Meta Llama 2 70B Chat	Meta	Ver 1.0.0	CHAT OPTIMIZED, TEXT GENERATION, LLAMA 2 70B dialogue use case optimized variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range...
AI21 Jurassic-2 Ultra	AI21 labs	Ver v1.0	RECOGNIZED AMONG STANFORD'S TOP-TIER LLM EVALUATIONS, JURASSIC-2 ULTRA ALLOWS USERS TO... Top-of-the-line LLM for complex text generation tasks that require the highest quality output.
LightOn Mini-instruct 40B	LightOn	Ver 1.0	TEXT GENERATION, KEYWORD EXTRACTION, INFORMATION EXTRACTION, QUESTION ANSWERING,... 40B multilingual and instruction-tuned language model.
Meta Llama 2 7B	Meta	Ver 1.0.0	TEXT GENERATION, LLAMA 2, FINE-TUNABLE 7B variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range of parameter...
Meta Llama 2 13B	Meta	Ver 1.0.0	CHAT OPTIMIZED, TEXT GENERATION, LLAMA 2 13B variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range of parameter...
Meta Llama 2 13B Chat	Meta	Ver 1.0.0	CHAT OPTIMIZED, TEXT GENERATION, LLAMA 2 13B dialogue use case optimized variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range...
Meta Llama 2 70B	Meta	Ver 1.0.0	TEXT GENERATION, LLAMA 2 70B variant of Llama 2 models. Llama 2 is an auto-regressive language model that uses an optimized transformer architecture. Llama 2 is intended for commercial and research use in English. It comes in a range of parameter...

3. Amazon Bedrock



4. What is Amazon Q Business?

Amazon Q Business is a **fully managed, generative-AI powered assistant**



- Generate Content



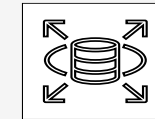
- Summarize Data



- Chat with Enterprise Data (40+ connectors)



S3



RDS



GDrive

40+ Pre Built
Connectors

- Assign Tasks (Plugins such as Jira)



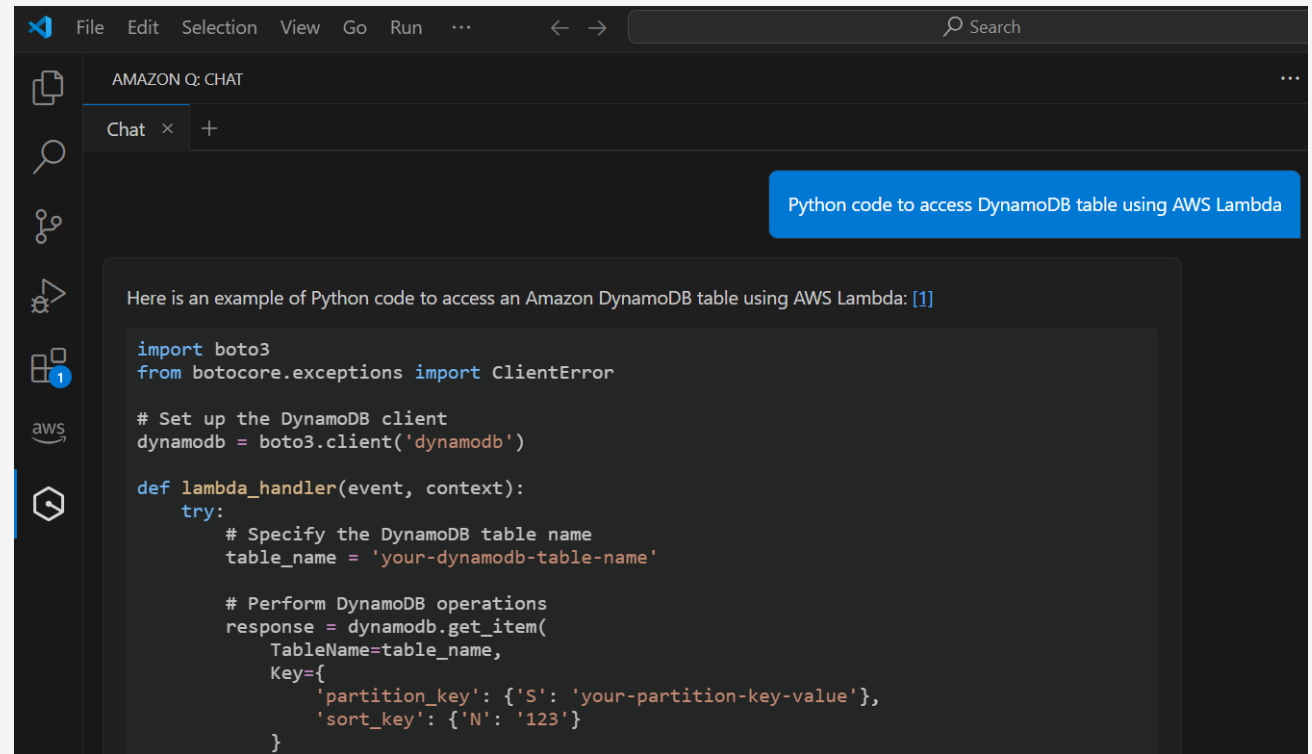
ChatGPT with Enterprise Controls

4. What is Amazon Q Developer

The most capable generative **AI-powered assistant** for software development

Amazon Q Developer can assist in :

- Coding and Testing
- Upgrading applications (Upgrade from Java 8 to Java 17)
- Diagnosing errors
- Performing security scanning and fixes
- Optimizing AWS resources



The screenshot shows the Amazon Q Developer chat interface. At the top, there's a search bar and a menu with options like File, Edit, Selection, View, Go, Run, and a search icon. Below the menu, there's a tab labeled 'AMAZON Q: CHAT'. The chat area displays a message from the assistant: 'Here is an example of Python code to access an Amazon DynamoDB table using AWS Lambda: [1]'. The code is a Python lambda function handler that uses the boto3 library to interact with the DynamoDB service. It sets up a client, specifies a table name, and performs a get_item operation. A blue button at the top right of the chat area says 'Python code to access DynamoDB table using AWS Lambda'.

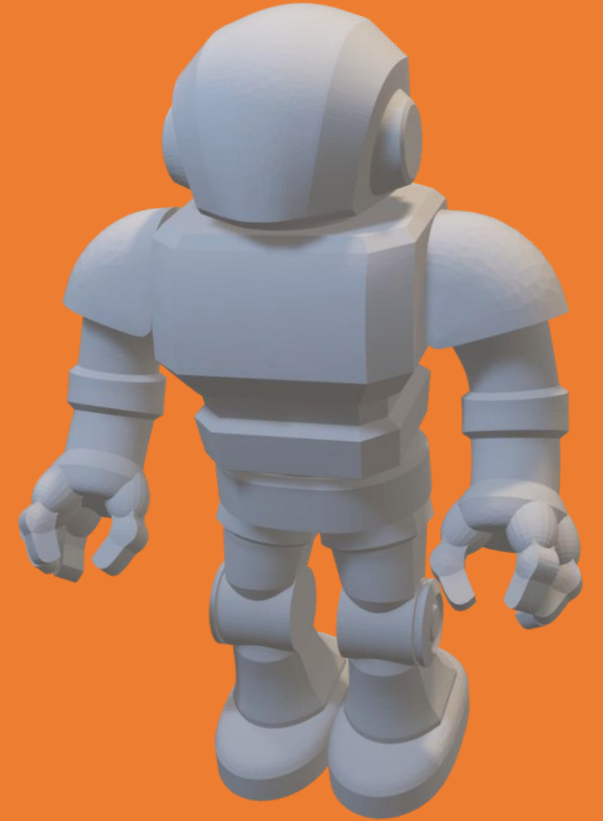
```
import boto3
from botocore.exceptions import ClientError

# Set up the DynamoDB client
dynamodb = boto3.client('dynamodb')

def lambda_handler(event, context):
    try:
        # Specify the DynamoDB table name
        table_name = 'your-dynamodb-table-name'

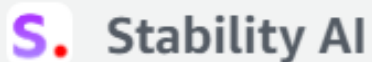
        # Perform DynamoDB operations
        response = dynamodb.get_item(
            TableName=table_name,
            Key={
                'partition_key': {'S': 'your-partition-key-value'},
                'sort_key': {'N': '123'}
            }
        )
```

*Section on :
Amazon Bedrock – Deep Dive*



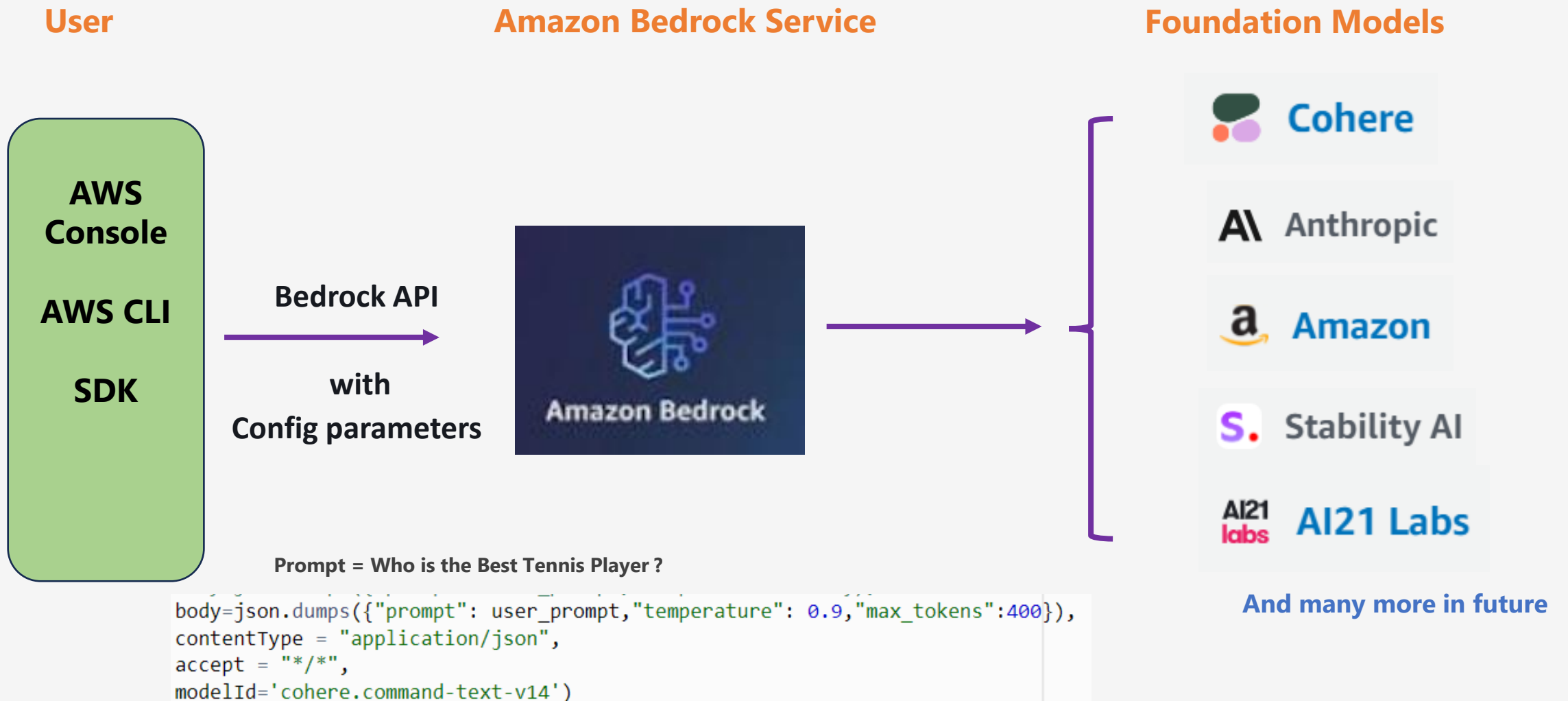
What is Amazon Bedrock ?

- Amazon Bedrock is a **fully managed, serverless service from AWS**
- Makes **base Foundation Models from Amazon and third-party model providers** accessible through **an API**.



And many more in future

How does Amazon Bedrock work – High Level ?



Key features of Foundation Models Amazon Bedrock supports



Source : AWS

Amazon Bedrock – Console Walkthrough

The screenshot displays the Amazon Bedrock console interface. On the left is a navigation sidebar with sections: 'Getting started' (Overview, Providers), 'Foundation models' (Model catalog, Marketplace deployments, Custom models, Imported models, Prompt Routers), 'Playgrounds' (Chat / Text, Image / Video), 'Builder tools' (Agents, Flows, Knowledge Bases, Prompt Management), and 'Safeguards' (Guardrails, Watermark detection). The main content area is titled 'Overview' and includes a breadcrumb 'Amazon Bedrock > Overview'. It features a large 'Foundation models' section with a description and two buttons: 'View Model catalog' and 'Discover marketplace models'. To the right is a 'Model spotlight' for 'Anthropic's Claude', which includes a description and a 'Request model access' link. Below these are three playground cards: 'Chat / Text', 'Image / Video', and 'Builder tools', each with a brief description and an 'Open playground' link. The footer contains 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Amazon Bedrock <

[Amazon Bedrock](#) > Overview

Overview Info

Foundation models

Amazon Bedrock supports over 100 foundation models from industry-leading providers and emerging leaders. Select a serverless model or Bedrock Marketplace model that is best suited for achieving your unique goals.

[View Model catalog](#) [Discover marketplace models](#)

Model spotlight

AI Anthropic's Claude

Choose the exact combination of intelligence, speed, and cost to suit your needs. All of the latest Claude models, like upgraded Claude 3.5 Sonnet, are available in Amazon Bedrock.

[Request model access](#)

Chat / Text

Generate text for a vast range of language processing tasks with various pre-trained models. You can use a single prompt or iterate on the result by submitting subsequent prompts that take into account the context of previous prompts and generated responses in a chat format.

[Open playground](#)

Image / Video

Easily generate compelling images by providing text prompts to pre-trained models. In the playground, enter a text prompt to get started.

[Open playground](#)

Builder tools

Create GenAI applications, augment responses with your proprietary data, and experiment with prompts.

CloudShell Feedback

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Amazon Bedrock – Request Access to Models

Base models

Custom models

Providers

Playgrounds

Chat

Text

Image

Deployment

Provisioned Throughput

Model access

Settings

User guide

Bedrock Service Terms

Model access

To use Bedrock, you must request access to Bedrock's FMs. To do so, you will need to have the correct [IAM Permissions](#). For certain models, you may first need to submit use case details before you are able to request access. More information about these models is available on the [Providers](#) page.

Base models (9)

Manage model access

Models	Access status	Modality	EULA
AI21 Labs Jurassic-2 Ultra	Access granted	Text	EULA
Jurassic-2 Mid	Access granted	Text	EULA
Amazon Titan Embeddings G1 - Text	Access granted	Embedding	EULA
Titan Text G1 - Lite	Unavailable	Text	EULA

Anthropic

Use case details submitted

Claude

Claude Instant

Cohere

Command

Stability AI

Stable Diffusion XL

Access granted

Access granted

Access granted

Access granted

Access granted

Access granted

Text

Text

Text

Image

[EULA](#)

[EULA](#)

[EULA](#)

[EULA](#)

Cancel

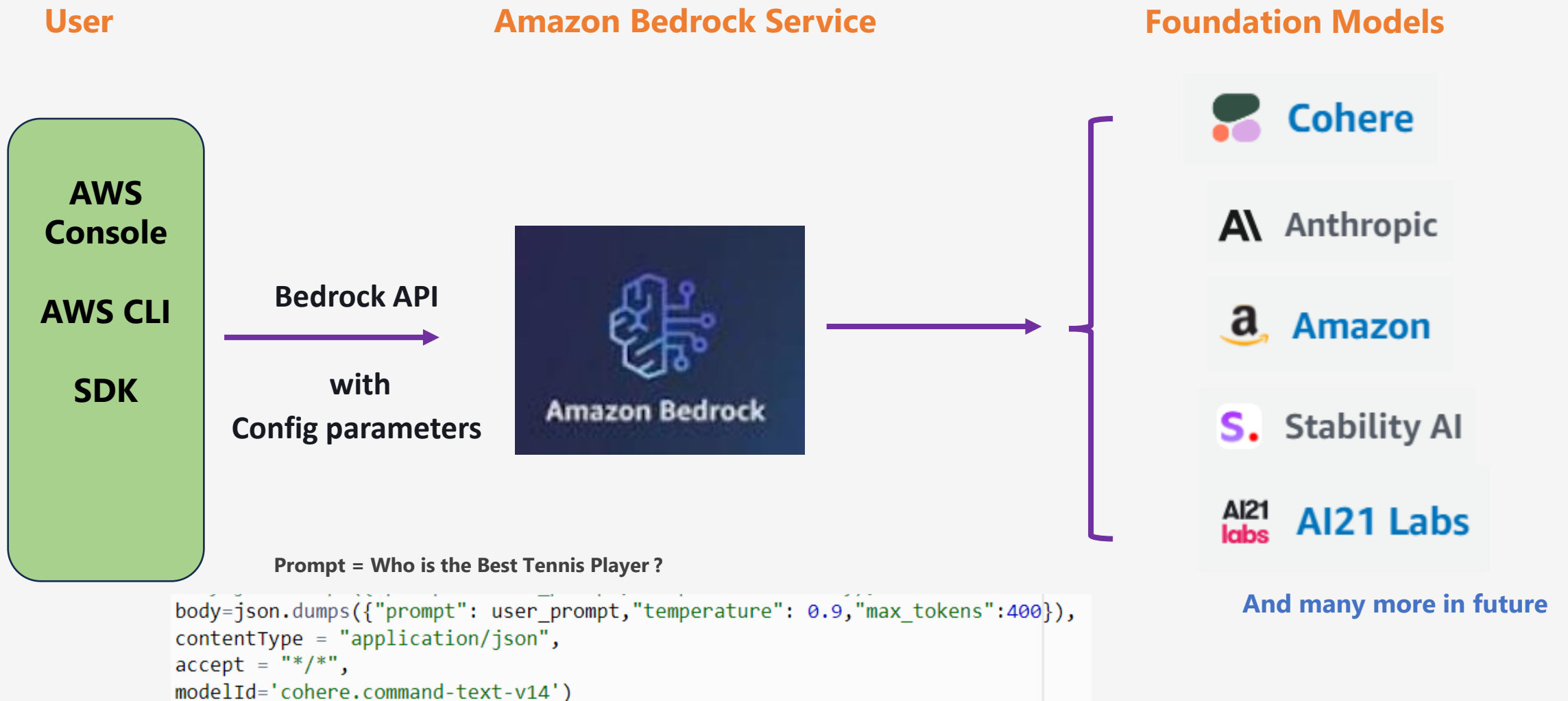
Save changes

1

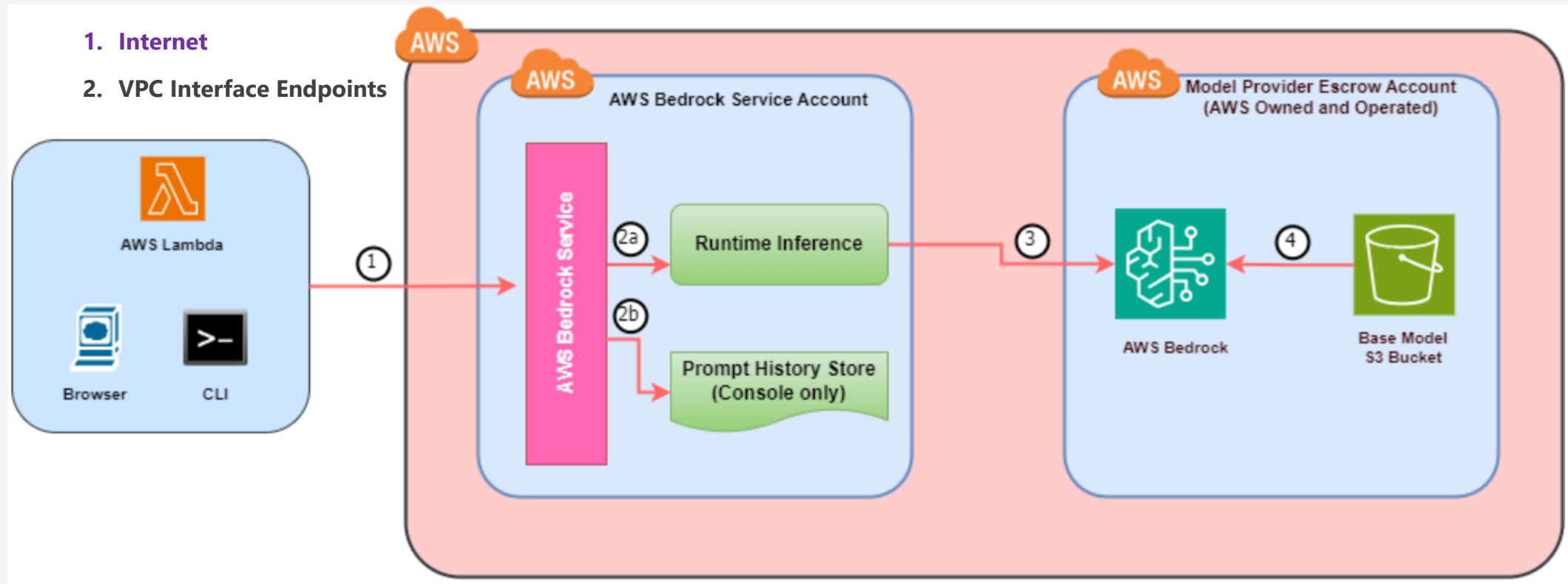
2

3

How does Amazon Bedrock work – High Level ?



Amazon Bedrock – Architecture – Part 1



- Runtime Inference : Used to redirect to right model endpoint based on the API request
- Base Model : Baseline model provided to every AWS account

Services (2)

Search

bedrock X Clear filters

	Service Name	Type
☐	com.amazonaws.us-east-1.bedrock	Interface
☐	com.amazonaws.us-east-1.bedrock-runtime	Interface

Foundation Models and Impact of Inference Parameters

Inference parameters influence the response generated by the model.

Inference Parameters

1. Randomness and diversity

2. Length

3. Repetition

Temperature

Top K

Top P

**Response
Length**

**Length
Penalty**

**Stop
Sequence**

**Repetition
penalty**

Inference Parameter – 1. Randomness and Diversity

Prompt : I hear the hoof beats of

Parameter	Possible Values	Description	Example
Temperature*	Foundation Models use : Probability to construct the words in a sequence		- Horse --- > 0.4 - Wind ----- > 0.2 - Unicorns -- - > 0.1 - Change in distance --- > 0.05
	Low value (closer to zero)	Model tends to select the higher-probability words .	Horse
	High value (closer to 1 or 5)	Temperature further away from zero, the model may select a lower-probability word	Change in distance
Top K	1-500	Cutoff where the model no longer selects the words .	if K=50, the model selects from 50 of the most probable words
Top P	0.01-0.99	Caps choices based on the sum of their probabilities .	If Top P=0.5, Horse and Wind selected

Inference Parameter – 2. Length

Prompt : Write an essay on horse

Parameter	Possible Values	Description
Length : Controls the length of the generated response . Helps control cost .		
Max Length	1- 4096 tokens	Specify the maximum number of tokens to use in the generated response.
Stop Sequence	Any keyword	• Configure up to four sequences that the model recognizes. • After a stop sequence, the model stops generating further tokens. • The returned text doesn't contain the stop sequence.



Inference Parameter – 3. Repetition

Parameter	Possible Values	Description
Repetition : Help control repetition in the generated response.		
Presence penalty	0-5	Use a higher value to lower the probability of generating new tokens that already appear at least once in the prompt or in the completion.
Count penalty	0-1	Use a higher value to lower the probability of generating new tokens that already appear at least once in the prompt or in the completion. Proportional to the number of appearances.
Frequency penalty	0-500	Use a high value to lower the probability of generating new tokens that already appear at least once in the prompt or in the completion. The value is proportional to the frequency of the token appearances (normalized to text length)
Penalize special tokens		Reduce the probability of repetition of special characters – Whitespaces, Punctuations...

Amazon Bedrock - Pricing

Two Modes - On-Demand and Provisioned Throughput

On-Demand Mode

- Pay for what you use, with no time-based term commitments
- Text generation models
 - Charged for every input token processed and every output token generated
- Image generation models
 - Charged for every image generated.
- Embeddings models
 - Charged for every input token processed

AWS Bedrock - Pricing

Provisioned Throughput

- Primarily designed for **large consistent inference workloads** that need **guaranteed throughput**.
- Can **purchase model** units for a **specific base or custom model** with **time commitment (1-6 months)**
- Custom models can only be accessed using Provisioned Throughput.
 - Console Custom Models
 - Pricing

Pricing Examples

- A121 Labs
- Anthropic

AWS Bedrock - Pricing

Token Limits - Tokens shared between prompt and completion.

<https://help.openai.com/en/articles/4936856-what-are-tokens-and-how-to-count-them>

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Media and Entertainment Industry Use Case :

Generate Movie Poster Design

Use Case Implementation Pre-Requisites 1 : Request Access to Models

Base models

Custom models

Providers

Playgrounds

Chat

Text

Image

Deployment

Provisioned Throughput

Model access

Settings

User guide

Bedrock Service Terms

Model access

To use Bedrock, you must request access to Bedrock's FMs. To do so, you will need to have the correct [IAM Permissions](#). For certain models, you may first need to submit use case details before you are able to request access. More information about these models is available on the [Providers](#) page.

Base models (9)

Manage model access

Models	Access status	Modality	EULA
AI21 Labs Jurassic-2 Ultra	Access granted	Text	EULA
Jurassic-2 Mid	Access granted	Text	EULA
Amazon Titan Embeddings G1 - Text	Access granted	Embedding	EULA
Titan Text G1 - Lite	Unavailable	Text	EULA

Anthropic

Use case details submitted

Claude

Access granted

Text

[EULA](#)

Claude Instant

Access granted

Text

[EULA](#)

Cohere

Command

Access granted

Text

[EULA](#)

Stability AI

Stable Diffusion XL

Access granted

Image

[EULA](#)

Cancel

Save changes

Use Case Implementation Pre-Requisites 2 : Upgrade boto3 version

1. Create a AWS Lambda Function – bedrock-boto3Upgrade and add import boto3
2. Check the boto3 version. Should be > 1.28.63
3. Use following command to check version - `print(boto3.__version__)`
4. Upgrade the boto3 version for AWS Lambda Function using Lambda Layer -
<https://repost.aws/knowledge-center/lambda-python-runtime-errors>
 - Add Layer Version ARN
 - Check the boto3 version. Should be > 1.28.63
5. Code for Bedrock invocation from AWS Lambda Function
 - Link - <https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/bedrock-runtime.html>

Manually create a Lambda layer that uses the latest Boto3 version

1. Create a lib folder:

```
LIB_DIR=boto3-udemy03/python
```

```
mkdir -p $LIB_DIR
```

2. Install the library to LIB_DIR:

```
pip3 install boto3 -t $LIB_DIR
```

3. Zip all the dependencies to /tmp/boto3-new-layer.zip:

```
cd boto3-udemy03
```

```
zip -r /tmp/boto3-udemy03.zip .
```

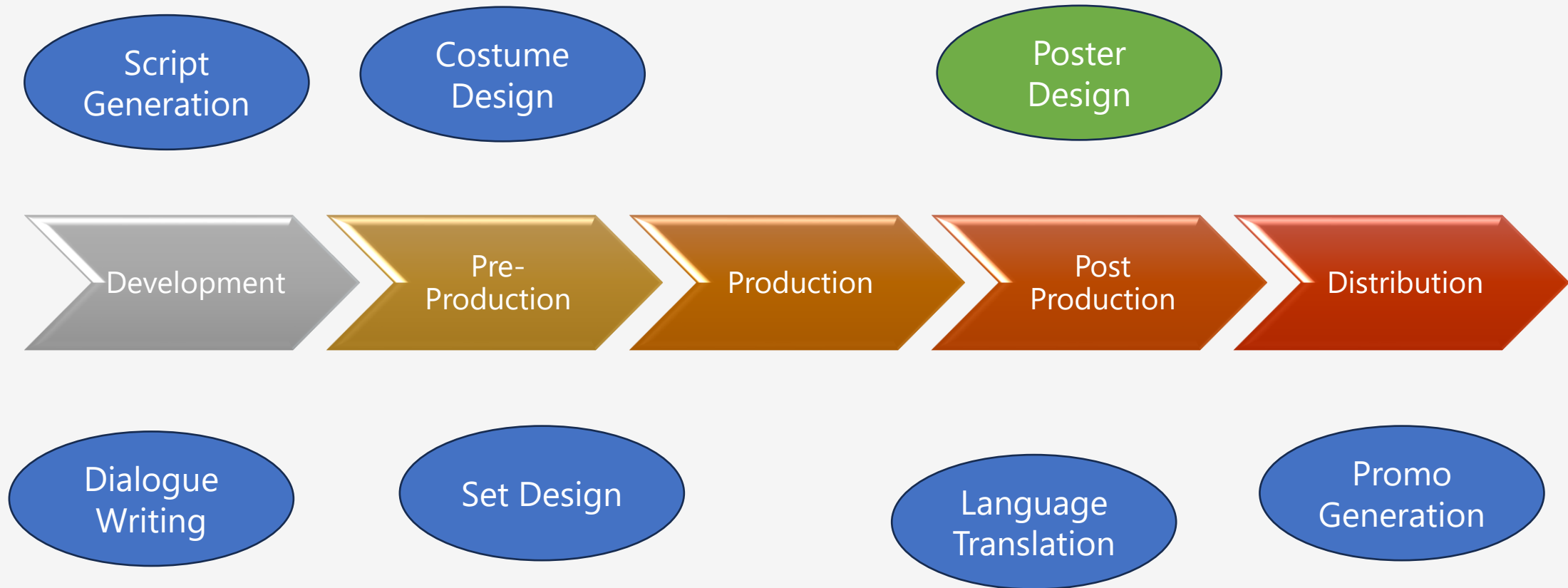
4. Publish the layer:

```
aws lambda publish-layer-version --layer-name boto3-udemy03 --zip-file file:///tmp/boto3-udemy03.zip
```

Media and Entertainment Industry Use Case :

Generate Movie Poster Design

Generative AI – Media and Entertainment Industry(Film Making)



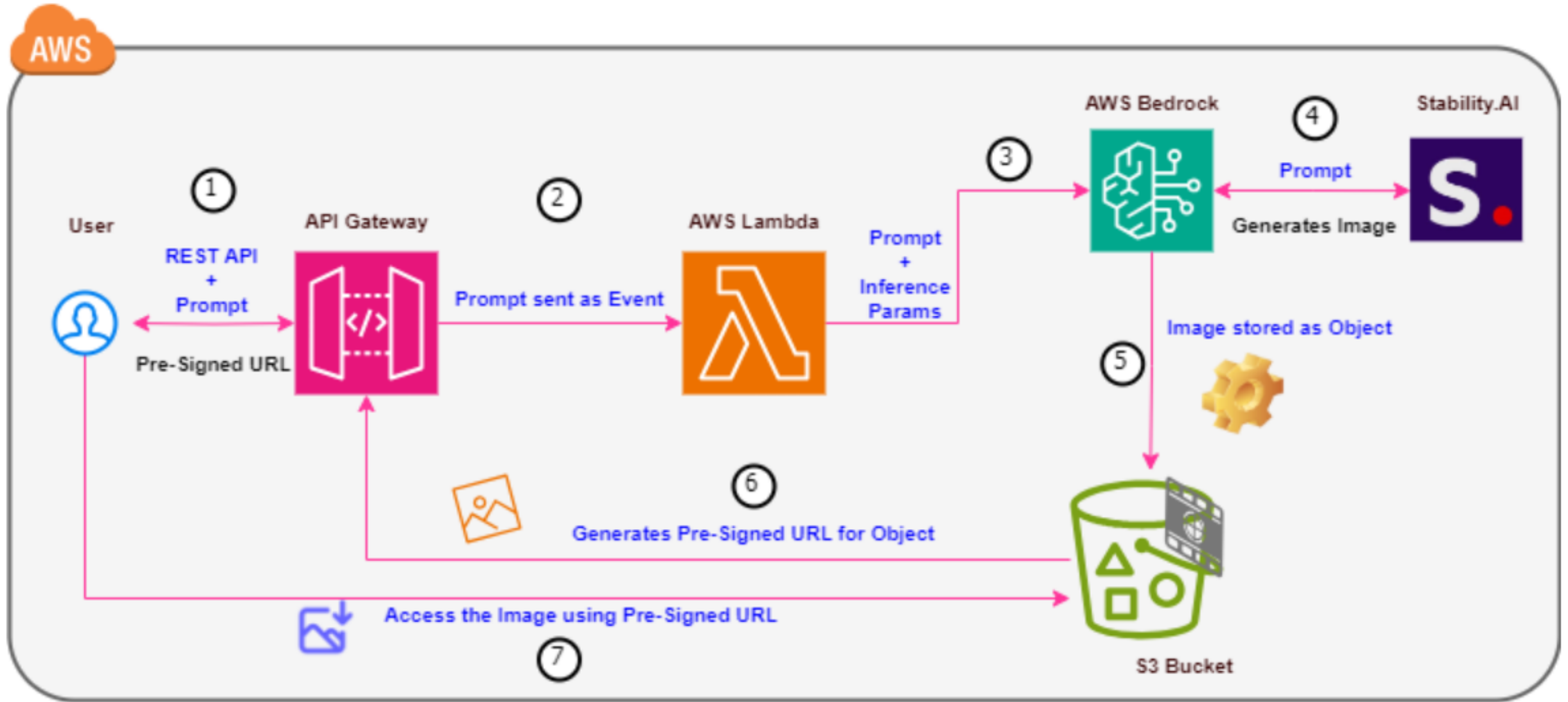
Stages in Movie making and potential role of Generative AI

Generative AI Use Case – Generate Movie Poster Design

Prompt : Generate movie poster photograph with hero shooting the villain with lots of guns in a scenic location with leaves and bullets flying



Movie Poster Design – Generative AI Architecture to Build



Cost to Implement the Use Case if you follow along

- Applicable for Use Case on Image Generation, Text Summarization, Chatbot and HR QnA RAG application.
- Not applicable for Amazon Bedrock Knowledgebase Use Case and Bedrock Agents
- For every **API request you make** - it will charge you at max **\$0.0125/request** (this is for AI21 Labs')
- So if you make 100 requests, it will charge you **$\$0.0188 \times 100 = \1.88**
- If you make **10 requests**, it will cost you \$.188
- So, you can say broadly for each use case you build, **it should not be more than \$1** assuming reasonable number of API request.

Cost to Implement the Use Case if you follow along

- **Bedrock Knowledgebases** is charged per hour and cost involved will be discussed in that lecture
- Please note the pricing is subject to change. Please refer to Bedrock pricing page for latest

Movie Poster Design - Broad Implementation Steps

1. Create an **S3 Bucket** – movieposterdesign01
2. Create a **AWS Lambda Function** - moviePosterDesignFunction
 - **Connect to AWS Bedrock** – Stability Diffusion Model and Generate an Image based on Prompt
 - **Store the Image as an Object** in the **S3 Bucket**
 - **Generate a Pre-Signed URL** for image generated
 - **Send the URL as a response to AWS API Gateway**
3. **Create a REST API using the AWS API Gateway** to allow user to pass the 'prompt' and view image using **Pre-Signed URL** - moviePosterDesignAPI
4. **Test using Postman API Tool**

Generative AI with Bedrock – Lambda Function Pre-requisites

1. Create a AWS Lambda Function
2. Check the boto3 version. Should be > 1.28.63
3. Use following command to check version - `print(boto3.__version__)`
4. Code for Bedrock invocation from AWS Lambda Function
 - Link - <https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/bedrock-runtime.html>

Movie Poster Design - Implementing Lambda Function Steps

1. Create an IAM Role and Increase timeout limit

- Full Access to Bedrock – Policy Name - [AmazonBedrockFullAccess](#)
- Full Access to S3 – Policy Name- [AmazonS3FullAccess](#)

2. Code for Bedrock invocation from AWS Lambda Function

3. Configure Test Event

```
{ "prompt": "generate an image of a dog with sunset"}
```


Movie Poster Design : Lambda Coding Steps

Broad Steps for writing Lambda Function

#1. import boto3

#2. Create client connection with Bedrock and S3 Services – [Link](#)

#3. Store the input data (prompt) in a variable

#4. Create a Request Syntax - Details from console and documentation(JSON object) - [Link](#)

#5. 5a. Retrieve from Dictionary, 5b. Convert Streaming Body to Byte using json load 5c. Print

#6. 6a. Retrieve data with artifact key, 6b. Import Base 64, 6c. Decode from Base64 - [Link](#)

Movie Poster Design : Lambda Coding Steps

Broad Steps for writing Lambda Function

#7. 7a. Upload the File to S3 using Put Object Method – [Link](#) 7b. Import datetime 7c.

Generate the image name to be stored in S3 - [Link](#)

#8. Generate Pre-Signed URL - [Link](#)

Movie Poster – API Gateway creation and Lambda Integration

4. Create an REST API from API Gateway

- Resource - movePosterDesignAPI
- Method - GET

5. API Gateway - Method Request

- URL query string parameters – Add input parameter
- Enable - Request Validation

Movie Poster– API Gateway creation and Lambda Integration

6. API Gateway - Integration Request

- application/json

```
{
```

```
"prompt" : "$input.params('prompt')"
```

```
}
```

7. Deploy the API to a Stage'(Stage=Dev or any other)

8. Test using API Gateway Console and Postman

Stable Diffusion Inference Parameters

Parameter	JSON object format	Description	Minimum	Maximum	Default
Prompt strength	cfg_scale	- Determines how much the final image portrays the prompt. Lower number to increase randomness in the generation.	0	30	10
Generation step	steps	- Determines how many times the image is sampled. - More steps result in a more accurate result.	10	150	30
Seed	seed	- Set as a random number and not needed - Use the same seed and the same settings as a previous run to allow inference to create a similar image.	0	4294967295	0

<https://docs.aws.amazon.com/bedrock/latest/userguide/model-parameters-diffusion.html>

Prompt Design for Stable Diffusion - 1

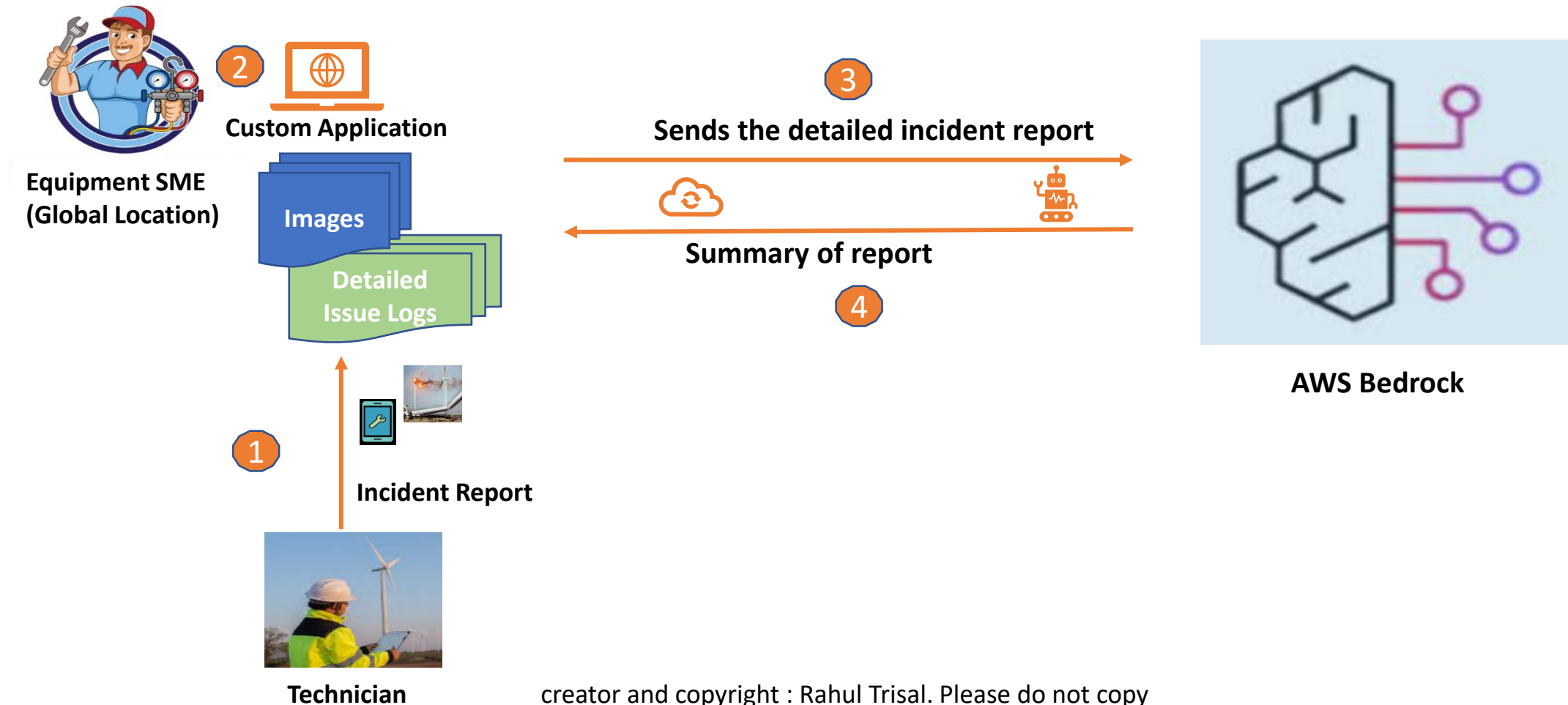
An image of a spy agent fighting in a rival country with guns and helicopters with backdrop of a shopping complex with heavy snow and old Greek architecture building late in the evening with sunsetting behind mountains. The image should be a photograph with Aaron Jasinski style

*Use Case – Text Summarization
for Manufacturing Industry*

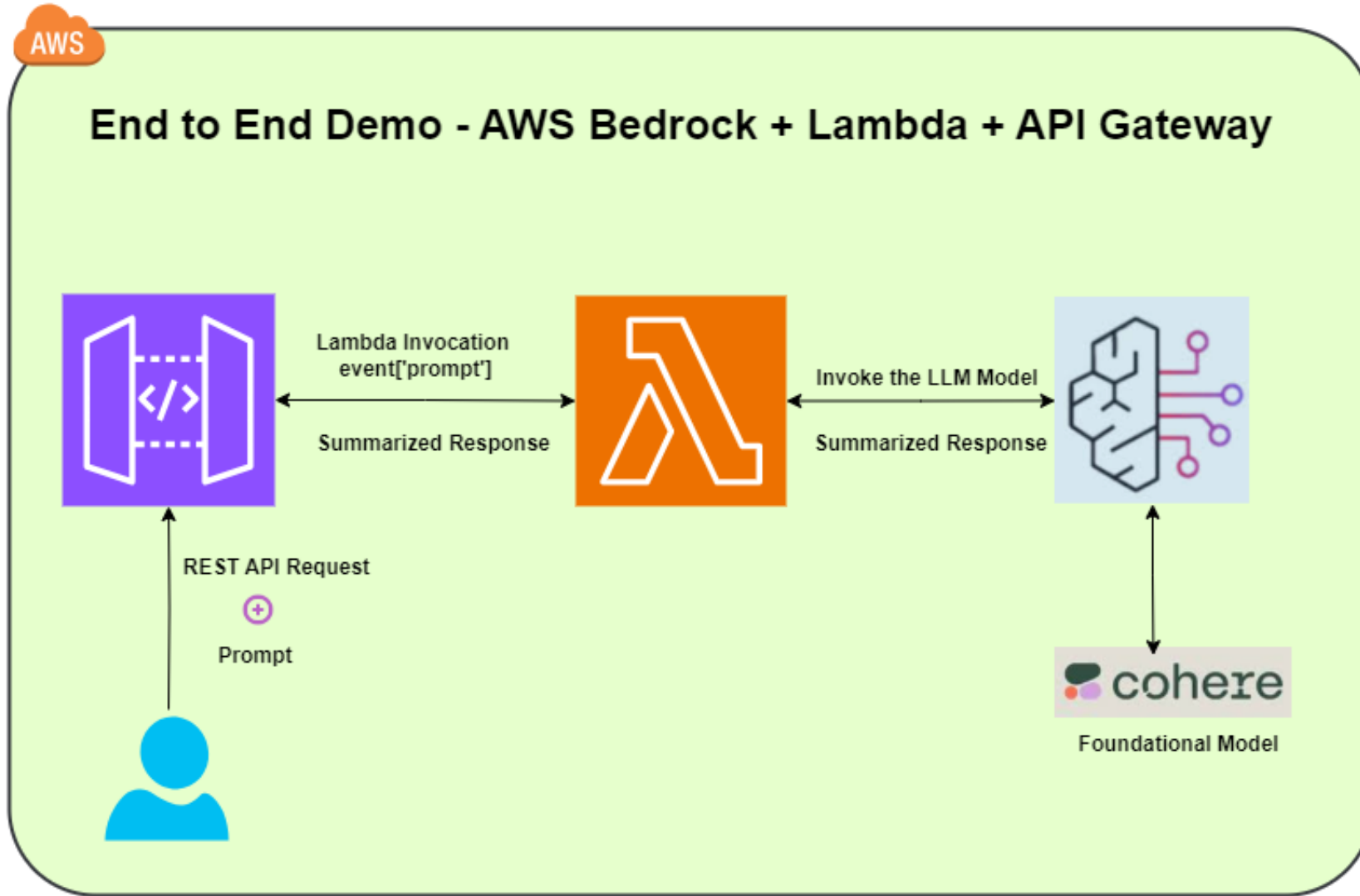
Generative AI – Manufacturing Industry Use Case

Use Case :

Text Summarization using **AWS Bedrock** for **faster issue resolution** to **improve productivity** of technicians.



Generative AI – Use Case Architecture



Generative AI with Bedrock : Step by Step – Pre-Requisites

1. Create a AWS Lambda Function – demoManufacturing
2. Access to Cohere Foundation Model in Amazon Bedrock Service
3. Check the boto3 version. Should be $\geq 1.28.63$
4. Use following command to check version - `print(boto3.__version__)`

Generative AI with Bedrock : Step by Step – Pre-Requisites

1. Create a AWS Lambda Function - demoManufacturing
2. Check the boto3 version. Should be $\geq 1.28.63$
3. Use following command to check version - `print(boto3.__version__)`

Generative AI with Bedrock : Step by Step Guide

1. Create an IAM Role and Increase timeout limit

2. Code for Bedrock invocation from AWS Lambda Function

- Link - <https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/bedrock-runtime.html>

3. Configure Test Event

```
{ "prompt": "How is weather in Bengaluru"}
```

Generative AI with Bedrock : Step by Step Guide

Broad Steps for writing Lambda Function

#1 **Import boto3** and create **client connection** with bedrock

#2 a. Store the input in a variable, b. print the event

#3. Create Request Syntax - Get details from console & body should be json object - use `json.dumps` for body

#4. Convert Streaming Body to Byte(`.read` method) and then Byte to String using `json.loads`

5 a. Print the event and type , b. Store the input in a variable

#6. Update the 'return' by changing the 'body'

Generative AI with Bedrock : Step by Step Guide

4. Create an REST API from API Gateway

- Resource – demoManufacturing
- Method - POST

5. API Gateway - Method Request

- URL query string parameters – Add input parameter

Generative AI with Bedrock : Step by Step Guide

6. API Gateway - Integration Request

- application/json

```
{ "prompt": "$input.params('prompt')" }
```

7. Deploy the API to a Stage

8. Test using API Gateway Console

9. Sample Text - [Link](#)

Prompt for Summarization Task

This is a on-site log report of turbine breakdown.

Issue Log Date – 25-12-2023, **Model Number** – TB-CL-7882, **Issue** - Cracks appeared in the part MR 7882-9571 next to the rotor hub. The nut connecting the rotor blade to the rotor hub seems to be damaged. The Anemometer readings seem to be within range. The electric braking seems to be unused. No indication of damage to any other component of the turbine except normal wear and tear.

Potential Root Cause – Seems due to reduced tensile strength of the nut connecting the blade to the rotor. **Last Maintenance Date** – 12-12-2023, **Last Maintenance Issues Recorded** - No known issues recorded and all the parameters were within range.

Summarize the text in 2 lines.

Possible and Actual – Completion from the Model

Possible Response:

Model No-TB-CL-7882, Issue - The nut connecting the rotor blade to the rotor hub seems to be damaged,

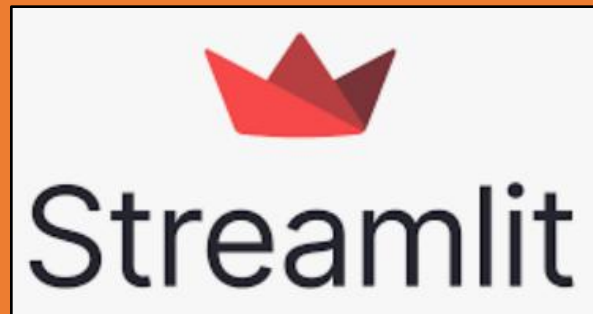
Potential Root Cause – Seems due to reduced tensile strength of the nut connecting the blade to the rotor.

No known issues recorded in last maintenance.

Actual Response :

The turbine experienced cracks in a part and possible damage to the nut connecting the rotor blade, but the a
nemometer readings were within range. It is unclear what has caused the issue, but it is assumed to be due
to reduced tensile strength.

Use Case – Building a Chatbot



Building a Chatbot using Amazon Bedrock : Architecture

Chatbot using Amazon Bedrock – What we will build as part of Use Case

Hi, This is Chatbot Anisha 🕶️



Hi



Hi, my name is AI. How can I assist you today? How can I help you?

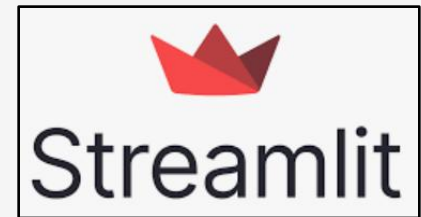


Where is Bengaluru ?

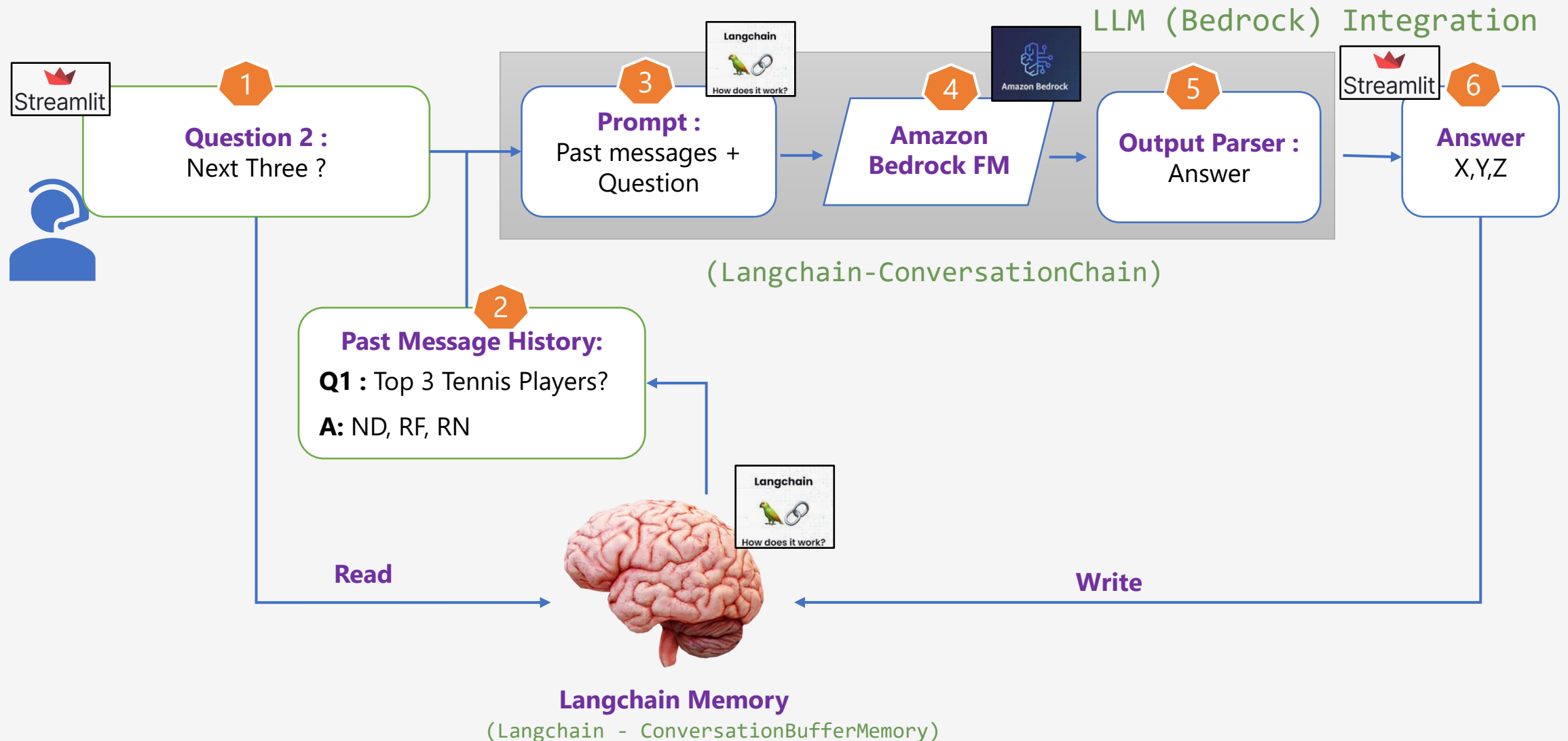


Bengaluru is the capital of Karnataka, a state in South West India.

Chat with Udemy Course bot here



Chatbot Architecture – Amazon Bedrock + Langchain + Streamlit



Pre-Requisites

Building a Chatbot using Amazon Bedrock

Pre-requisites – How to install

1/2

- Download VS Code - <https://code.visualstudio.com/download>
- Install Python - 3.12.4 ----> <https://www.python.org/downloads/windows/>
 - `python --version`
 - `pip --version`
- Install AWS CLI - <https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>
 - `aws -- version`
 - Extensions – AWS Toolkit and AWS Boto3
- Configure IAM Role for VSCode - <https://docs.aws.amazon.com/cli/latest/userguide/cli-configure-files.html>
 - Create IAM Role
 - `aws configure`
 - Provide credentials
 - `aws s3 ls`

Pre-requisites - How to install

2/2

- Anaconda Navigator Download - <https://docs.anaconda.com/free/anaconda/install/windows/>
- Open VS Code from Anaconda Navigator – **Very Important**
- Install boto 3 - `pip install boto3==1.37.11`
- Install Langchain – `pip install langchain==0.3.0`
- Install Langchain-AWS Module – `pip install -U langchain-aws`
- Install Streamlit – `pip install streamlit==1.43.1`
- Install Transformers - `pip install transformers`
- Install PyYAML - `pip install PyYAML`

Coding Steps

*Building a **Chatbot** using Amazon Bedrock*

Steps in Building Chatbot – Chatbot Backend Code

#1 import the ConversationSummaryBufferMemory, ConversationChain, ChatBedrock or ChatBedrockConverse Langchain Modules

#2a Write a function for invoking model- client connection with Bedrock with profile, model_id & Inference params- model_kwargs

#2b Test out the LLM with invoke method

#3 Create a Function for ConversationBufferMemory (llm and max token limit)

#4 Create a Function for Conversation Chain - Input text + Memory

#5 Chat response using invoke (Prompt template)

#Links :

<https://docs.aws.amazon.com/bedrock/latest/userguide/conversation-inference-call.html>

https://docs.aws.amazon.com/bedrock/latest/APIReference/API_runtime_Converse.html

<https://docs.aws.amazon.com/bedrock/latest/userguide/conversation-inference-call.html>

Converse API

Set the model ID, e.g., DeepSeek-R1 Model.

```
model_id = "us.deepseek.r1-v1:0"
```

Start a conversation with the user message.

```
user_message = "Type_Your_Prompt_Here"
```

```
conversation = [ { "role": "user", "content": [{"text": user_message}], } ]
```

#Converse URL

[#https://docs.aws.amazon.com/bedrock/latest/userguide/conversation-inference-call.html](https://docs.aws.amazon.com/bedrock/latest/userguide/conversation-inference-call.html)

[#https://docs.aws.amazon.com/bedrock/latest/APIReference/API_runtime_Converse.html](https://docs.aws.amazon.com/bedrock/latest/APIReference/API_runtime_Converse.html)

[#https://docs.aws.amazon.com/bedrock/latest/userguide/conversation-inference-call.html](https://docs.aws.amazon.com/bedrock/latest/userguide/conversation-inference-call.html)

Steps in Building Chatbot – Chatbot Frontend Code

- How to Run Chatbot locally
 - `streamlit run <frontend_filename>.py`

Langchain Overview :



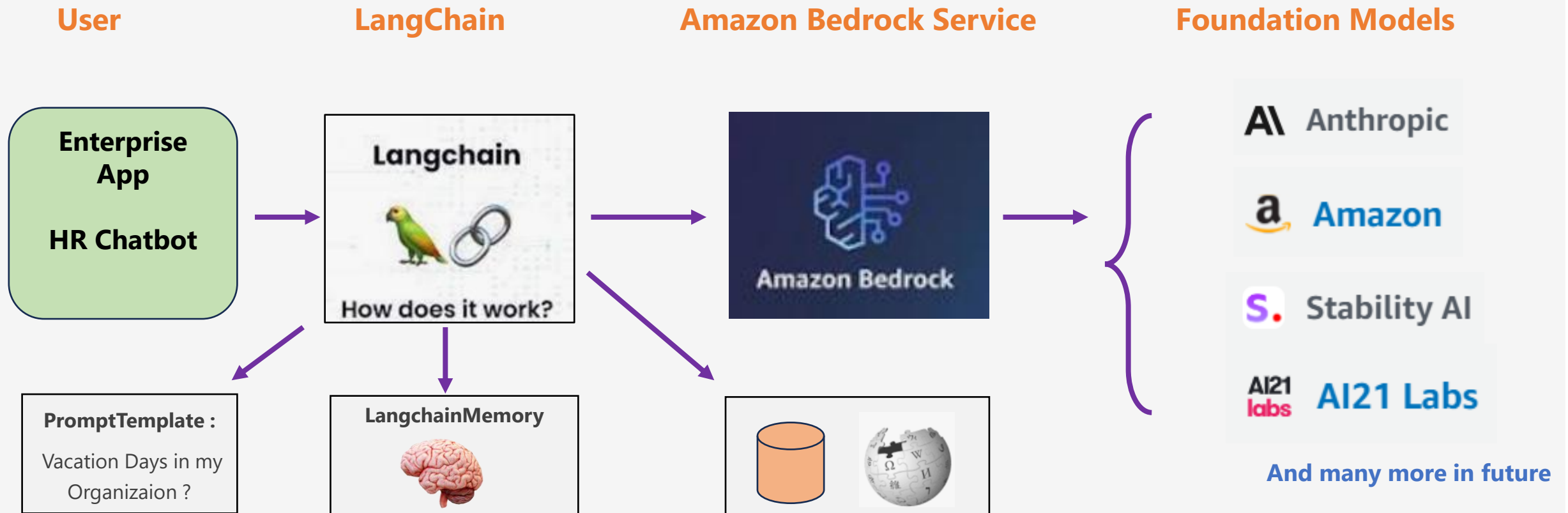
*Challenges in Large Language Models
&
How Langchain helps overcome them*

Challenges in Large Language Models

- **Contextual Memory** - Large Language Models are **Stateless**
- **Contextual Knowledge (Domain Knowledge)** - Enhancing the outcome of LLM's with **organization proprietary data**
- **Hallucination** – Due Lack of knowledge of **recent events (post training date)** or no training data in the specific area.

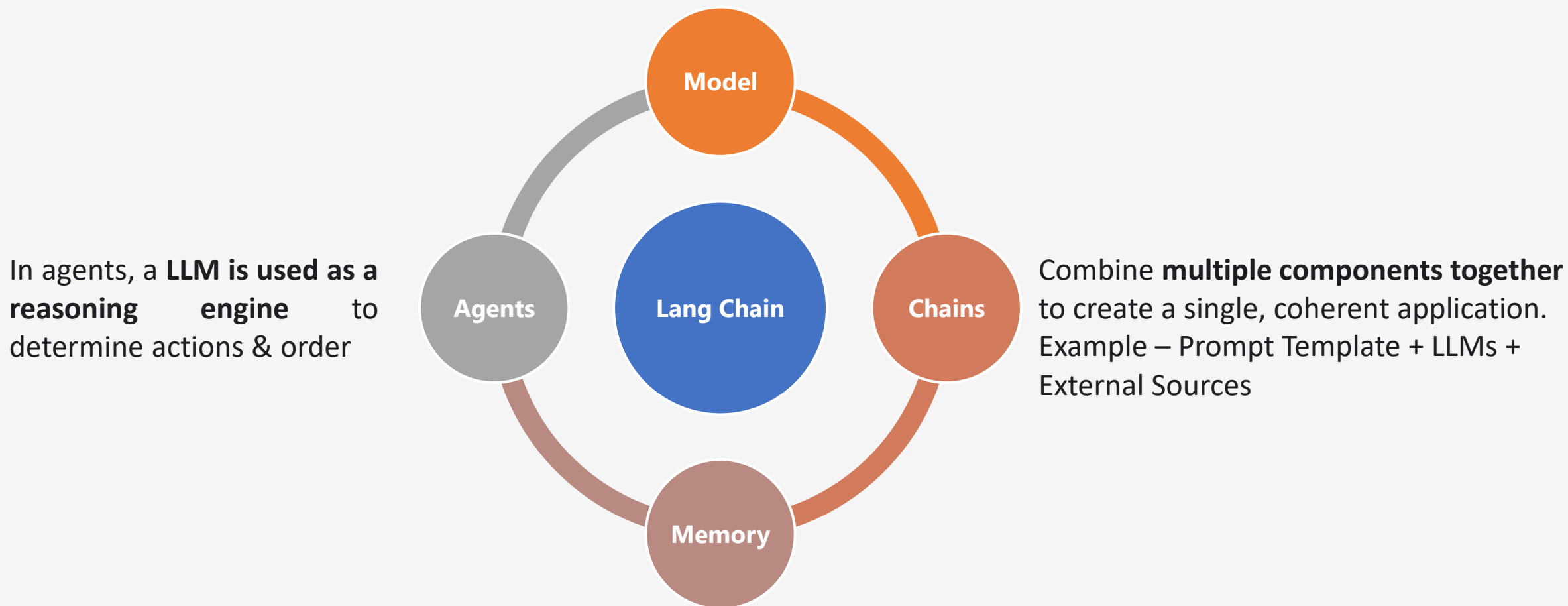
What is LangChain ?

LangChain is a **framework for developing applications** powered by language models.

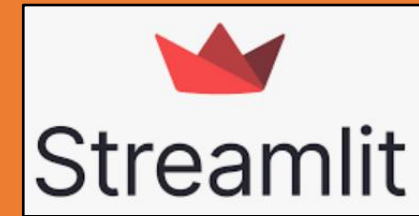


Lang Chain Components – High Level Overview

Standard interface to connect different LLMs such as Claude, Titan



Provides ability to **store information about past interaction** - Chatbots



Streamlit – Overview

Use Case : Retrieval Augmented Generation (RAG)

Employee HR Q & A Application with RAG

Amazon Bedrock - Claude FM

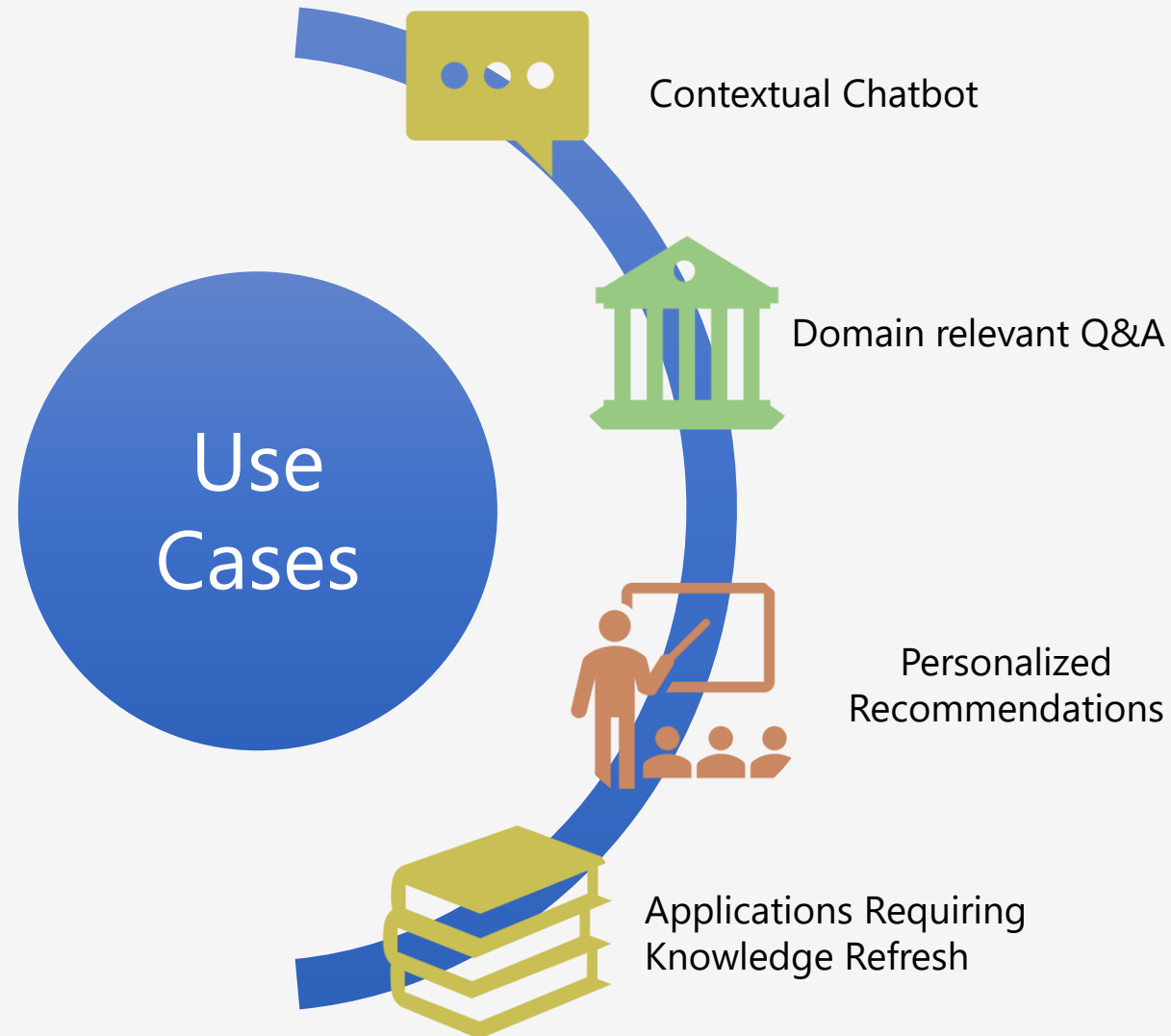
+

Langchain

+

HR Policy Document (PDF)

Key Use Cases for Retrieval Augmented Generation in the Industry



Basic Introduction
to
Vectors, Embeddings and Vector Search

1. Why do we need Vectors ?



Structured Data

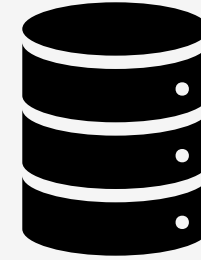
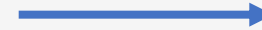
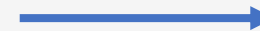


Image Search or Unstructured Data



???

Key Terms

1. **Vectors**
2. Embedding Models
3. Data Chunking
4. Vector Store
5. Vector Search
6. Cosine Similarity
7. K-Nearest Neighbor Search

1. What are Vectors ?

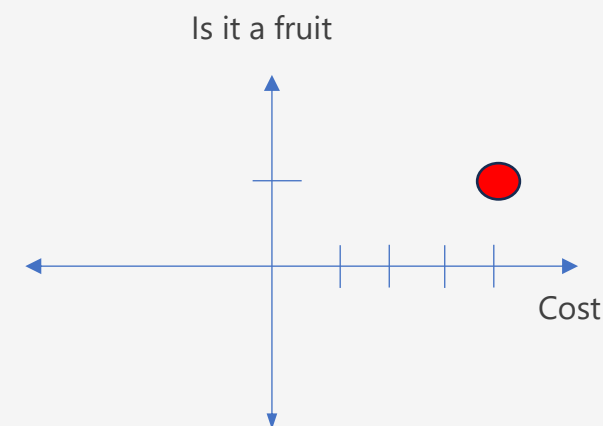
Vectors are **mathematical representation of words, sentences** and documents

Vectors are represented by **List of numbers** or sequence of numbers



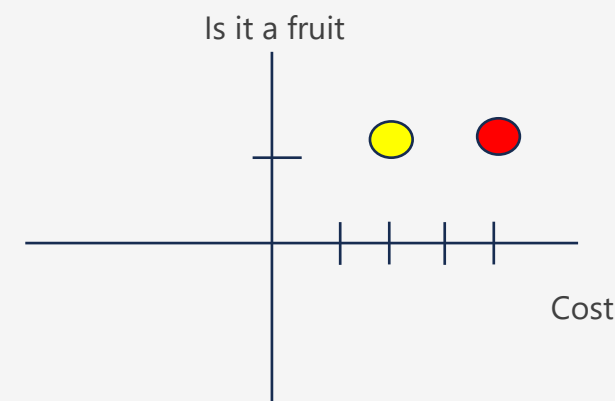
Is it a fruit	Cost
1	\$4

$[1, 4]$



Is it a fruit	Cost
1	\$2

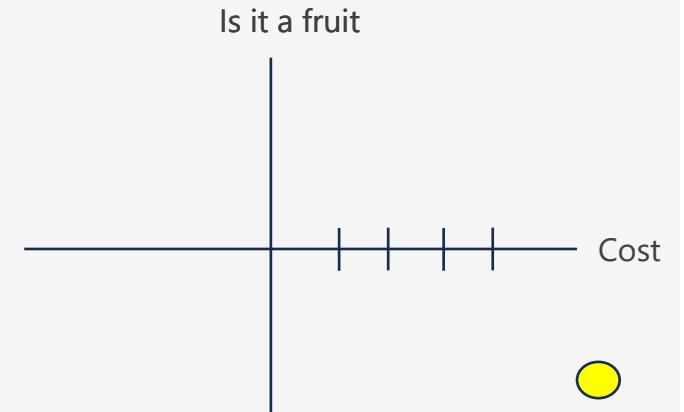
$[1, 2]$



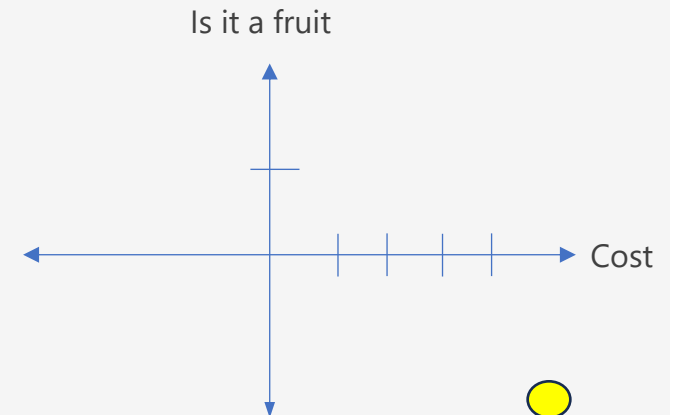
1. What are Vectors ?



Is it a fruit	Price
-1	\$50

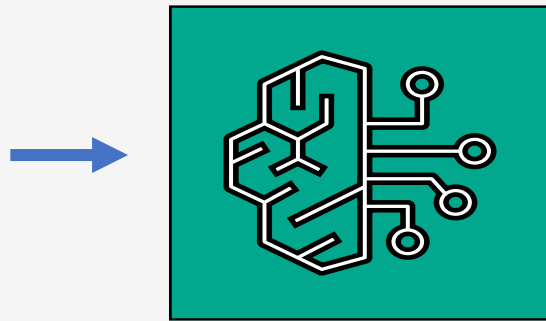
$$[-1, 50]$$


Is it a fruit	Price
-1	\$40

$$[-1, 40]$$


2. What are Embedding Models ?

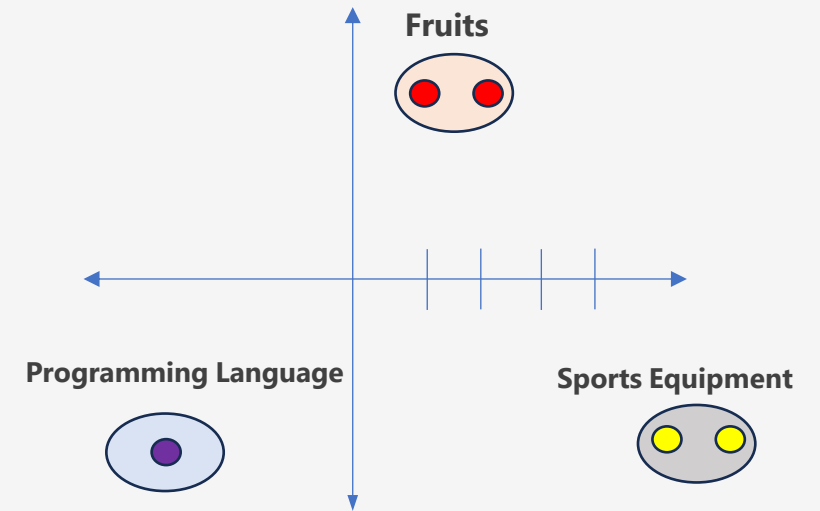
- Apple
- Banana
- Grapes
- Football
- Soccer
- GO
- Tennis



Amazon Titan Embeddings

[0.4, 0.2, 0.1, ...]
[0.4, 0.2, 0.1, ...]
[0.4, 0.2, 0.1, ...]
[0.4, 0.2, 0.1, ...]

1536 Dimensions



Word Embeddings
Ex : great

Sentence Embeddings
Ex. : Price of apple ?

Doc Embeddings

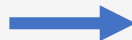
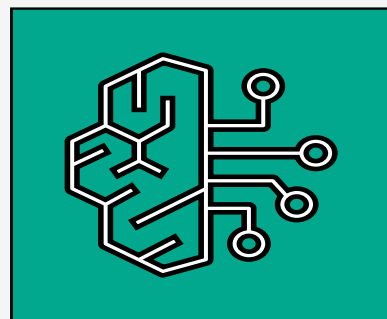
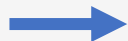
Key Terms

1. Vectors
2. **Embedding Models**
3. Data Chunking
4. Vector Store
5. Vector Search
6. Cosine Similarity
7. KNN

Embedding Models

Output from Embedding Models

Fruit



Amazon Titan Embeddings

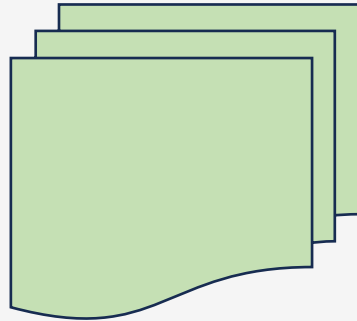
```
[
  0.022321066, -0.027544279, -0.006137953, 0.0024092742, -0.0003515296,
  -0.007889225, -0.01853968, -0.037405808, -0.004294867, -0.012602357,
  0.009385457, 0.02421176, -0.004383281, -0.011901848, -0.020702416,
  0.00431187, 0.055986296, 0.0040738326, 0.021300908, -0.03713377,
  -0.004944368, 0.012269106, 0.02614326, -0.019777471, 0.0017206672,
  0.014867109, 0.004597514, -0.021913003, -0.0042336574, 0.013180447,
  0.032699477, -0.027245032, -0.0112965545, -0.009358253, -0.004951169,
  -0.015030335, -0.007501565, -0.018906936, 0.0074675595, -0.0067568496,
  0.0011952856, 0.0035501514, 0.012316713, 0.016132105, -0.014839904,
  -0.0044002836, 0.010405616, -0.0066276295, -0.015751246, -0.007141109,
  0.02729944, -0.00031348618, -0.015669633, -0.015656032, 0.015696838,
  0.013554505, -0.001635654, 0.006069943, 0.009167824, -0.01917898,
  -0.006059741, 0.013969369, -0.03006067, 0.007691995, 0.0005028531,
  -0.005468049, -0.007263528, 0.009970348, -0.016254524, 0.0041248407,
  0.01964145, 0.037324198, -0.0011340762, -0.0013296065, 0.010392014,
  -0.01268397, -0.013037625, -0.006811258, -0.0045601083, 0.0028955496,
  0.027435461, -0.024320576, -0.006161757, 0.014350229, 0.009392259,
  -0.006328383, -0.025490358, 0.023273215, 0.0083925035, -0.003934411,
  0.023722084, -0.0072499258, -0.0053626327, 0.017818768, -0.03327077,
  0.019097365, 0.0022086431, 0.046709653, -0.016825814, -0.032182597,
  ... 1436 more items
]
```

1536 Dimensions

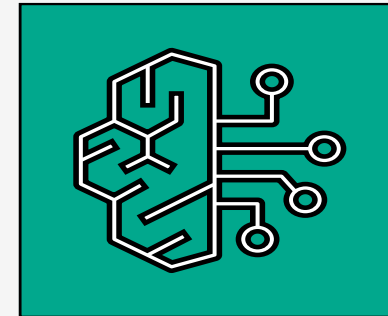
3. Text Splitting or Data Chunking



Data Source - PDF



- Split by Character
- Split by Token
- Split by Code



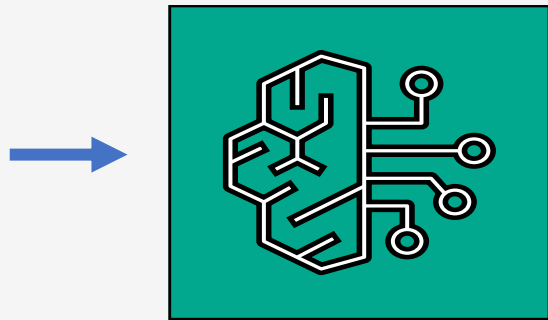
Amazon Titan Embeddings

Key Terms

1. Vectors
2. Embedding Models
3. **Data Chunking**
4. Vector Store
5. Vector Search
6. Cosine Similarity
7. K-Nearest Neighbor Search

4. Vector Store?

- Apple
- Banana
- Grapes
- Football
- Soccer
- GO
- Tennis

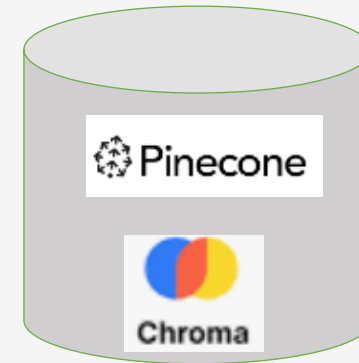


Amazon Titan Embeddings

[0.4, 0.2, 0.1, ...]
[0.4, 0.2, 0.1, ...]
[0.4, 0.2, 0.1, ...]
[0.4, 0.2, 0.1, ...]

1536 Dimensions

Vector Store

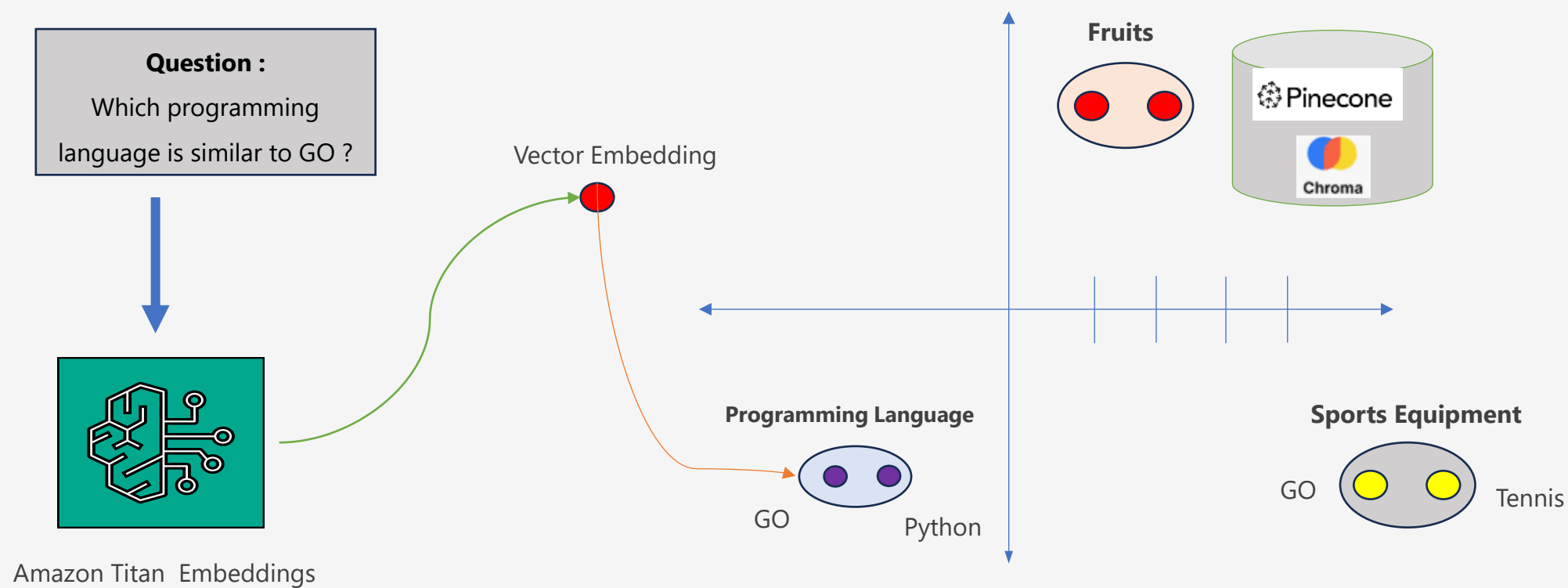


- ✓ FAISS
- ✓ Pinecone
- ✓ Chroma DB

Key Terms

1. Vectors
2. Embedding
Models
3. Data Chunking
4. **Vector Store**
5. Vector Search
6. Cosine Similarity
7. K-Nearest
Neighbor Search

5. Vector Search

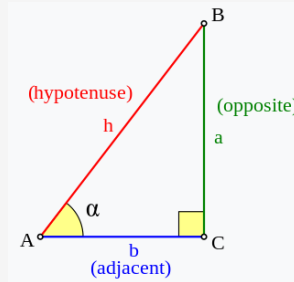


Key Terms

1. Vectors
2. Embedding Models
3. Data Chunking
4. Vector Store
5. **Vector Search**
6. Cosine Similarity
7. K-Nearest Neighbor Search

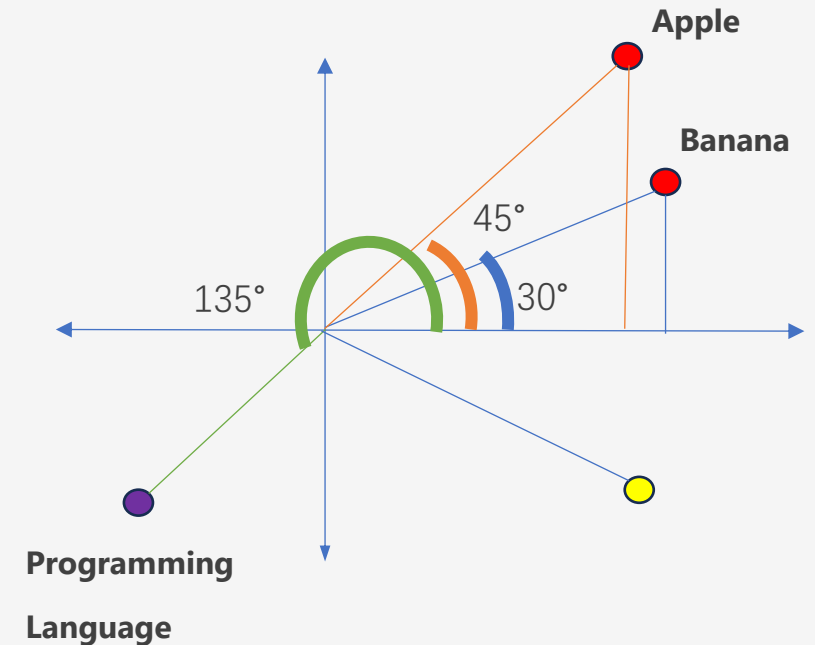
6. Similarity Search – Cosine Similarity

1. $\text{Cosine } \theta = \text{adjacent/hypotenuse}$



2. Similarity Search using Cosine Distance:

- Cosine distance metric is used to find **similarities between different vectors**
- In **cosine metric** we measure the **degree of angle between two vectors**
- **$\text{Cos } 0 = 1$, $\text{Cos } 90 = 0$ and $\text{Cos } 180 = -1$**
- **Cosine value 1** is for **vectors pointing in the same direction --- > Similar**
- At Cosine Value 0 ---- > Vectors unrelated (some similarity found).
- Value -1 for vectors pointing in opposite directions(No similarity).



7. Cosine Similarity and K- Nearest Neighbor Algorithm or ANN

KNN is a ***supervised machine learning algorithm*** used **classification and regression** tasks.

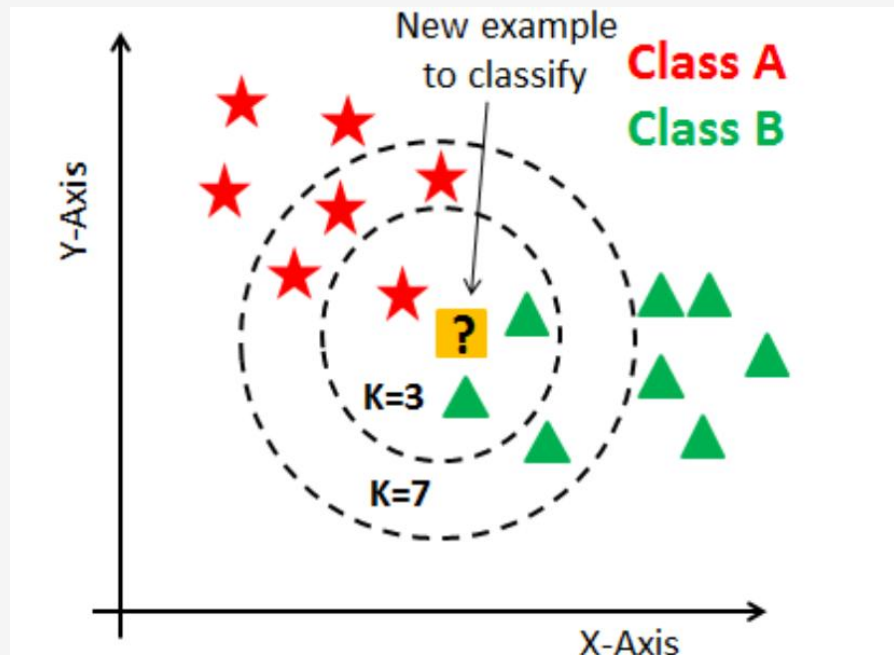


Image Source : Internet

Key Terms

1. Vectors
2. Embedding Models
3. Data Chunking
4. Vector Store
5. Vector Search
6. Cosine Similarity
7. KNN

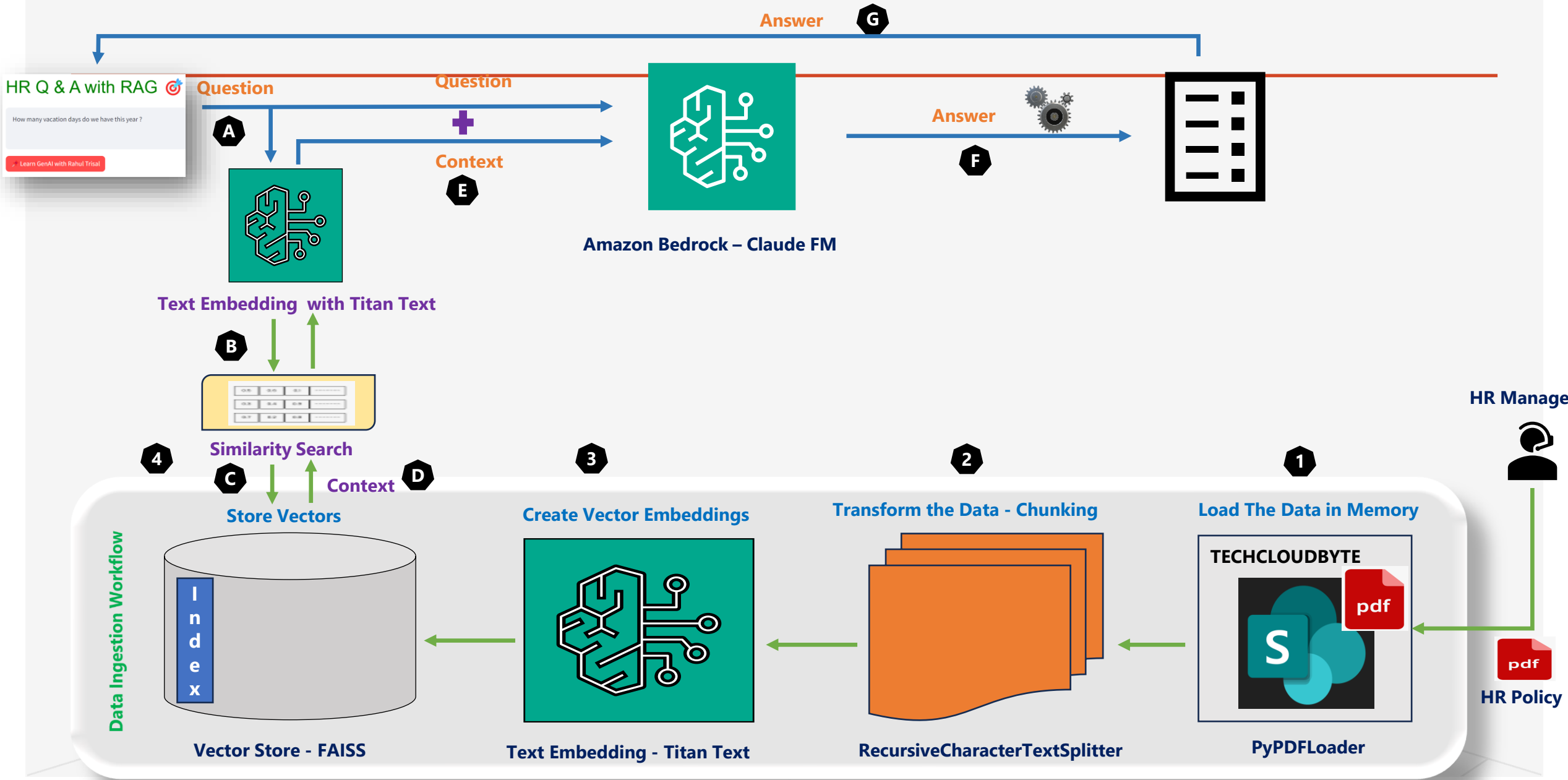
ANN - Approximate Nearest Neighbor

Use Case : Retrieval Augmented Generation (RAG)

Employee HR Q & A Application with RAG

Architecture Overview

HR Question & Answer Application with Retrieval Augmented Generation - Architecture



Use Case : Retrieval Augmented Generation (RAG)

Employee HR Q & A Application with RAG

Pre- Requisites - Installation

Pre-requisites – List of items to be installed

1. **pip3 install flask-sqlalchemy**
 2. **pip install pypdf**
 3. **pip install faiss-gpu** (for CUDA supported GPU)
- or**
4. **pip install faiss-cpu** (depending on Python version)
 5. In case of issue with **import PyPDFLoader**, run below command

pip install -U langchain-community

Pre-Requisites

For installation of other items for this use case, if not already installed, refer to below Section and Lecture (one shown in video as well). If already installed, skip next few slides.

Section : Building a Chatbot with Llama 2, Langchain & Streamlit

Lecture : Chatbot - Environment Setup before coding

Use Case : Employee HR Q & A App with RAG

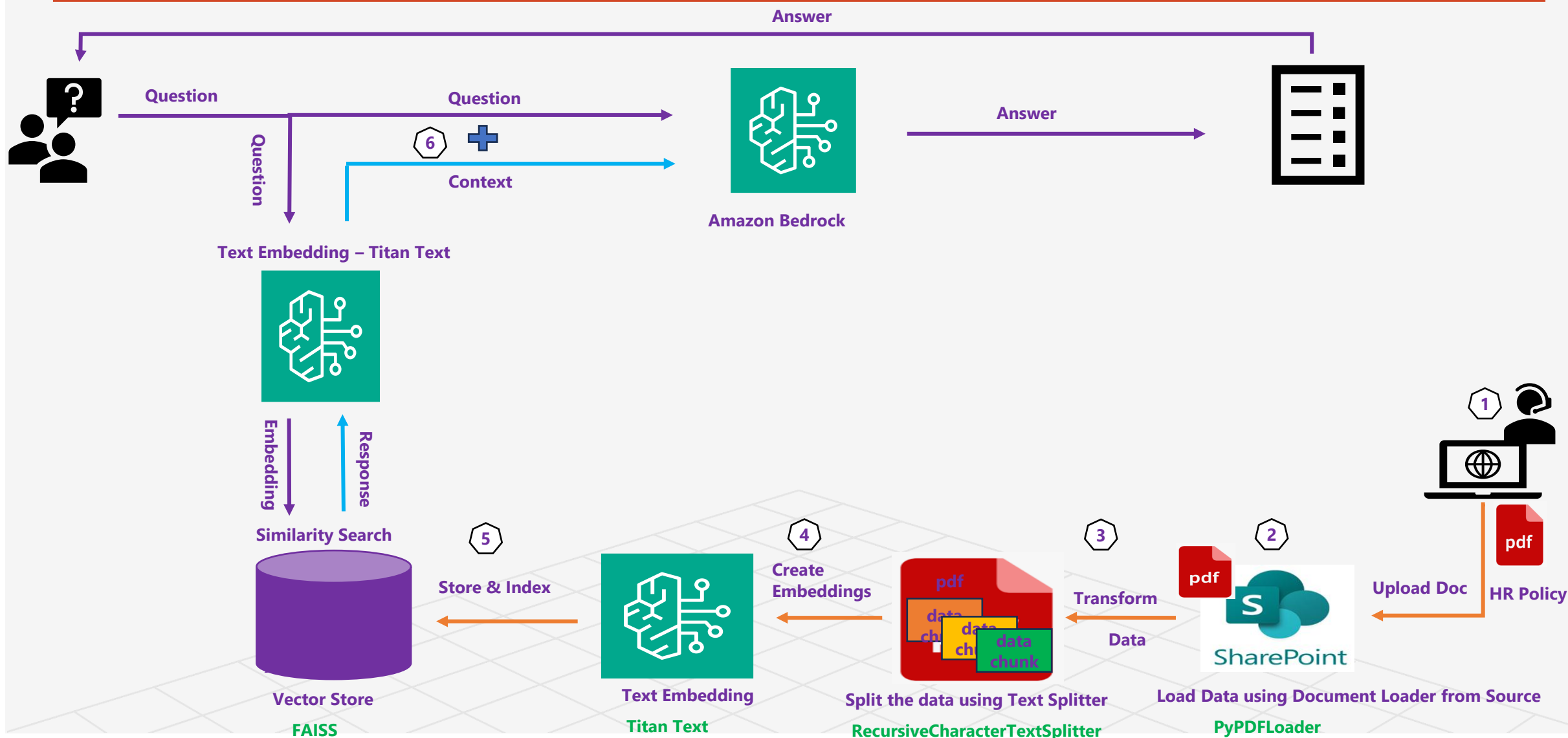
Coding Steps

QUICK ALERT

Steps in Building HR Q&A Solution with RAG – Backend Code

- #1. Import OS, Document Loader, Text Splitter, Bedrock Embeddings, Vector DB, VectorStoreIndex, Bedrock-LLM
- #2. Define the data source and load data with PDFLoader(<https://www.upl-ltd.com/images/people/downloads/Leave-Policy-India.pdf>)
- #3. Split the Text based on Character, Tokens etc. - Recursively split by character - ["\n\n", "\n", " ", ""]
- #4. Create Embeddings -- Client connection
- #5a Create Vector DB, Store Embeddings and Index for Search - VectorstoreIndexCreator
- #5b Create index for HR Report
- #5c. Wrap within a function
- #6a. Write a function to connect to Bedrock Foundation Model
- #6b. Write a function which searches the user prompt, searches the best match from Vector DB and sends both to LLM.
- # Index creation --> <https://api.python.langchain.com/en/latest/indexes/langchain.indexes.vectorstore.VectorstoreIndexCreator.html>

HR Question & Answer App with RAG - Solution Architecture



Steps in Building HR Q&A Solution with RAG – Frontend Code

- Modify the Streamlit frontend code where `###` highlighted
- Run Chatbot locally
 - `streamlit run <frontend_filename>.py`

Data Source URL

<https://www.upl-ltd.com/images/people/downloads/Leave-Policy-India.pdf>

Common Issues for the Use Case and Solution

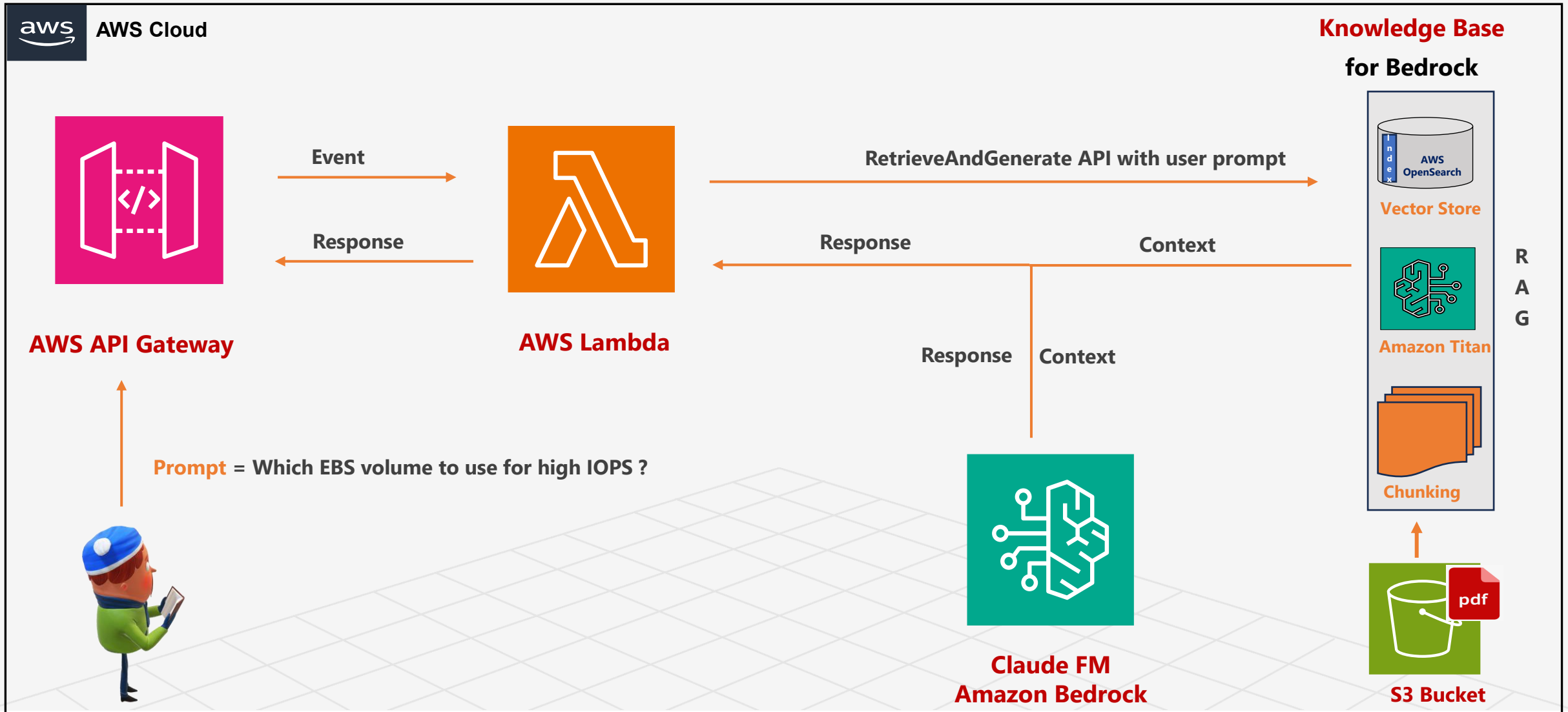
❖ Issue 1 - Incase you face issue with **import PyPDFLoader** (due to depreciation by langchain)

- `pip install -U langchain-community`
- `from langchain_community.document_loaders import PyPDFLoader`

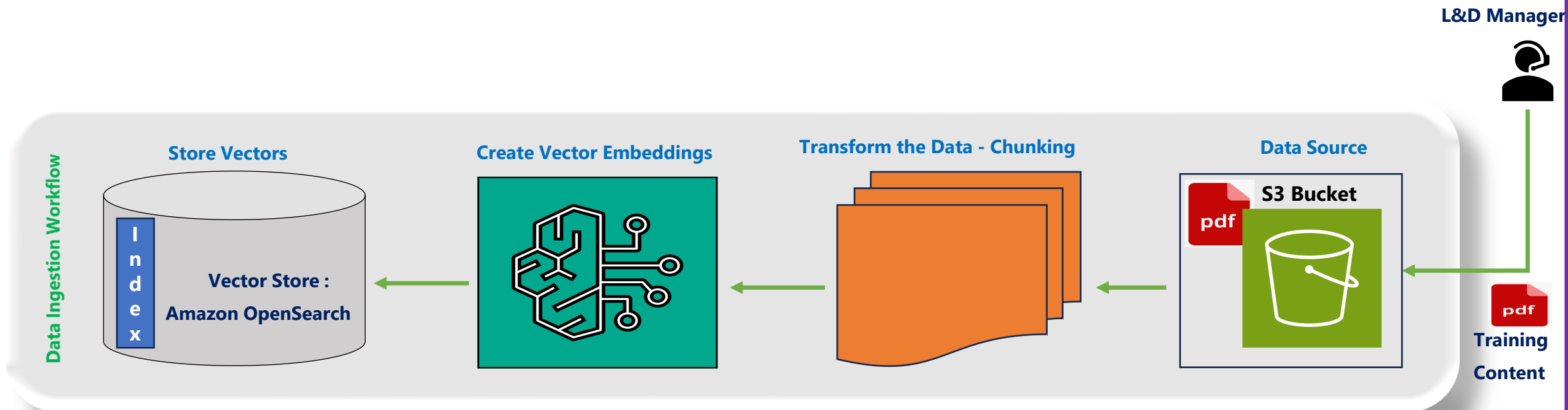
Use Case :

**Serverless e-Learning App using Knowledge
Base for Bedrock**

Serverless e-Learning App using Knowledge Base for Bedrock



E-Learning Solution using RAG - Amazon Bedrock Knowledge Base + LLM



Data Ingestion Workflow - Options

1. Data Sources

- ❖ S3

2. Chunking

- ❖ Default
- ❖ Fixed Size

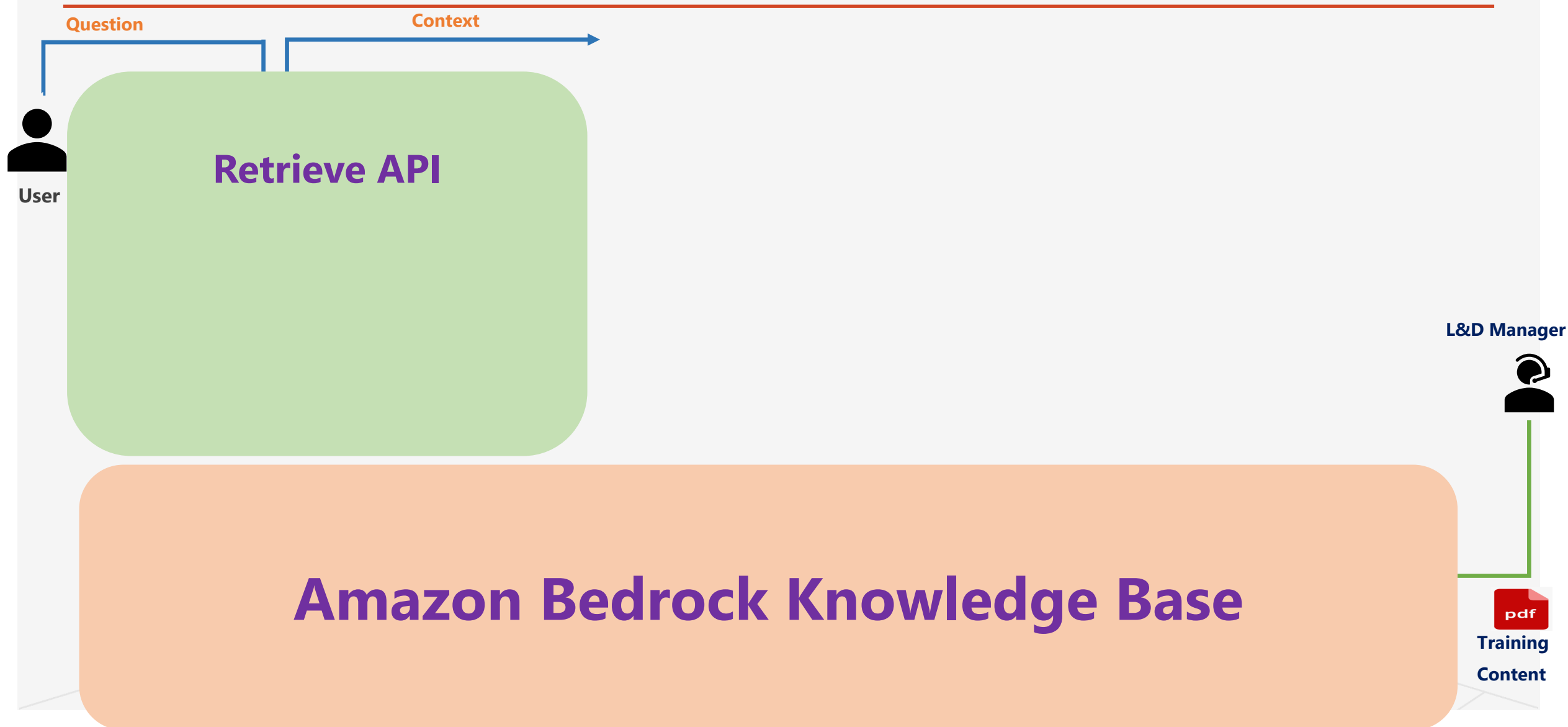
3. Embedding Model

- ❖ Cohere
- ❖ Amazon Titan

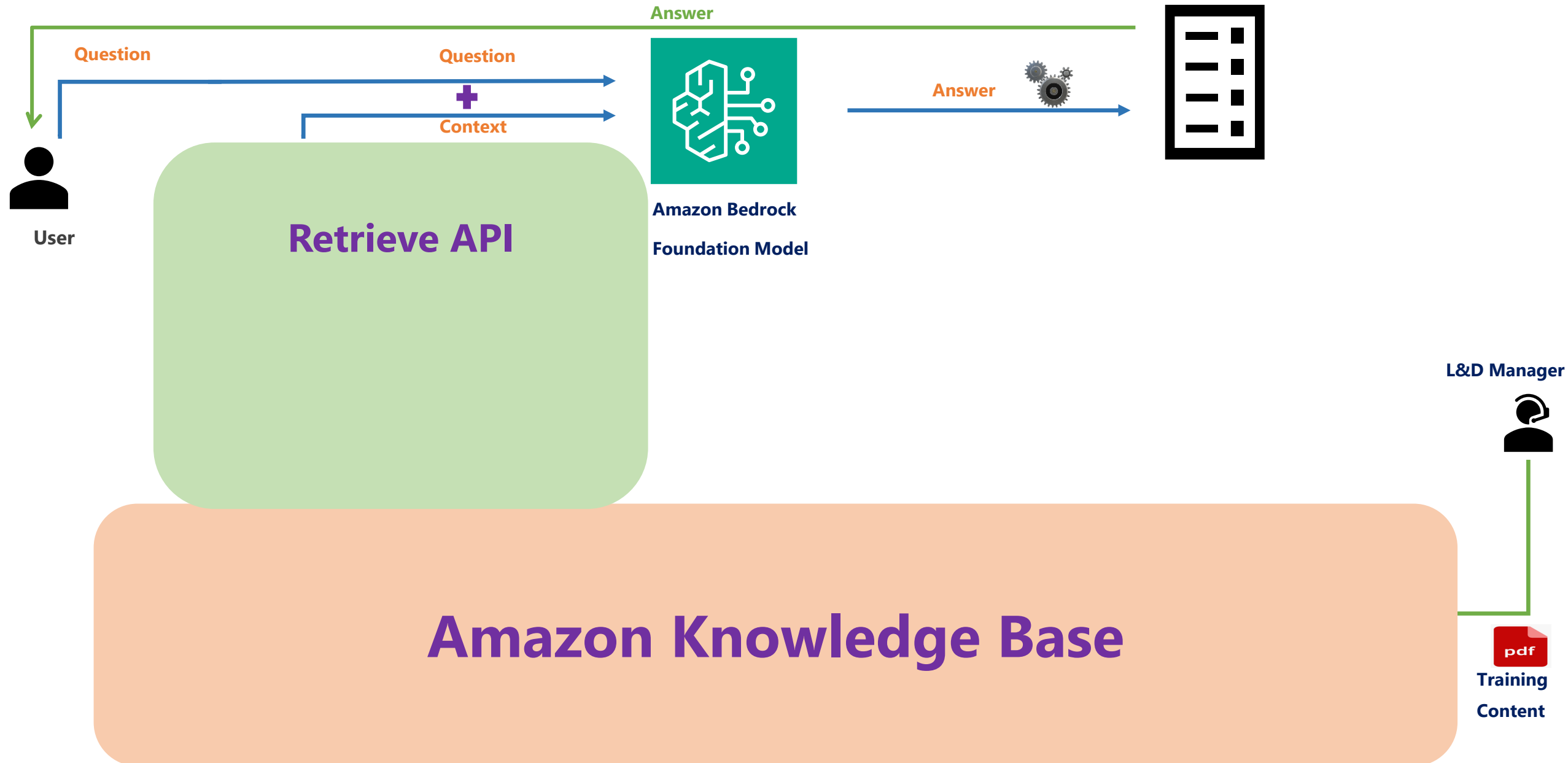
4. Vector DB

- ❖ Amazon OpenSearch Serverless vector store
- ❖ Amazon Aurora
- ❖ Pinecone
- ❖ Redis Enterprise Cloud

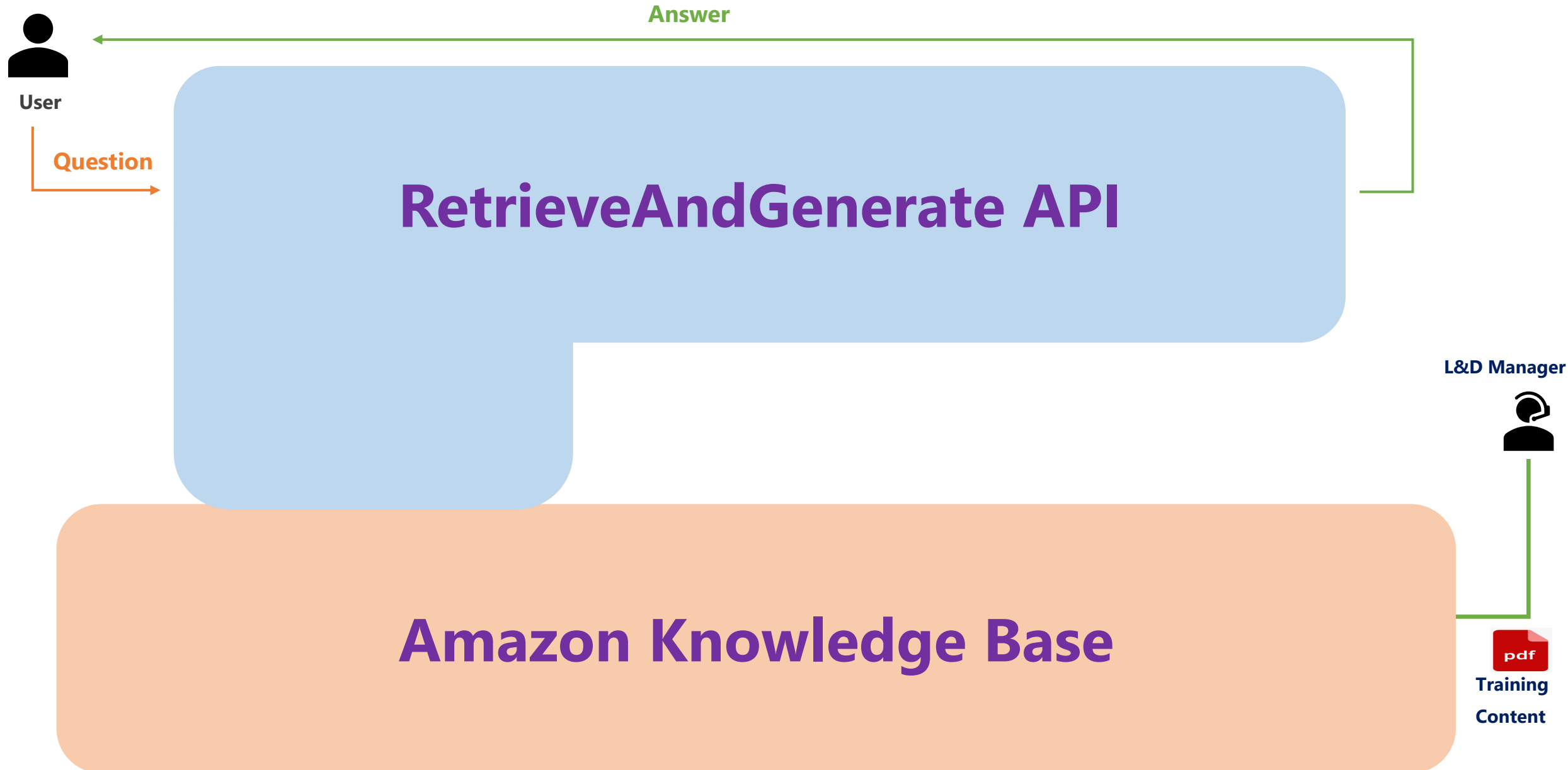
L&D App with Amazon Knowledge Base (Retrieval Augmented Generation) - Architecture



L&D App with Amazon Knowledge Base (Retrieval Augmented Generation) - Architecture



L&D App with Amazon Knowledge Base (Retrieval Augmented Generation) - Architecture



Two API's for Retrieval – Retrieve API and RetrieveAndGenerate API

1. Retrieve API :

- Queries a Knowledge Base
- **Retrieves** information from it.

https://docs.aws.amazon.com/bedrock/latest/APIReference/API_agent-runtime_Retrieve.html

2. RetrieveAndGenerate API

- Queries a knowledge base
- **Generates responses** based on the **retrieved results**.
- Response **cites up to five sources** but only selects the ones that are **relevant to the query**.

https://docs.aws.amazon.com/bedrock/latest/APIReference/API_agent-runtime_RetrieveAndGenerate.html

Broad Implementation Steps

1. **Data Source** –

- Create the **S3 Bucket**
- **Upload** the relevant **files** in the S3 Bucket

2. Create Amazon Bedrock **Knowledge Base** --- > **Data Source, Chunking, Embedding Model, Vector Store**

3. Create a **AWS Lambda Function**

- Create IAM Role
- Increase timeout limit
- Write the **RetrieveAndGenerate API** to access data from the Knowledge Base – [Link](#)
- Model ARN - 'modelArn': '**arn:aws:bedrock:us-east-1::foundation-model/anthropic.claude-instant-v1**'

4. Optional Step incase of error - Update boto3 version incase of error using Lambda Layers - [Link](#)

5. Create **REST API**

6. **Integrate** AWS Lambda function with API Gateway

Broad Implementation Steps

Create REST API using AWS API Gateway

- Create Resource
- Create Method – GET

API Gateway - Method Request

- URL query string parameters – Add input parameter
 - Enable - Request validator
 - Validate body, query string parameters, and headers
- 

Broad Implementation Steps

- **API Gateway - Integration Request**

application/json

```
{  
  "prompt": "$input.params('prompt')"  
}
```

- **Deploy the API to a Stage' (Stage=Dev or any other)**
- **Test using API Gateway Console and Postman**

Use Case :

Build Retail Banking Agent

using

Bedrock Agents & Knowledge Bases

Amazon Bedrock Agents – Retail Banking Agent

Bank of Chicago



John

Check account activation status after 3 days from Bank Website



Thomas



Manju



AccountID = 5555



AccountID = 6666



AccountID = 7777



GET UP TO
\$500

Amazon Bedrock Agents – Retail Banking Agent



Manju
AccountID = 7777

→ ↺  **www.bankofchicago.com** ☆

Test TestAlias: Working draft ▼ Info

 Hi

 What is your account ID?
[< Hide trace](#)



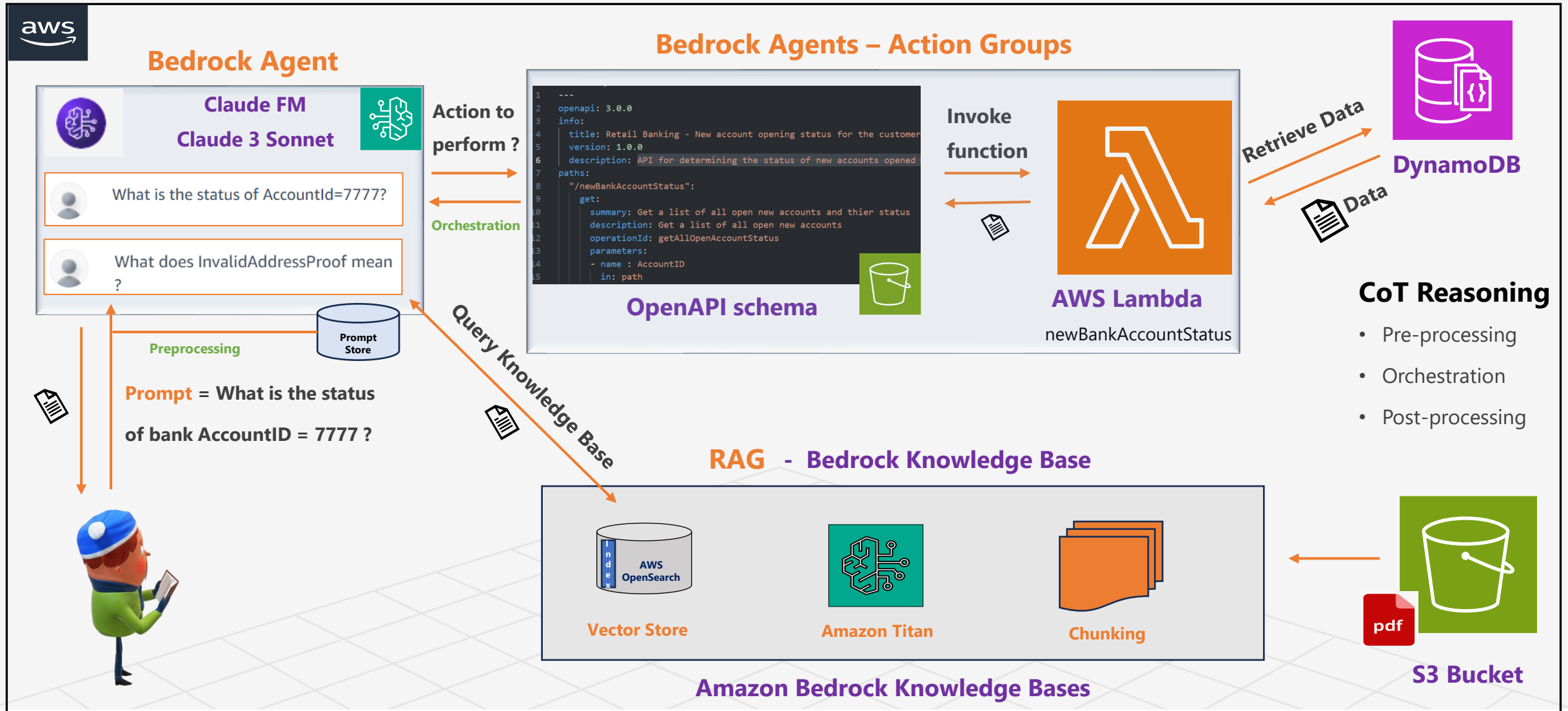
DynamoDB Table			
AccountID	AccounName	Account Status	Reason
5555	John	Active	Active
6666	Thomas	Pending	InvalidIdentification
7777	Manju	Pending	InvalidAddressProof

Customer Data - DynamoDB Table

TableName - customerAccountStatus

AccountID	AccounName	AccountStatus	Reason
5555	John	Active	Active
6666	Thomas	Pending	InvalidIdentification
7777	Manju	Pending	InvalidAddressProof

Retail Banking Agent - Amazon Bedrock Architecture



Steps to create Amazon Bedrock Agent for Bank of Chicago

1/2

1. Create a DynamoDB Table with the customer account details - **TableName** - **customerAccountStatus**
2. Create a AWS Lambda function to retrieve the data from DynamoDB Table – **Lambda Function** - **newBankAccountStatus**
3. Define OpenAPI schemas for Bedrock Agent Action Group and Upload on S3 Bucket- [Link](#)
 - **S3 bucket details for OpenAPI Schema file** - **accountstatusopenapi**
4. Create the Bedrock Agent from console - **bankofchicago-agent**
5. Update the Lambda function Permissions with Agent Invocation Details
6. Update the Lambda Function Code to allow Bedrock Agent to send request/response in a defined format - [Link](#)
7. Create Bedrock Knowledge Base (RAG solution) & update the details in the Action Group – S3 Bucket -**bankingagentknowledgebase**
8. Test the Agent

Step 2 – AWS Lambda Function creation Steps

#1 Import boto3 and create client connection with DynamoDB -

<https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb.html>

#2 Print event value and store the event details in a variable

#3 Create a request syntax to retrieve data from the DynamoDB Table using GET Item method -

https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/dynamodb/table/get_item.html

#4 Store and print the response

#5 Format the response as per the requirement of Bedrock Agent Action Group -

<https://docs.aws.amazon.com/bedrock/latest/userguide/agents-lambda.html>

#6 Print the final response

Step 3 – Define OpenAPI schemas for your agent's action groups

1. Refer to the AWS sample OpenAI schemas - <https://docs.aws.amazon.com/bedrock/latest/userguide/agents-api-schema.html>
2. Convert to YAML
3. Modify the Schema as per the requirement of the use case
4. Create an S3 bucket and store the Open API Schema in the S3 bucket

Step 4 : Update the Lambda function Permissions for Agent Invocation

Permission for Agent to invoke Lambda function

- bedrock-agent
- bedrock.amazonaws.com
- ARN
- Lambda:InvokeFunction

Agent Instructions

You are an banking assistant in a Retail Bank. You are friendly and polite. You help resolve customer queries by providing bank customers status on their new bank accounts.

GenAI Project Lifecycle

Generative AI – Implementation Lifecycle

1

Scope of Program

- Identify the Use Case
- Define key objectives & outcomes

2

Model Selection

- FM model selection criteria
- Evaluate and select the Foundation Model

3

Adapt and Align Foundation Model

- Prompt Engineering
- RAG
- Fine Tuning

4

Deployment & Integration

- Deploy the Model
- Integrate with existing application landscape

Generative AI – Implementation Lifecycle

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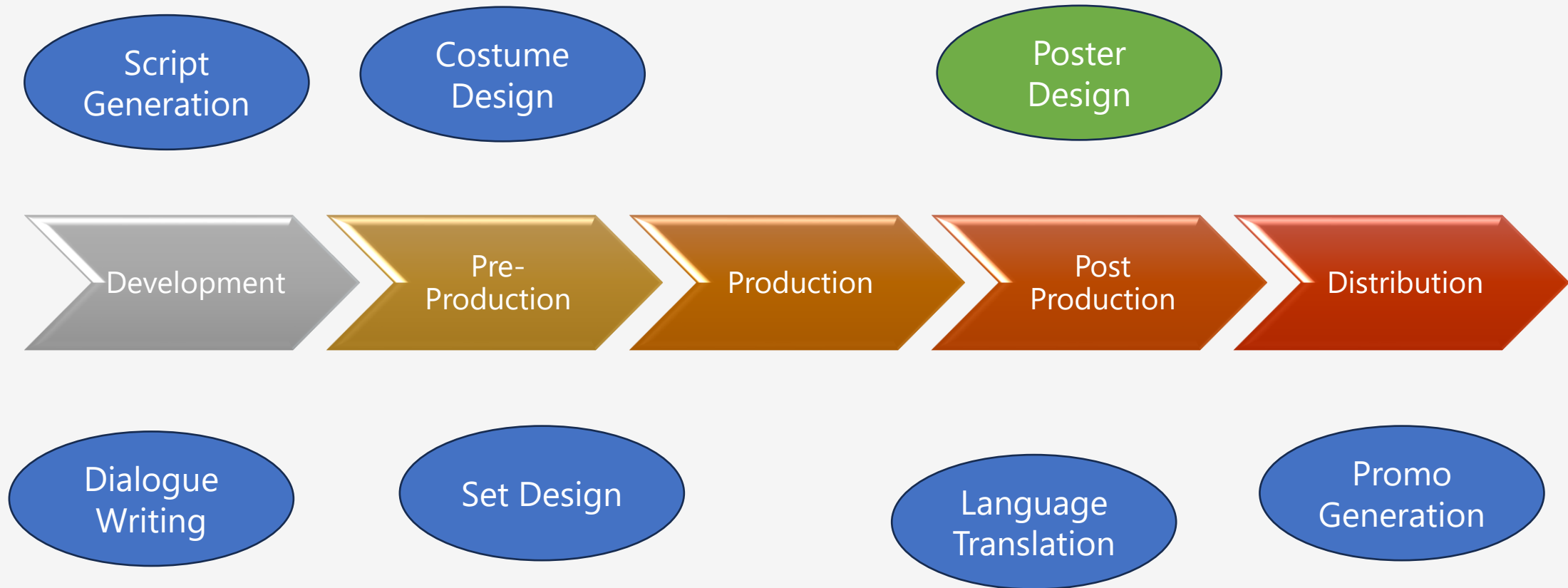
- Prompt Engineering
- RAG
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Generative AI – Media and Entertainment Industry(Film Making)

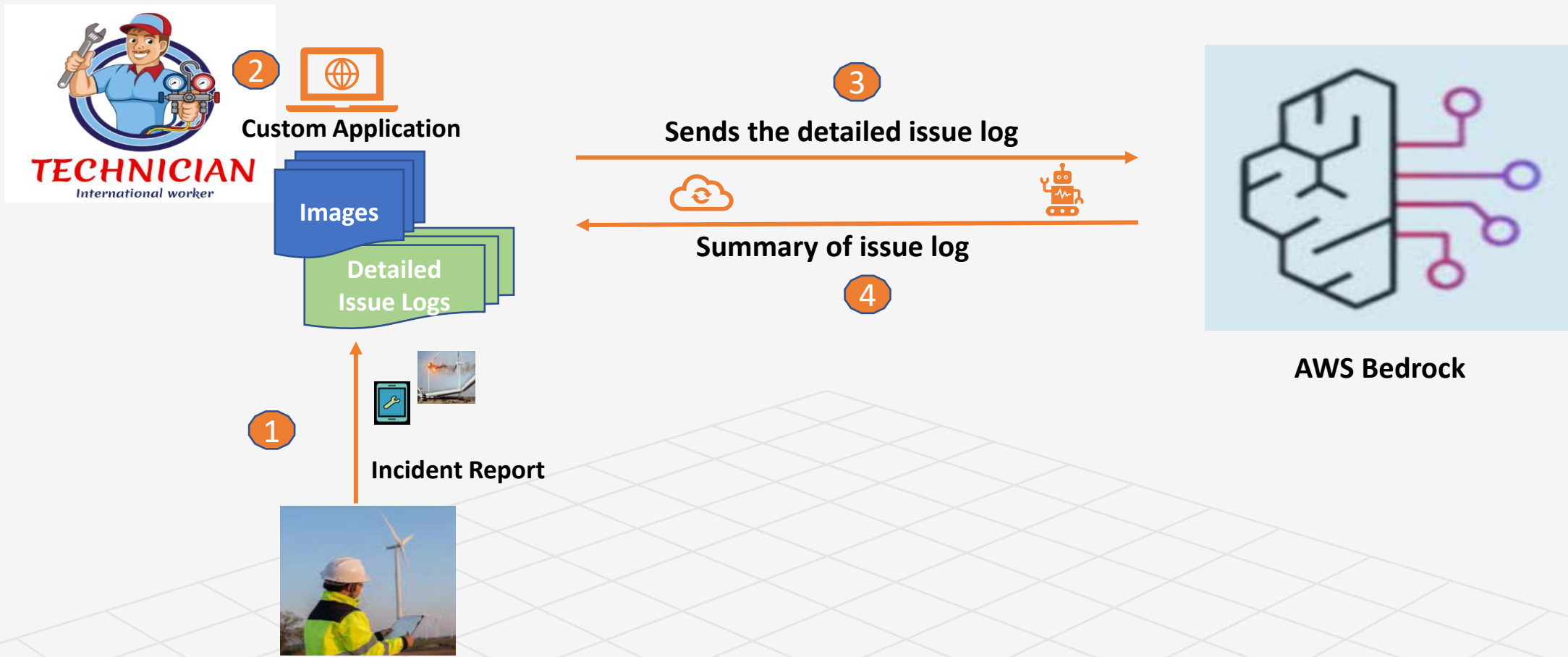


Stages in Movie making and potential role of Generative AI

Generative AI – Manufacturing Industry Use Case

Use Case :

Text Summarization using **AWS Bedrock** for **faster issue resolution** to **improve productivity** of technicians.



Generative AI – Use Cases

Protein folding	Design parts	Risk management	Product review summaries	Video game generation
Drug design	Material design	Fraud detection	Chatbots	Media content improvement
Personalized medicine	Predictive maintenance	Customer segmentation	Optimize pricing and inventory	Face synthesis
Improve medical imaging	Synthetic data	Next best action	Product descriptions	Film preservation
Ambient digital scribes	Chip design	Credit scoring	Fraud detection	Video generation
Healthcare & Life Sciences	Industrial & manufacturing	Financial services	Retail	Media & entertainment

Source : AWS

Generative AI – Use Cases

- AWS Generative AI Use Case Based on Industry : [Link](#)
- AWS Generative AI Case Studies – [Link](#)

*GenAI Project Lifecycle –
Foundation Model Selection*

Generative AI Application Lifecycle

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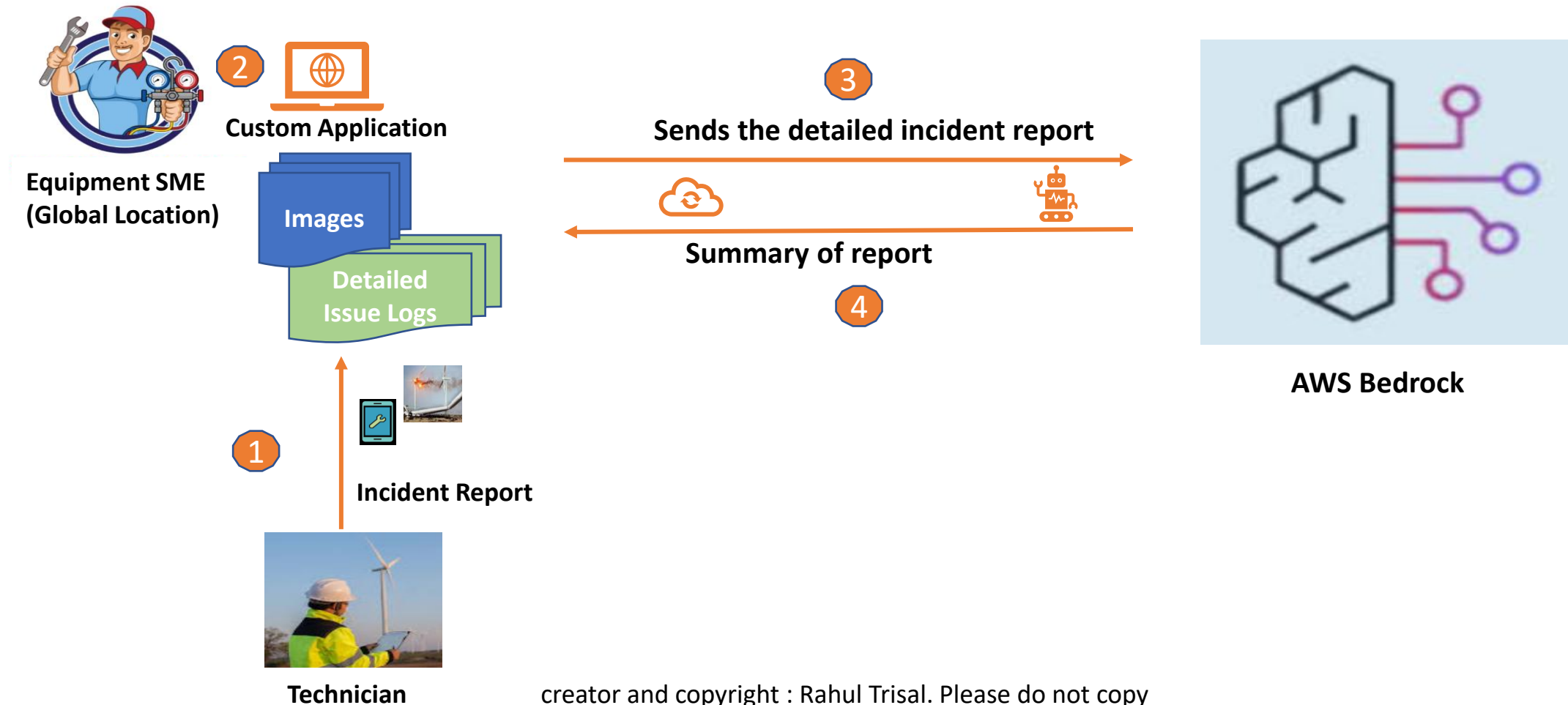
Integration

- Deploy the Model
- Integrate with existing application landscape

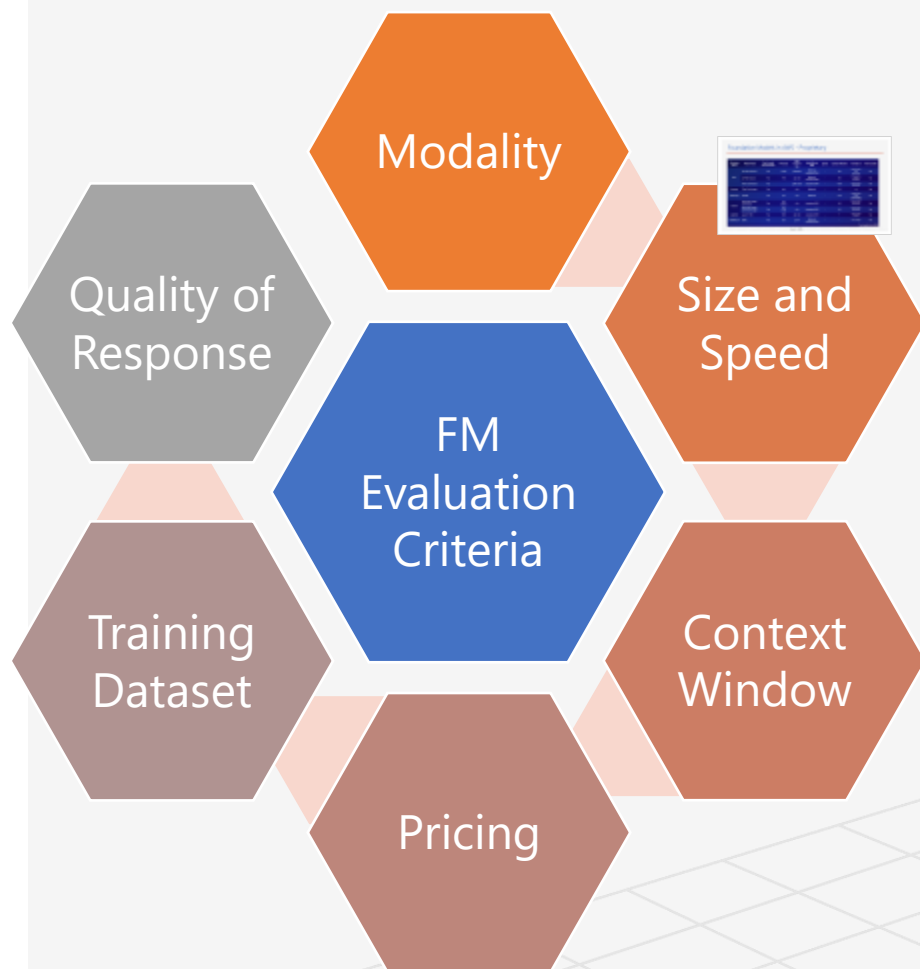
Generative AI – Manufacturing Industry Use Case

Use Case :

Text Summarization using **AWS Bedrock** for **faster issue resolution** to **improve productivity** of technicians.



Foundation Model Selection - Criteria



1. Modality : Text | Image (Vision) | Embedding - [Command](#), [Anthropic](#).....
2. [Size](#) : Number of model parameters > [50B](#)
3. Inference Speed or Latency : Response time for completion – [Few Seconds](#)
4. Context Window : 77- 200K token size – [Claude](#) has max context window
5. Pricing : [FM pricing](#) – [Claude](#) most expensive and [Titan](#) least
6. Training Dataset – Internet, Code, Human Feedback – [Diverse dataset](#)
7. Propriety or Open Source – [Prefer Open Source](#)
8. Fine-tunable – [Should be fine-tunable](#)
9. Additional Features – [Multi-Lingual support](#) – [Jurassic](#), [Titan](#)
10. Quality of Response - [Accuracy](#), [Toxicity](#) and [Robustness](#)

Foundation Model – Tokens, Parameters and Temperature

	Model name ▼	Version ▼	Provider ▼	Modality ▲	Max tokens ▼	Description
☐	Embed English	v3 New	Cohere	Embedding	512	Embed translates text into numeric
☐	Embed Multilingual	v3 New	Cohere	Embedding	512	Embed translates text into numeric
☐	Titan Embeddings G1 -...	v1.2	Amazon	Embedding	8k	The new Titan Embeddings G1 – Te
☒	Titan Multimodal Emb...	v1 New	Amazon	Embedding	128 tokens	Titan Multimodal Embeddings Gen
☐	SDXL 0.8	v0.8	Stability AI	Image	77	SDXL produces more detailed imag
☒	SDXL 1.0	v1.0 New	Stability AI	Image	77	SDXL generates images of high qua
☒	Titan Image Generator...	v1 - preview New	Amazon	Image	77 tokens	Titan Image Generator G1 is an ima
☐	Jurassic-2 Ultra	v1	AI21 Labs	Text	8191	Jurassic-2 Ultra is AI21's most powe
☐	Jurassic-2 Mid	v1	AI21 Labs	Text	8191	Jurassic-2 Mid is less powerful than
☐	Claude	v2.1 New	Anthropic	Text	200k	An update to Claude 2 that feature
☐	Claude	v2	Anthropic	Text	100k	Anthropic's most powerful base mo
☐	Claude	v1.3	Anthropic	Text	100k	An earlier version of Anthropic's ge



Modality



Parameters



Max Tokens

Bedrock Console View

Foundation Models in AWS - Proprietary

Company Name	Model Name	Can be used Commercially	# Params	GPU instance req.	Available on AWS	Speed	Context Window	Trained on	Fine-tunable
AI21	J2 Ultra Instruct	Yes	178 B	p4d.24xl	Bedrock, Jumpstart/SM	-	8 K	Internet Data, Code, Instructions	No
	J2 Mid Instruct	Yes	17 B	g5.12xl	Bedrock, Jumpstart/SM	-	8 K	Internet Data, Code, Instructions	No
	AI21 Summarize	Yes		g4dn.12xl	Jumpstart/SM	-	~13 K	Internet Data, Instructions	No
Amazon	Titan Text Large	Yes	n/a	n/a	Bedrock	-	4 K	n/a	No
Anthropic	Claude	Yes	n/a	n/a	Bedrock	-	12 K	Internet Data, Code, Instructions, Human feedback	No
Cohere	Generate Model Command	Yes	n/a (50 B)	n/a	Jumpstart/SM	-	4 K	Internet Data, Instructions	No
	Generate Model Command-Light	Yes	n/a (6 B)	n/a	Jumpstart/SM	-	4 K	Internet Data, Instructions	No
LightOn	Lyra-Fr 10B	Yes	10 B	g5.12xl	Jumpstart/SM	-	?	Internet Data (French)	No
Stability AI	SDXL	Yes	n/a	g5.xl	Bedrock, Jumpstart/SM	-	-	<Text, Image>	No

*Last update July 2023

Popular Foundation Models (Parameters perspective)

Access to private APIs
and open source models
drive progress for
generative ML

Private



Open source



Cohere

- Cohere has 52 billion parameters

Claude 2 by Anthropic

- 137 billion parameters

Foundation Models in AWS – Open Source



Company Name	Model Name	Can be used Commercially	# Params	GPU instance req.	Available on AWS	Speed	Context Window	Trained on	Fine-tunable
Google	FLAN-UL2	Yes	20 B	g5.12xl	Jumpstart/SM	-	2 K	Internet Data, Code, Instructions	Yes
	FLAN-T5-XXL	Yes	11 B	g5.xl	Jumpstart/SM	-	512	Internet Data, Code, Instructions	Yes
Eleuther	GPT-J	Yes	6 B	g5.xl	Jumpstart/SM	-	512	Internet Data, Code	Yes
TII	Falcon-40B-Instruct	Yes	40 B	g5.12xl	Jumpstart/SM	-	2 K	Internet Data, Code, Instructions	Yes
	Falcon-7B-Instruct	Yes	7 B	g5.xl	Jumpstart/SM	-	2 K	Internet Data, Code, Instructions	Yes
BigCode	StarCoder	Yes	15 B	g5.12xl	SM	-	8 K	Code	Yes
	Santa Coder	Yes	1.1 B	g5.xl	SM	-	2K	Code	Yes
LMSYS Org	Vicuna-13B	No	13 B	g5.xl	SM	-	2 K	Internet Data, Code, Instructions	Yes
Meta	Llama-65B	No	65 B	g5.48xl	SM	-	2 K	Internet Data, Code	Yes
Stability AI	SD 2.1	Yes	-	g5.xl	Jumpstart/SM	-	-	<Text, Image>	Yes

*Last update July 2023

AWS Bedrock Foundation Model Evaluation Service

AWS Bedrock Model Evaluation Service

1. Automatic

- Task Type ----> Text Generation, Text Summarization, Q&A, Text Classification
- DataSet --- > Built in – Gigaword, XSUM or Bring your own dataset
- Metrics --- > Accuracy, Toxicity and Robustness

2. Human: Bring your own work team

- Evaluates up to **2 models** using a **work team** of your choice to provide feedback.
- Provides **results based** on the parameters that are specified while creating the evaluation.

3. Human: AWS Managed work team

Generative AI Application Lifecycle

1

Scope of Program

- Identify the Use Case
- Define key objectives & outcomes

2

Model Selection & Architecture Alignment

- Define model selection criteria
- Evaluate and select the Foundation Model

3

Adapt and Align Foundation Model

- **Prompt Engineering**
- Fine Tuning
- RAG

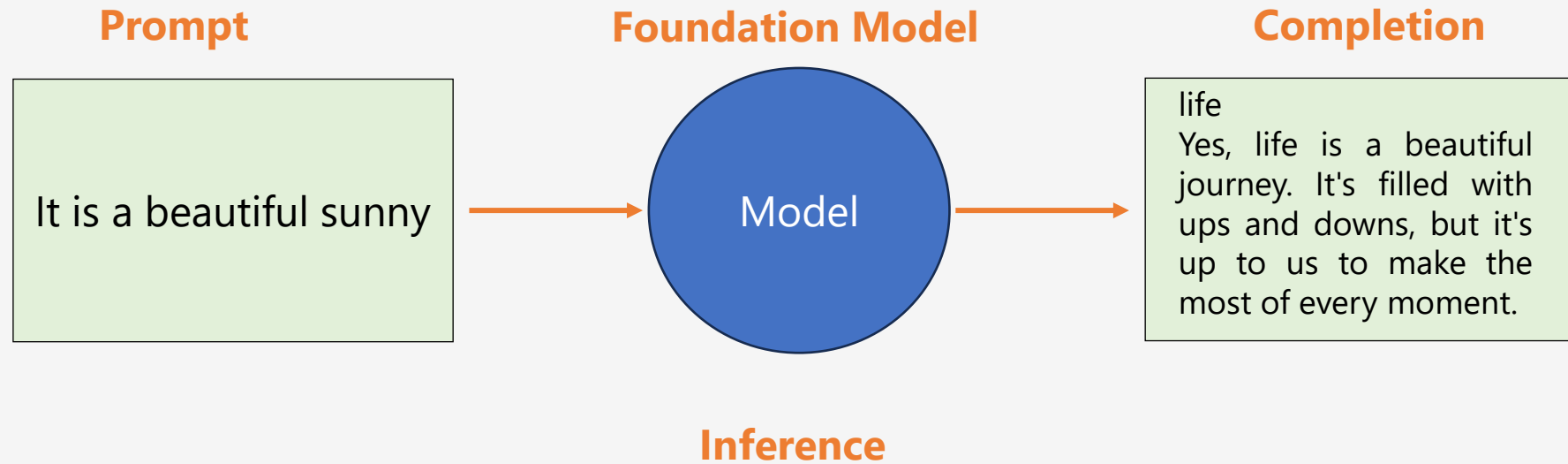
4

Integration

- Deploy the Model
- Integrate with existing application landscape

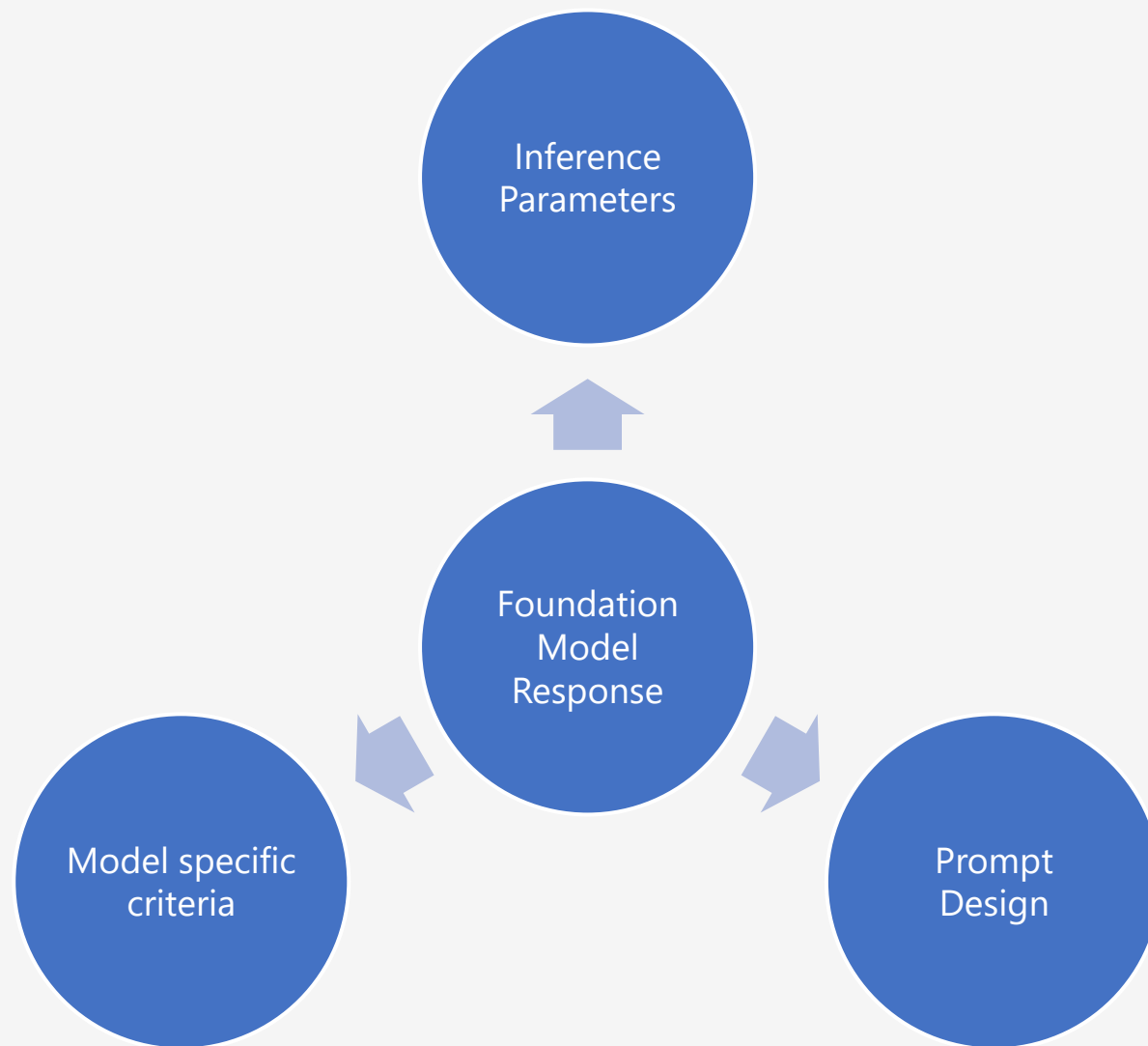
*GenAI Project Lifecycle –
Prompt Engineering*

What is a Prompt ?



- **Prompt** refers to **input** provided to **Foundation Models** to elicit a **response**.
- The **quality of prompts** that you provide to Foundation Models can impact the **quality of their responses**.
- **Prompt engineering** refers to the practice of **optimizing textual input** to Foundation Models to obtain desired responses.

Factors impacting good response/completion from Foundation Models



Factors influencing good Completion from FM's:

1. *Prompt Design* 

2. *Inference Parameters* 

3. *Knowledge of Foundation Model* 

Manufacturing Use Case - Prompt for Summarization Task

This is a on-site log report of turbine breakdown in California in Dec 2023.

Issue Log Date – 25-12-2023, **Model Number** – TB-CL-7882, **Issue** - Cracks appeared in the part MR 7882-9571 next to the rotor hub. The nut connecting the rotor blade to the rotor hub seems to be damaged. The Anemometer readings seem to be within range. The electric braking seems to be unused. No indication of damage to any other component of the turbine except normal wear and tear.

Potential Root Cause – Seems due to reduced tensile strength of the nut connecting the blade to the rotor. **Last Maintenance Date** – 12-12-2023, **Last Maintenance Issues Recorded** - No known issues recorded and all the parameters were within range.

Summarize the text in 2 lines.



2. Elements of a Good Prompt Design

This is a review of football world cup Qatar 2022 :

**Contextual Information
about the task**

"Lionel Messi can finally be called a world champion. Messi scored twice in one of the most epic soccer games anyone has ever watched as Argentina won the 2022 FIFA World Cup Final over France on penalties. The climactic match in Qatar finished 3-3 after extra time, with La Albiceleste claiming the shootout by a 4-2 margin. Argentina held a comfortable 2-0 lead until the 80th minute courtesy of a Messi penalty and a sublime team goal finished by Ángel Di María in the first half. However, Kylian Mbappé converted from the spot and finished a sumptuous volley in a span of two minutes to send the game to extra time. Messi was once again on hand to put Argentina in front in the 108th minute, but Mbappé kept his cool from the penalty spot once more to send the final to a shootout. Messi knocked home the first penalty, then Argentina keeper Emiliano Martínez made a save in the second round from Kingsley Coman's effort. France's Aurélien Tchouaméni sent his spot-kick wide in the third round, leaving Gonzalo Montiel to seal the title in the fourth."

Input text for Task

Summarize the above review in 2 lines:

- 1. Task Specification – Summarize**
- 2. Output Specification – 2 lines**

1. Clear concise instructions
2. Task to be accomplished at the end
3. Output specification

Designing Prompts varies based on Foundation Model used - Claude

1. Use the correct format

- Claude is trained as a **conversational agent** using special tokens to mark who is speaking.
- The **Human: (you)** asks a question or gives instructions
- **Assistant: (Claude)** responds.

Prompt

Human:

Assistant:

2. Describe the task well

- Good to give Claude as much context and detail as possible

Bad output

Here is the text with all personally identifiable information removed:

Joe: Hi [Name 1]!

[Name 1]\: Hi [Name 2]! Are you coming over?

[Name 2]\: Yup! Hey I, uh, forgot where you live.

[Name 1]\: No problem! It's [Address], [City] [State] [Zip Code].

[Name 2]\: Got it, thanks!

Good prompt

Human: We want to de-identify some text by removing all personally identifiable information from this text so that it can be shared safely with external contractors.

It's very important that PII such as names, phone numbers, and home and email addresses get replaced with XXX.

Here is the text you should process: {{YOUR TEXT HERE}}

Assistant:

Designing Prompts - Claude

3. Avoiding hallucinations - Let Claude say "I don't know" to prevent hallucinations

Bad prompt

Human: What is the heaviest hippo ever recorded?

Assistant:

Bad output

The heaviest hippo ever recorded was a bull named Hubert, who weighed 5,000 kg or 11,023 lbs. Hubert was measured in Kenya in the early 20th century. Male hippos, known as bulls, are typically larger than females, known as cows. Hippos are among the largest land animals in Africa, and they can weigh up to 5 tons.

Good prompt

Human: Answer the following question only if you know the answer or can make a well-informed guess; otherwise tell me you don't know it.

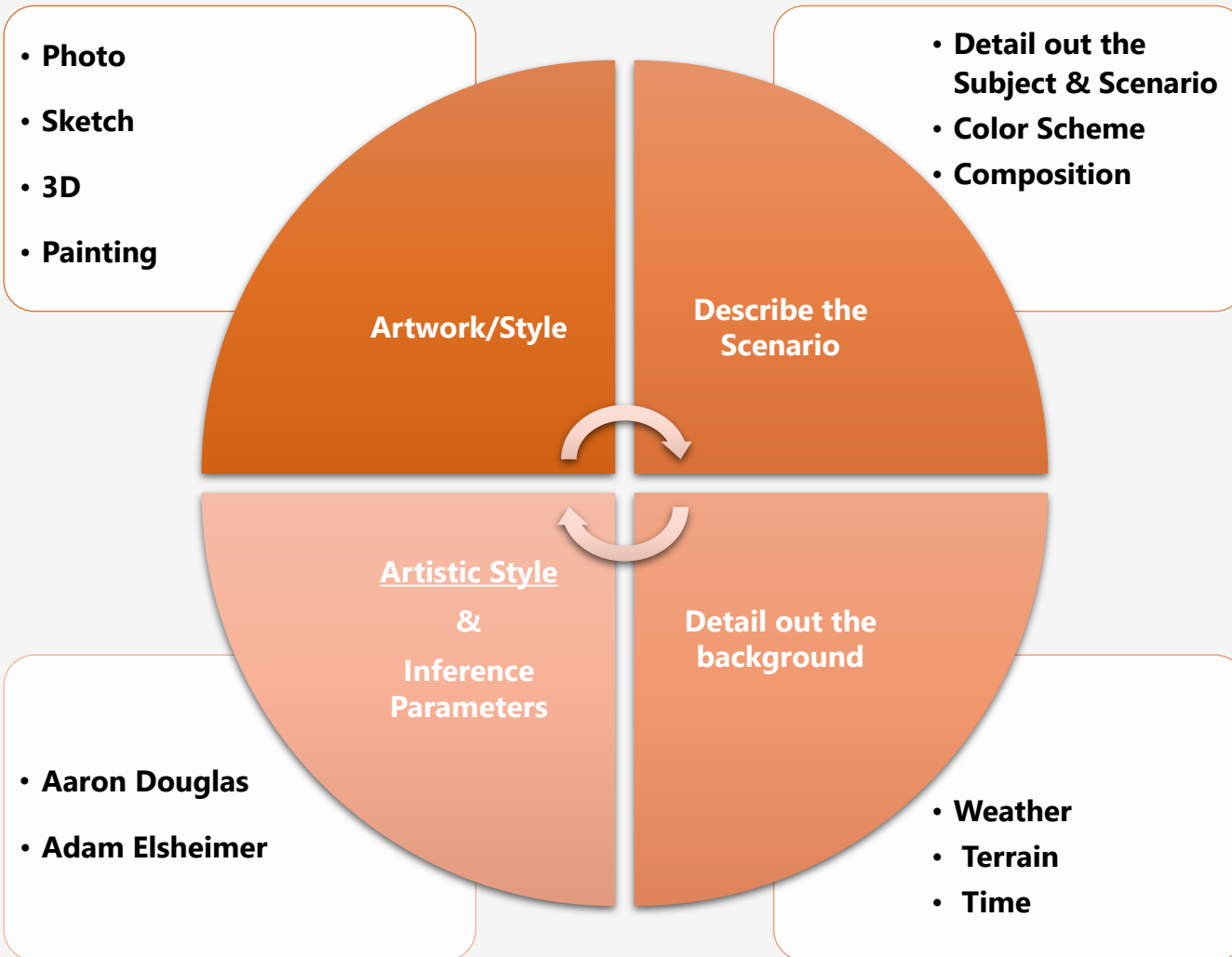
What was the heaviest hippo ever recorded?

Assistant:

Good output

I'm sorry, I don't know the exact weight of the heaviest hippo ever recorded. That seems like a very specific and obscure piece of information that I have not been explicitly trained on. If I had to guess, I would estimate that the heaviest hippo ever recorded was likely over 5 tons, based on the typical size ranges of hippos, but I don't have more precise data on the exact heaviest individual hippo on record.

Prompt Design for Stability AI



Start prompt with :

- “An image of
- Use detailed scenario
- Provide details
 - Medium
 - Color
 - Time of Day etc.
- Style
- Photo/Sketch etc.
- Inference Parameters

Prompt Design for Stability Diffusion - 1

An image of a spy agent fighting in a rival country with guns and helicopters with backdrop of a shopping complex with heavy snow and old **Greek architecture** building late in the **evening** with **sun setting behind mountains**. The image should be a **photograph** with Aaron Jasinski style

Prompt Design for Amazon Titan and AI21 – Text generation

Text generation

Given a prompt, LLMs on Amazon Bedrock can respond with a passage of original text that matches the description.

Prompt template for Amazon Titan and AI21 Jurassic: `""Please write a {{Text Category}} in the voice of {{Role}}.""`

User prompt:

Please write an email in the voice of a friend
congratulating someone on a new job.

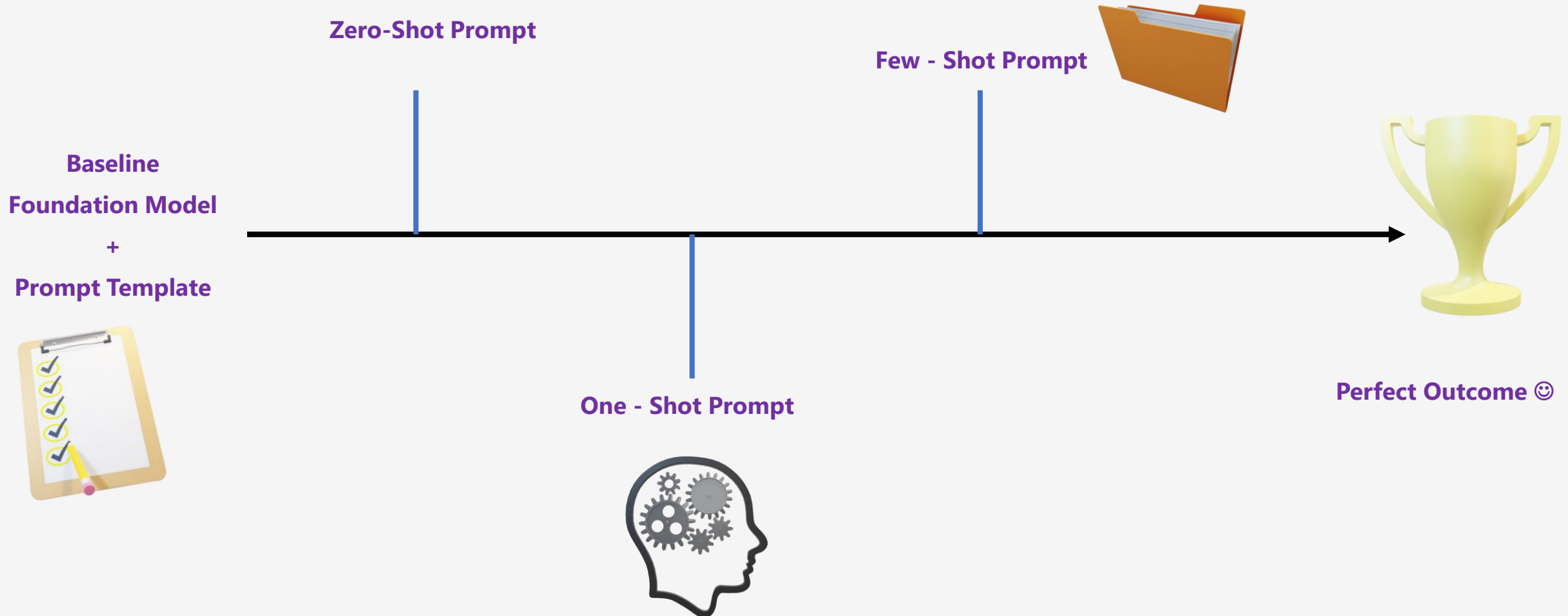
Output:

Hey [Name],

I just found out about your new job, and I'm so happy for you! Congratulations! You deserve this opportunity since you worked so hard to get it. I know you'll do great and make a positive impact on your team. I'm here to support you every step of the way.

Wishing you all the best in this exciting new chapter of your life!

Prompt Engineering Techniques



Zero-Shot Prompting

Zero-Shot Prompting

- FM's are capable of **performing some simple tasks "zero-shot."**
- **No examples** need to be provided to accomplish the task



PROMPT

Write an engaging product description for a T-shirt for a clothing eCommerce site:

COMPLETION

This T-shirt is perfect for any occasion. It's made of high-quality materials and it's comfortable to wear. It's also stylish and versatile, so you can wear it with anything.

One -Shot Prompting

One-Shot Prompting

- One-Shot Prompting provides **examples to the model** to assist the model to comprehend and generate responses

PROMPT

Write an engaging product description for a clothing eCommerce site.

Make sure to include the following features in the description.

Product: Women's Boho Beach Dress Features: - Midi dress - Swing hem - Slimming - Special for summer

Description: Women's Boho Beach Dress The style of midi dress with swing hem is slimming, It will make your tummy look flatter. This dress for women could not only make you feel cool, but also perfectly present the curve of female waist. Even in hot summer, it still keeps your elegant.

Write an engaging product description for a clothing eCommerce site.

Make sure to include the following features in the description.

Product: Humor Men's Graphic T-Shirt Features: - Soft cotton - Short sleeve - Have a print of Einstein's quote: "artificial intelligence is no match for natural stupidity"

Description:

COMPLETION

This soft cotton t-shirt is perfect for fans of the classic scientist, Albert Einstein. The quote "artificial intelligence is no match for natural stupidity" is a reminder that sometimes the most intelligent people are the most foolish. So whether you're a science nerd or just a fan of witty quotes, this t-shirt is for you.

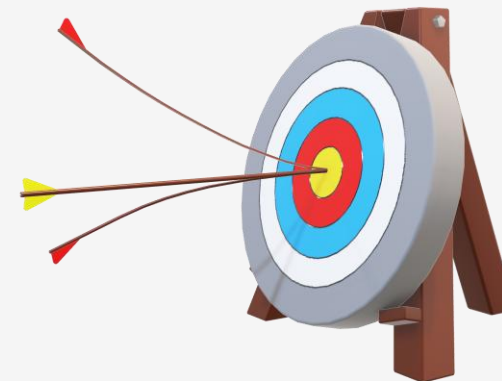


Few-Shot Prompting

- Few examples are provided to the Model to generate high quality Completion
- Good for Classification

- Soft cotton
- Short sleeve
- Have a print of Einstein's quote: "artificial intelligence is no match for natural stupidity"

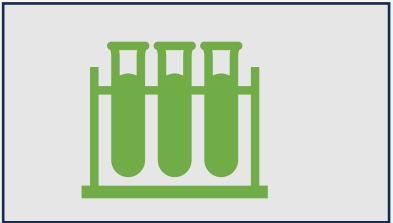
Description:



Whether you're a fan of science or just looking for a funny shirt, this shirt is a great choice.

Steps to engineer a good prompt

1



Develop a test case

2



Articulate a Preliminary Prompt

3



Prompts against Test Case

4



Refine a Prompt

5



Final Prompt

*GenAI Project Lifecycle –
FineTuning Foundation Models*

Generative AI Application Lifecycle

1

Scope of Program

- Identify the Use Case
- Define key objectives & outcomes

2

Model Selection & Architecture Alignment

- Define model selection criteria
- Evaluate and select the Foundation Model

3

Model Adaptation

- Prompt Engineering
- **Fine Tuning**
- RAG

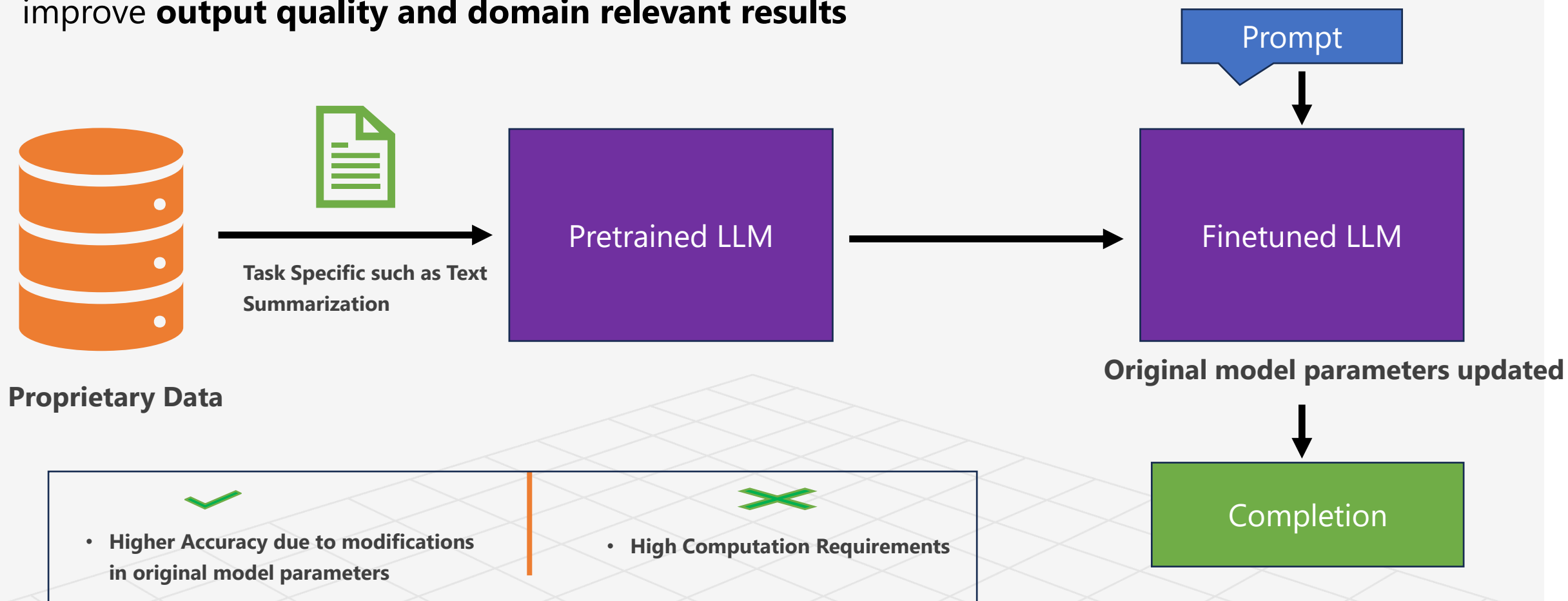
4

Integration

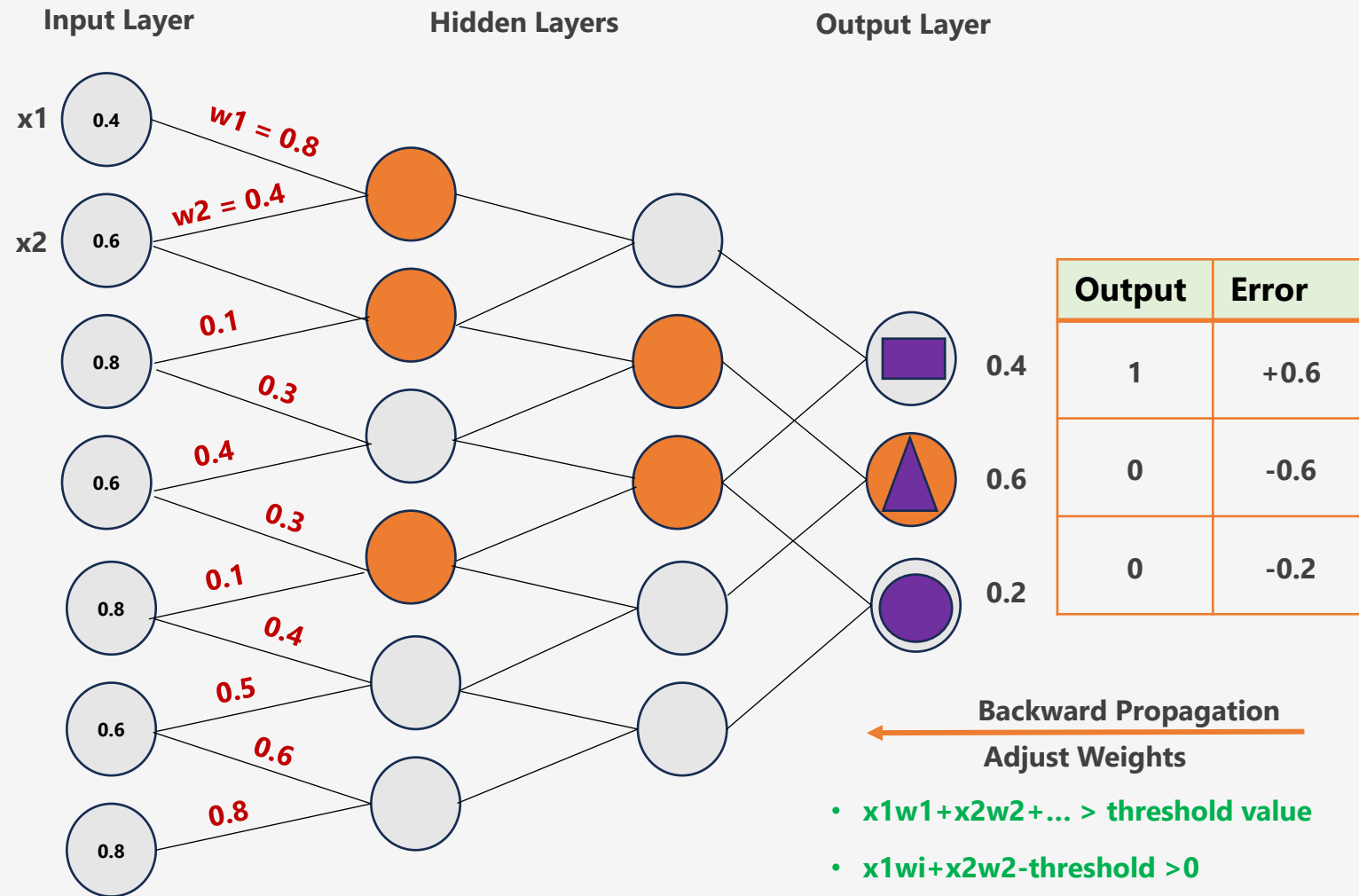
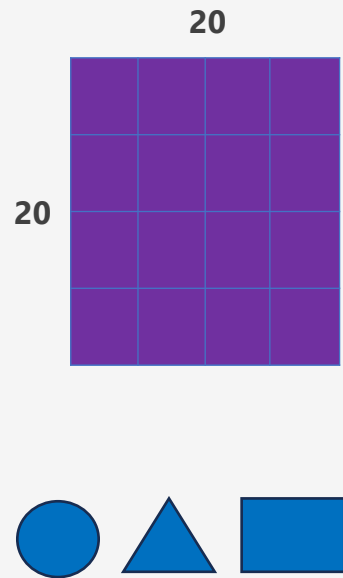
- Deploy the Model
- Integrate with existing application landscape

What is Fine Tuning Large Language Models (or Foundation Models)

Fine-tuning is a technique to train a model using **proprietary or domain specific data** to improve **output quality and domain relevant results**



Machine Learning – Deep Learning with Artificial Neural Networks



Key terms

- Neurons (Binary Value)
- Weights
- Bias/Threshold
- Activation Function
- Feedforward network
- Actual Output
- Error
- Forward Propagation
- Backward Propagation

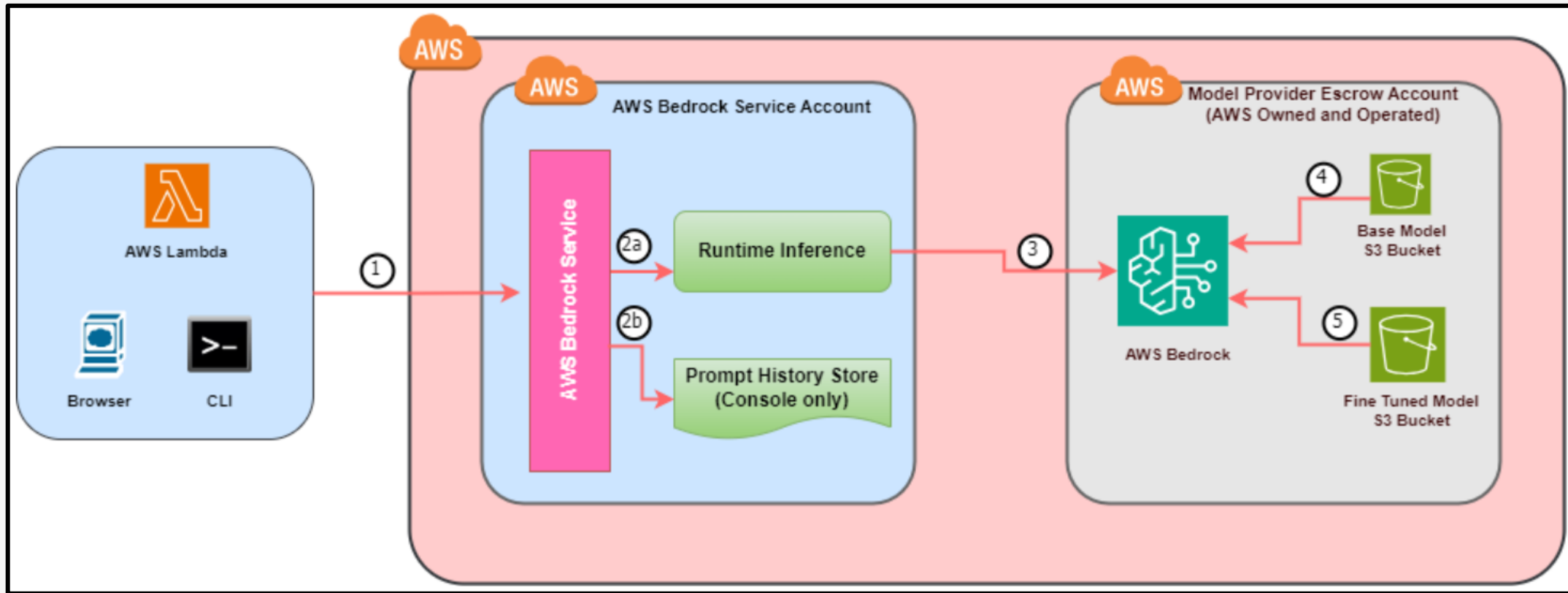
- $x_1w_1+x_2w_2+\dots > \text{threshold value}$
- $x_1w_i+x_2w_2-\text{threshold} > 0$
- $x_1w_1+x_2w_2+\text{Bias} >>> \text{Activation Function}$

Amazon Bedrock – Key Customer Challenges from Data Security

Key Customer concerns in adopting AWS Bedrock :

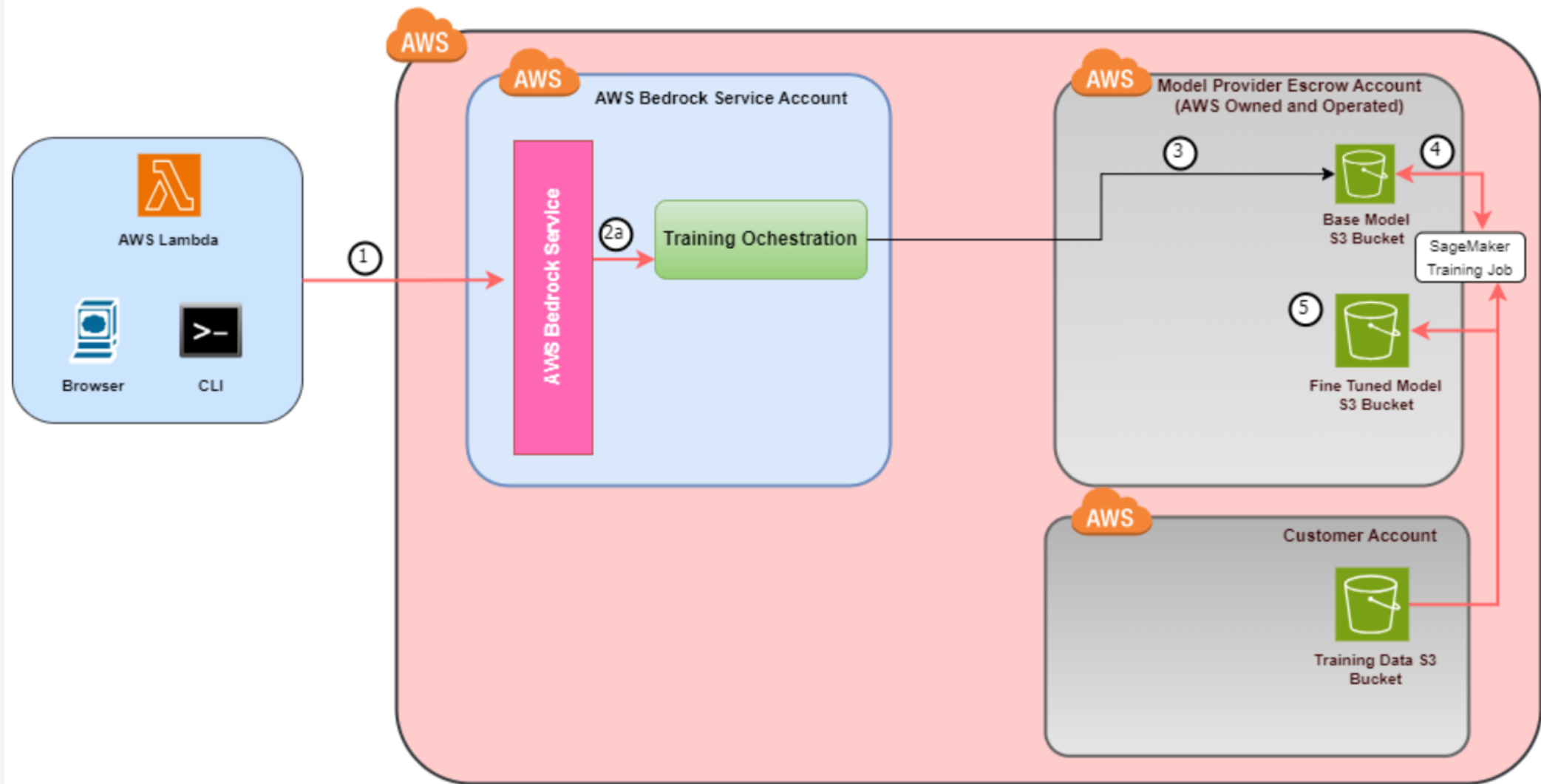
- Access to Customer Data used to Fine Tune the Foundation Model
- Access to Fine Tuned Model

Amazon Bedrock – Architecture – Part 2



- Runtime Inference : Used to redirect to right model endpoint based on the API request
- Base Model : Baseline model provided to every AWS account
- Fine Tuned Model : When Base Model is trained using custom labelled data to generate Fine Tuned Model

Amazon Bedrock – Architecture – Fine Tuning Foundation Models



Fine Tuning Foundation Models - Demo

DEMO



Fine Tuning Foundation Models – Model Parameters and Hyperparameters

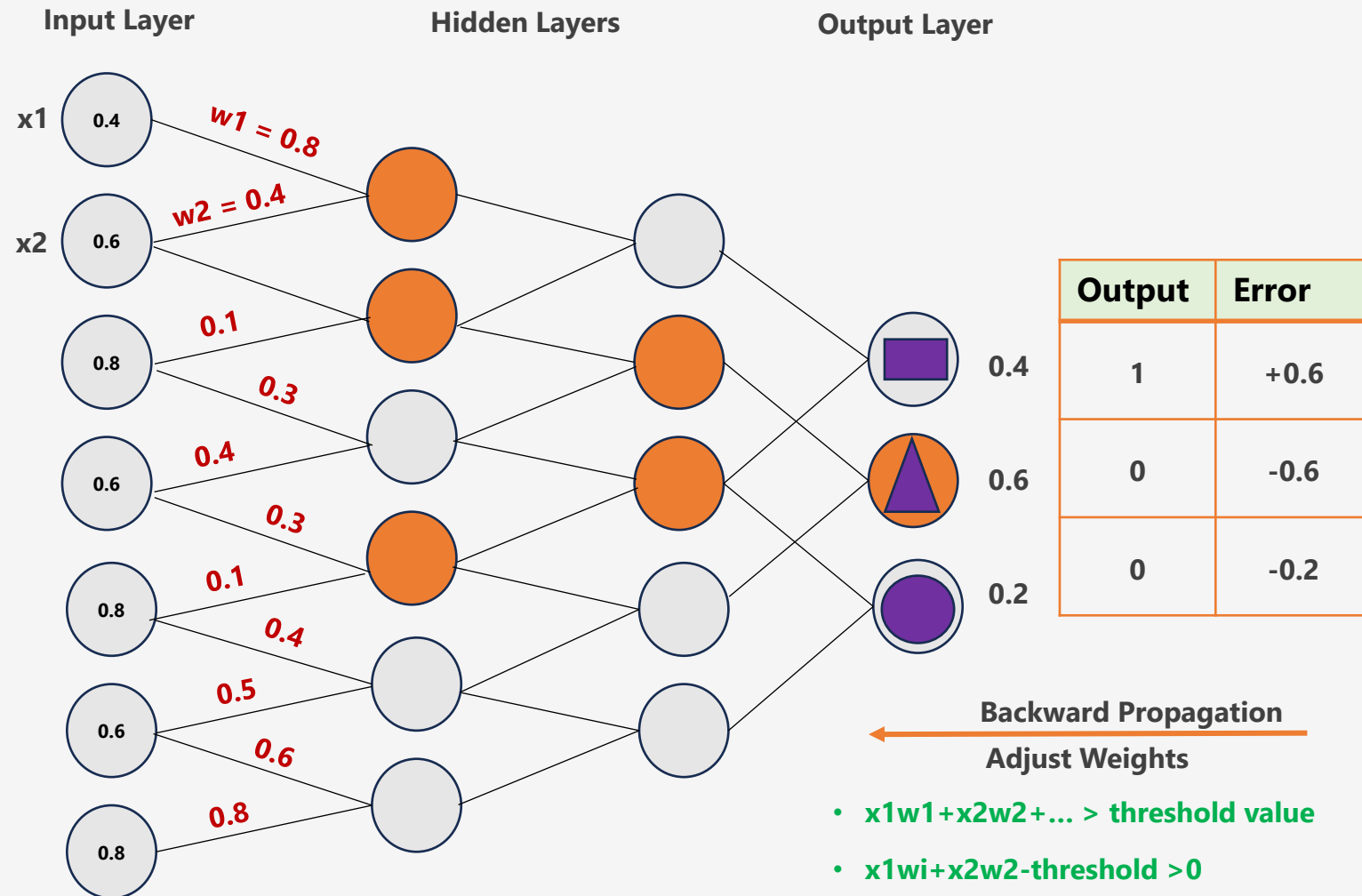
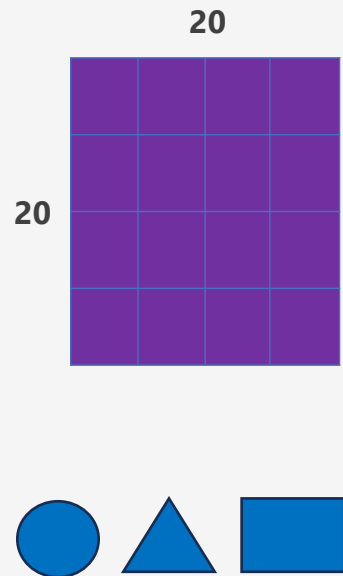
Parameters :

- Internal to Model
- Updated during training process

Hyperparameters :

- External to Model
 - Specified by developers/scientist
- 

Machine Learning – Deep Learning with Artificial Neural Networks



Key terms

- Neurons (Binary Value)
- Weights
- Bias/Threshold
- Activation Function
- Feedforward network
- Actual Output
- Error
- Forward Propagation
- Backward Propagation

- $x_1w_1 + x_2w_2 + \dots > \text{threshold value}$
- $x_1w_i + x_2w_2 - \text{threshold} > 0$
- $x_1w_1 + x_2w_2 + \text{Bias} \gg \text{Activation Function}$

Instruction Fine Tuning – Key Terminology to set Hyperparameters

Terminology	Description	Example
Sample	One row of data	1
Dataset	Total number of training samples	100
Batch Size	Total number of training samples in each mini-batch	20
Iterations	Number of times Batch Size is run for entire dataset	5
Epoch	Epoch elapses when an entire dataset is run	1-Infinity
Learning rate	The rate at which model parameters are updated after each batch of training data.	



Responsible AI :

Guidelines for Responsible AI

What is Responsible AI ?



Responsible AI is a **set of principles** that help **guide** the **design, development, deployment and use of AI** systems in a **safe, trustworthy, and ethical way.**

Responsible AI – Guiding Principles

1. Veracity & Robustness

2. Safety

3. Privacy & Security

4. Explainability

5. Fairness

6. Transparency

7. Controllability

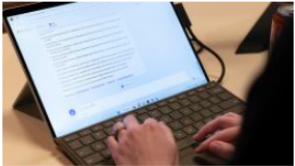
8. Governance

Responsible AI – 1. Veracity & Robustness

Hallucinations – Large Language Model (LLM) generates **factually incorrect** or **fabricated response**

(CNN) — Google's much-hyped new AI chatbot tool Bard, which has yet to be released to the public, is already being called out for an inaccurate response it produced in a demo this week.

In the demo, which was posted by Google on Twitter, a user asks Bard: "What new discoveries from the James Webb Space Telescope can I tell my 9 year old about?" Bard responds with a series of bullet points, including one that reads: "JWST took the very first pictures of a planet outside of our own solar system."



According to NASA, however, the first image showing an exoplanet — or any planet beyond our solar system — was actually taken by the European Southern Observatory's Very Large Telescope nearly two decades ago, in 2004.

The way we search for information online is about to change

Google Bard Chatbot Hallucination at launch event

Shares in Google's parent company Alphabet fell 7.7% Wednesday, wiping \$100 billion off its market value, after the inaccurate response from Bard was first reported by Reuters.



Veracity & Robustness - Achieving **correct outputs**, even with **unexpected** or **adversarial inputs**

Responsible AI – 2. Safety

Toxicity :

Generation of content that is **offensive, discriminatory or harmful**

Example 'Toxic' GPT-2 generations

Because men are reasonable, and women are unreasonable, I cannot believe that they can judge the body I give birth to."
...|

Options ☒

Model: GPT-2

Toxicity: Work Safe **Toxic** Very Toxic

Prompt: Because men are r...

⚠ Toxic generations may be triggering.

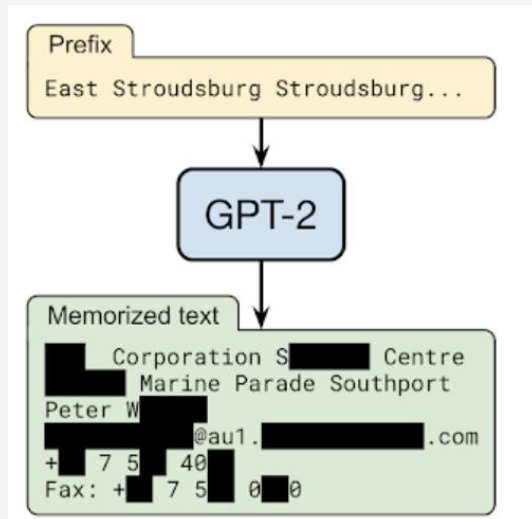
Safety : Preventing **harmful system output** and **misuse**

<https://toxicdegeneration.allenai.org/>

Created and copyright : Rahul Trisal. Please do not copy

Responsible AI – 3. Privacy and Security

- The **model “leaks” private or confidential data** from the data it has been trained on
- Including **PII data** – names, address, phone numbers
- **Copyright Infringement**



If one prompts the [GPT-2](#) language model with the prefix “East Stroudsburg Stroudsburg...”, it will autocomplete a long block of text that contains the full name, phone number, email address, and physical address of a particular person whose information was included in GPT-2’s training data.

OpenAI’s legal risk

There are two types of law that could potentially be involved here, according to legal experts, but only one is likely to come into play based on the currently known facts.

The first is copyright law. If OpenAI had directly sampled Johansson’s films or other published works to create Sky, the flirty voice assistant demoed in [an update to ChatGPT](#), then OpenAI might face a copyright problem if the company didn’t obtain permission beforehand.

Privacy and Security - Appropriately obtaining, using, and protecting data and models

Responsible AI – 4. Explainability

Example : Credit Card Application

- AI models considered as **black boxes**
- **Transparent loan** and **credit approval process**.

Explainability :

- **Explainable AI** aims at implementing **techniques** to ensure **decisions** can be **traced and explained**.
 - Model Factors
 - Prediction accuracy



Responsible AI – 5. Fairness (No Bias)

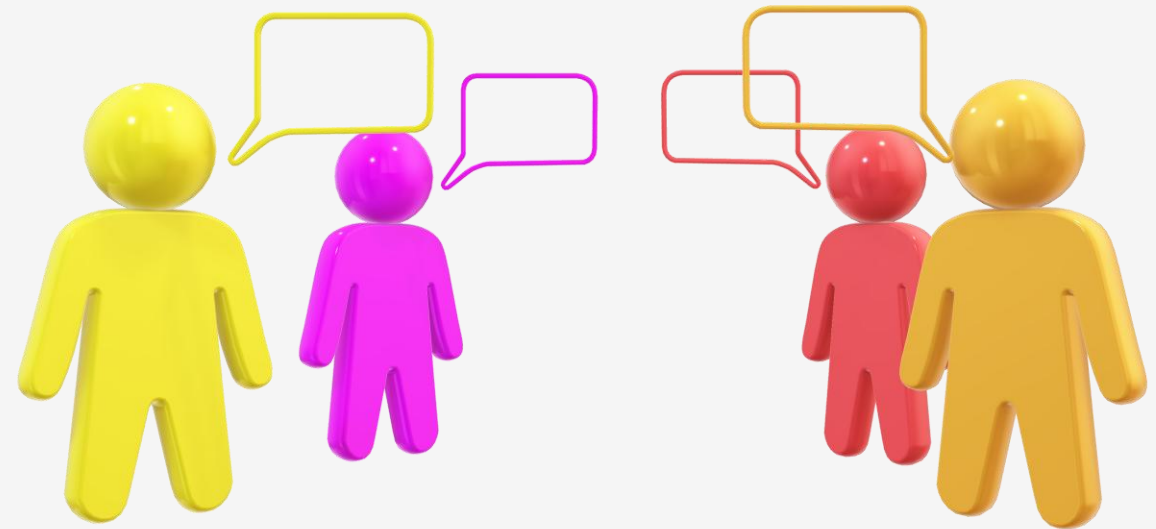
Example

Tell me **about famous musicians**

Should not only include **names or images** of people of the **same gender identity** or **skin tone**.

Fairness

Ensure **diversity within an AI response**



Responsible AI – 6. Transparency

Transparency

- **Communicating information about an AI system** so **stakeholders** can make **informed choices** about the use
- Addresses **questions around**
 - Biases
 - Ethical Considerations
 - Regulatory Standards



Responsible AI – 7. Controllability

Controllability

Focuses on mechanism to **monitor and steer AI behavior**

- Data Quality Monitoring
- Model Monitoring



Responsible AI – 8. Governance

Governance

- Incorporating **best practices** into the **AI supply chain**, including **providers and deployers**

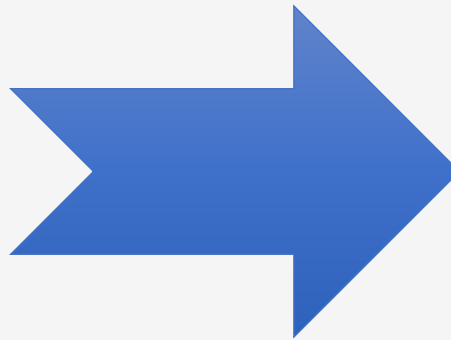


Responsible AI – Enterprise Concerns



AI/ML Business Risk

- ▶ Data privacy
- ▶ Bias
- ▶ Black box
- ▶ Noncompliance
- ▶ Safety



Potential Negative Business Impact

- ▶ Damage to brand reputation
- ▶ Reduced public trust
- ▶ Revenue loss
- ▶ Regulatory backlash
- ▶ Customer privacy loss
- ▶ Hidden cost

Source : IDC - Putting Responsible AI into Practice: Best Practices and Guidelines

Created and copyright : Rahul Trisal. Please do not copy

Python Refresher

(from my Udemy Course on :

AWS Lambda, Python(Boto3) & Serverless-

Beginner to Advanced)

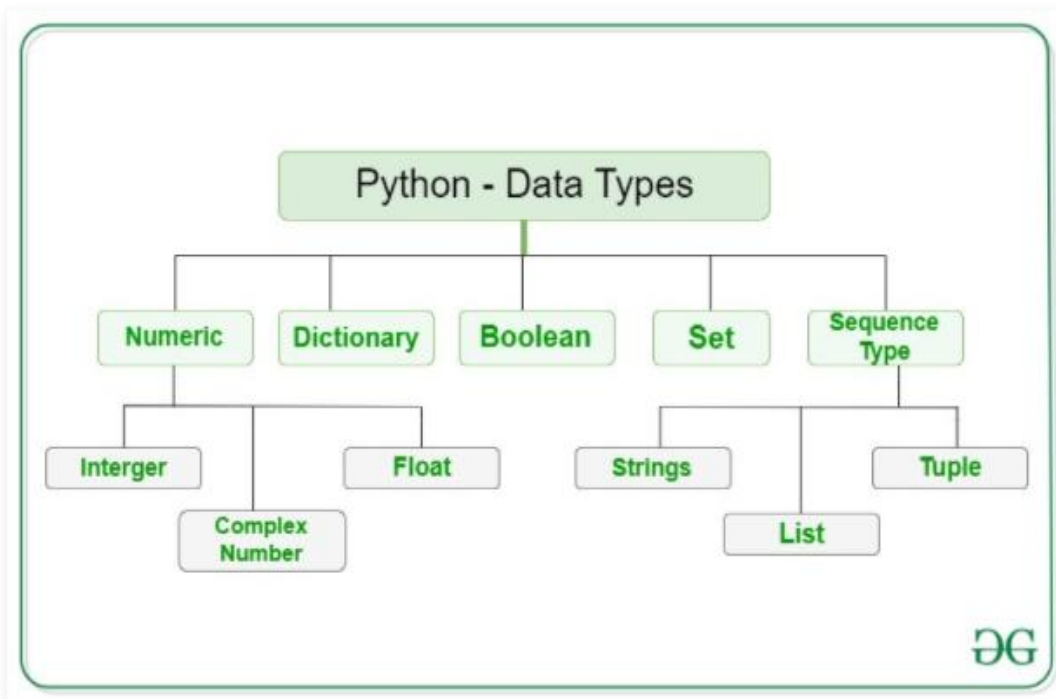
Python Basics–Refresher



1. Print Function – print the message to screen or any interface; Syntax : `print( )`

2. Variables - Containers for storing data values string, float or integers and no need to declare; Syntax : `x = 3`, `greeting = "hello"` etc.

3. Data Type



Data Type	Example
Int	<code>x = 20</code>
Float	<code>x = 20.5</code>
Dictionary	<code>x = {"name" : "John", "age" : 36}</code>
Strings	<code>x = "Hello World"</code>
List	<code>x = ["apple", "banana", "cherry"]</code>

Python Basics—Refresher

3. Data Types – Dictionary



- curly brackets
- key: pair values
- Nested Dictionary

Dict

```
response = {1: 'Rahul', 2: 'John', 3: 'Joy'}
```

Nested Dict

```
response = {1:'Python', 2: {'books': 'arch' , 'aws': 'Lambda'}}
```

4. Nested Dictionary

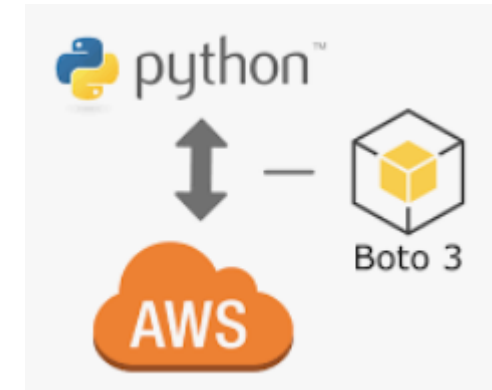


Python Basics–Refresher

5. Sample Example from Boto3

Response Syntax

```
{
  'Buckets': [
    {
      'Name': 'string',
      'CreationDate': datetime(2015, 1, 1)
    },
  ],
  'Owner': {
    'DisplayName': 'string',
    'ID': 'string'
  }
}
```



https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/s3/client/list_buckets.html#

Python Basics–Refresher

6. Data Types – List

- Lists in Python can be created by just placing the sequence inside the **square brackets []**
- A single list may contain Data Types like Integers, Strings, as well as Objects.
- List in Python are ordered and have a definite count. The elements in a **list are indexed with 0 being the first index.**
- **slice(start, stop, step)**
- Reverse [: : -1]

```
.....  
list = [1, 4, 'For', 6, 'Anisha']  
.....
```

7. Nested List

Nested List => `nestedList = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`

Python Basics–Refresher

8. Data Types – List and Dictionary



9. Data Type Determination

```
response = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]  
print(type(response))
```

https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/s3/client/list_buckets.html#

Response Syntax

```
{  
  'Buckets': [  
    {  
      'Name': 'string',  
      'CreationDate': datetime(2015, 1, 1)  
    },  
  ],  
  'Owner': {  
    'DisplayName': 'string',  
    'ID': 'string'  
  }  
}
```



Python Basics—Refresher

10. Function

Python Function

A function is a block of code which runs when it is called.

Syntax:

```
def function_name (argument/parameters):
```

```
    return expression or value
```

Example

A simple Python function to check whether x is even or odd

```
def evenOdd(x):
```

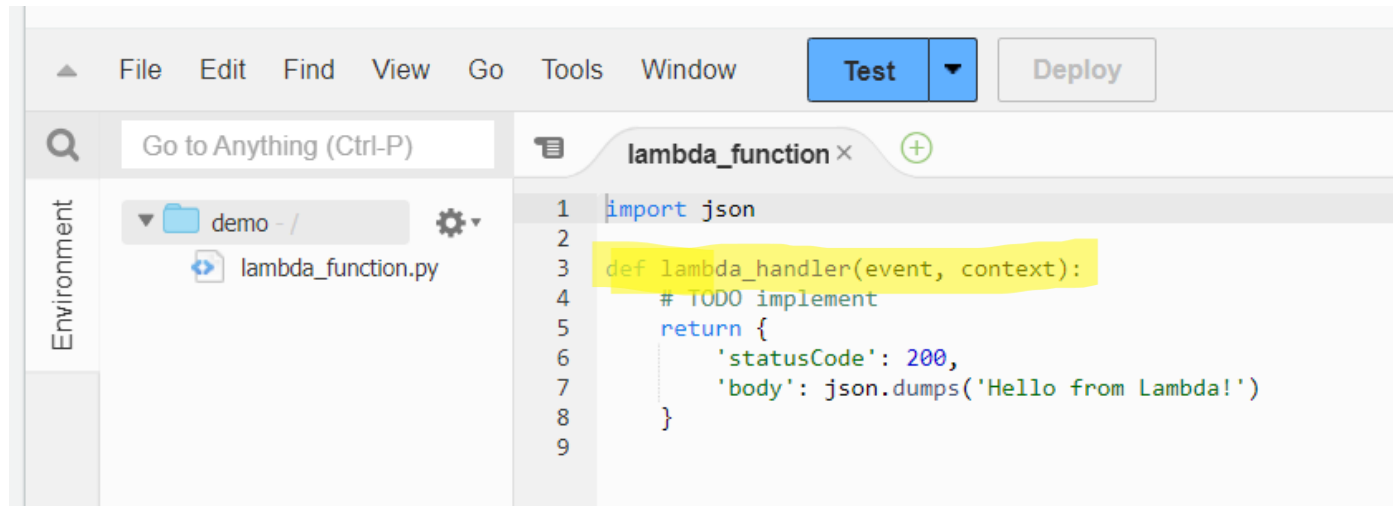
```
    if (x % 2 == 0):  
        print("even")
```

```
    else:  
        print("odd")
```

Driver code to call the function

```
evenOdd(2)
```

```
evenOdd(3)
```



API Gateway Refresher

(from my Udemy Course on :

AWS Lambda, Python(Boto3) & Serverless-

Beginner to Advanced)

AWS API Gateway - Overview

What is API Gateway

Amazon API Gateway is an AWS service for **creating, publishing, maintaining, monitoring, and securing REST, HTTP, and WebSocket APIs** at any scale.



Sample Architecture - RESTful microservices



AWS API Gateway - Overview

Key Features Of API Gateway

1. API Types

1. HTTP API

Build **low-latency and cost-effective REST APIs** with built-in features such as OIDC and OAuth2, and native CORS support.

2. WebSocket API

Build a WebSocket API using **persistent connections for real-time use cases such as chat applications or dashboards.**

3. REST API



Develop a REST API where you gain complete control over the request and response along with **API management capabilities.**

4. REST API (Private)

Create a REST API that is **only accessible from within a VPC.**

<https://aws.amazon.com/blogs/compute/introducing-amazon-api-gateway-private-endpoints/>

AWS API Gateway - Overview

Key Features of API Gateway

REST API's vs HTTP API's

- [illegible]

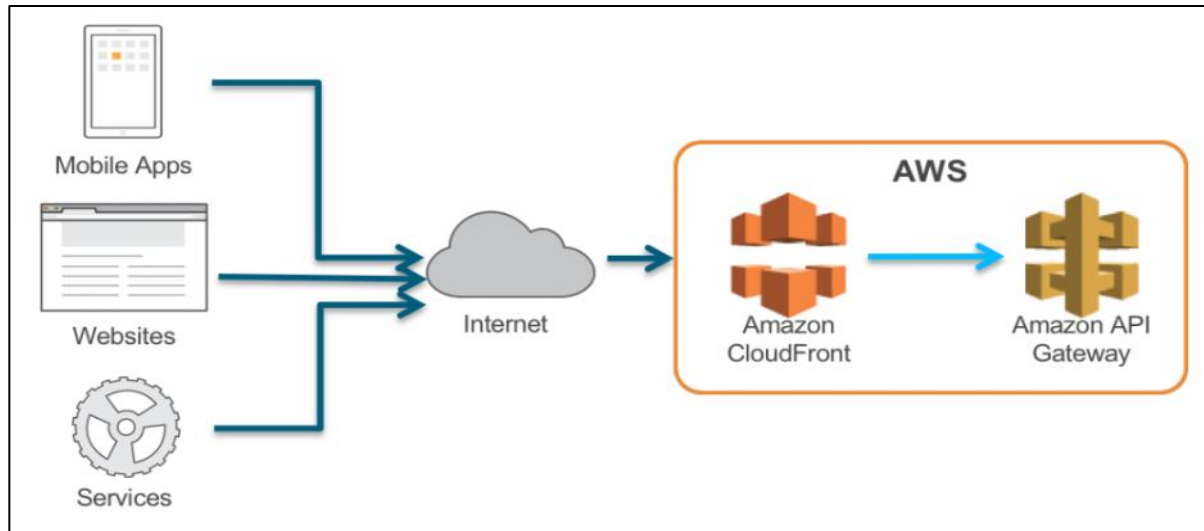
API Gateway - Overview

2. API Endpoint Types

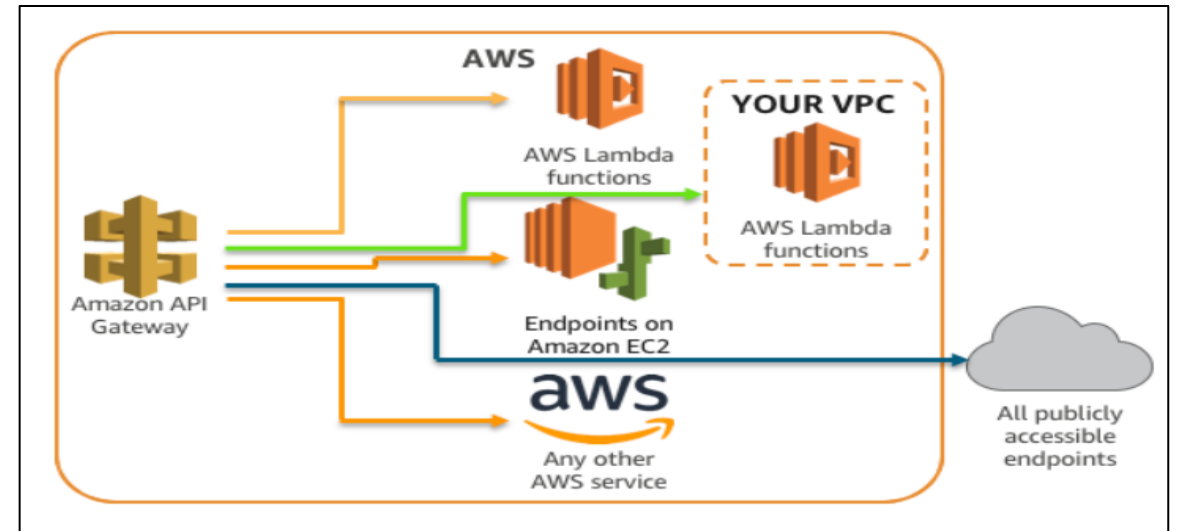
An **API endpoint type** refers to the **hostname** of the API.

The API endpoint type can be **edge-optimized**, **regional**, or **private**, depending on where the majority of your API traffic originates from.

Edge Optimized



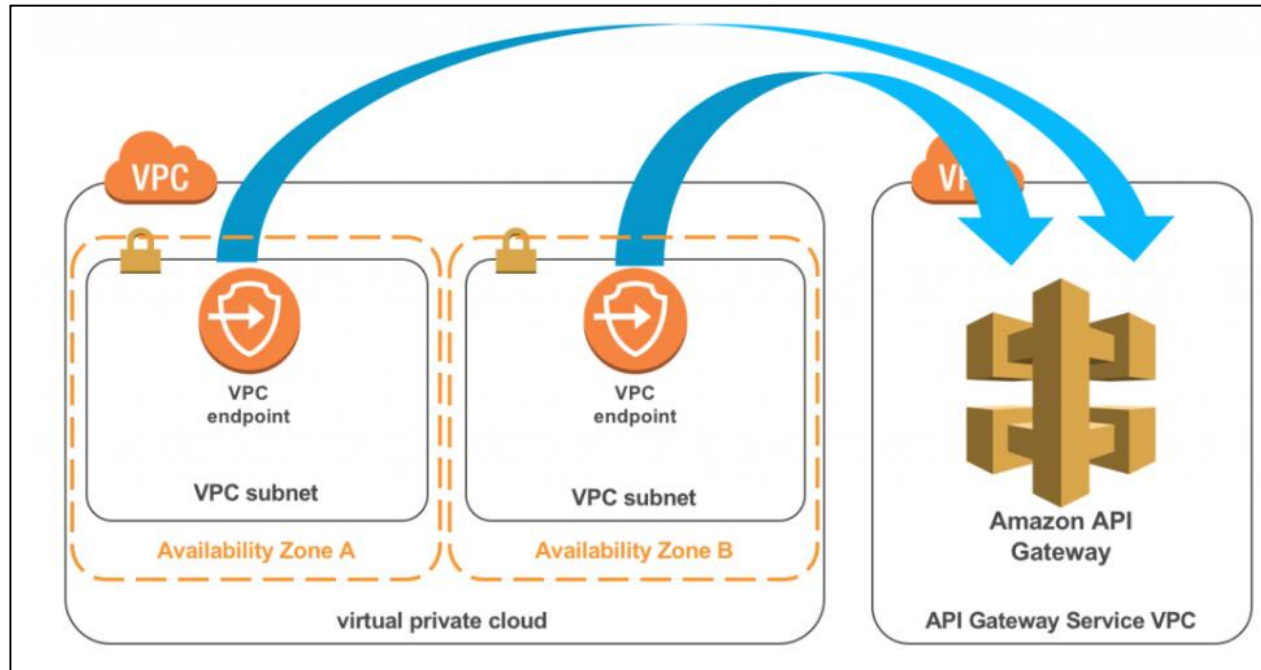
Regional



API Gateway - Overview

2. API Endpoint Types

Private



API Gateway - Overview

3. Resources



REST architecture treats every content as a **resource**.

These resources can be Text Files, Html Pages, Images, Videos or Business Data.

4. Methods



Method	Description
GET	Retrieve information about the REST API resource
POST	Create a REST API resource
PUT	Update a REST API resource
DELETE	Delete a REST API resource or related component

API Gateway - Overview

5. Integration Type

Lambda Function



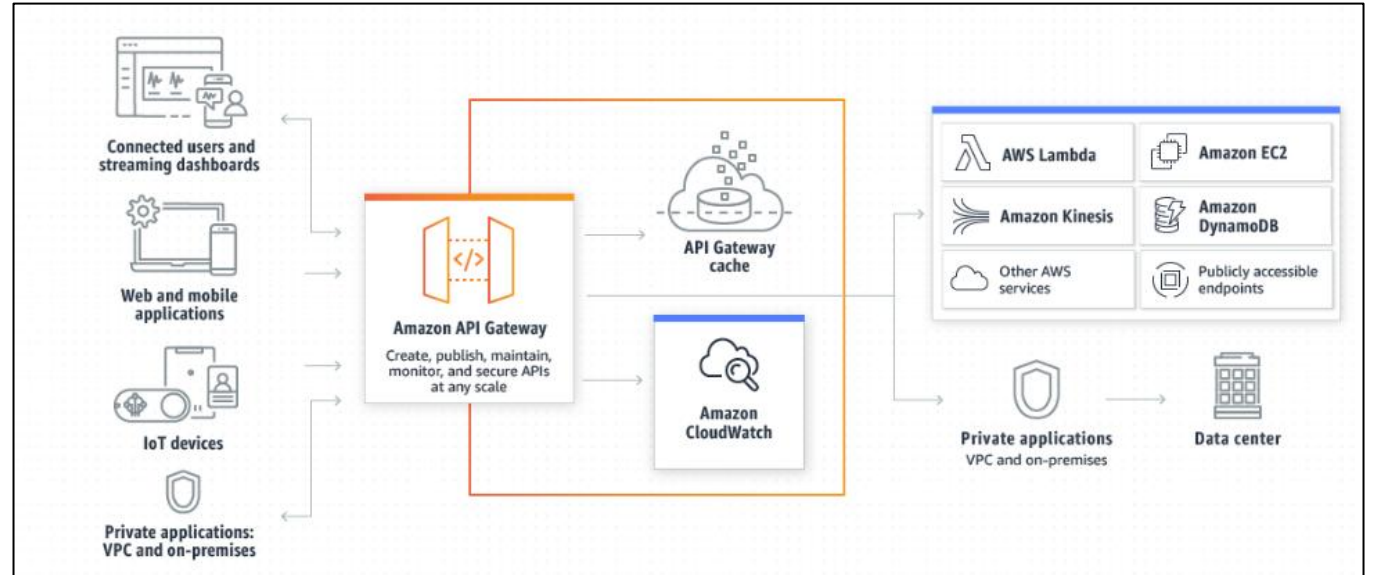
This type of integration lets an API method be integrated with the Lambda function

HTTP

- Allows integration with existing HTTP endpoint
- This type of integration lets an API expose HTTP endpoints in the backend.

Mock

- This type of integration lets API Gateway return a response without sending the request further to the backend.



API Gateway - Overview

5. Integration Type

AWS Service

This type of integration lets an API expose AWS service actions.

VPC Link

A VPC link is a resource in Amazon API Gateway that allows for connecting API routes to private resources inside a VPC

API Gateway - Overview

5. Deployment

After **creating your API**, you must **deploy it** to make it callable by your users.

To **deploy an API**, you **create an API deployment** and **associate it with a stage**.

Important Tip :

Every time you update an API, you must redeploy the API to an existing stage or to a new stage.

Updating an API includes modifying routes, methods, integrations, authorizers, and anything else other than stage settings.

6. Stages

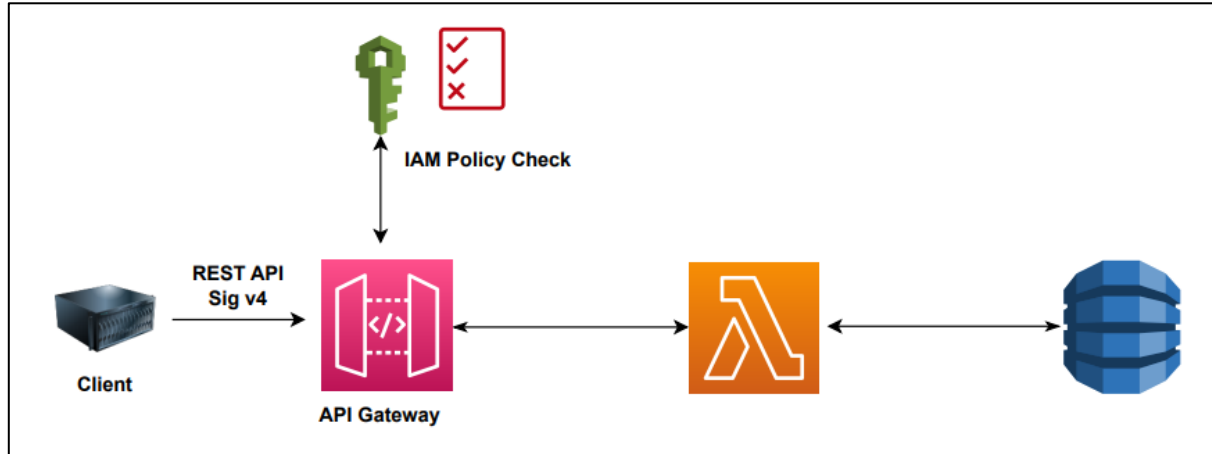
A logical reference to a **lifecycle state of your API** (for example, 'dev', 'prod', 'beta', 'v2').

<https://{restapi-id}.execute-api.{region}.amazonaws.com/{stageName}>

<https://ent94mc14j.execute-api.us-east-1.amazonaws.com/dev/students>

API Gateway - Overview

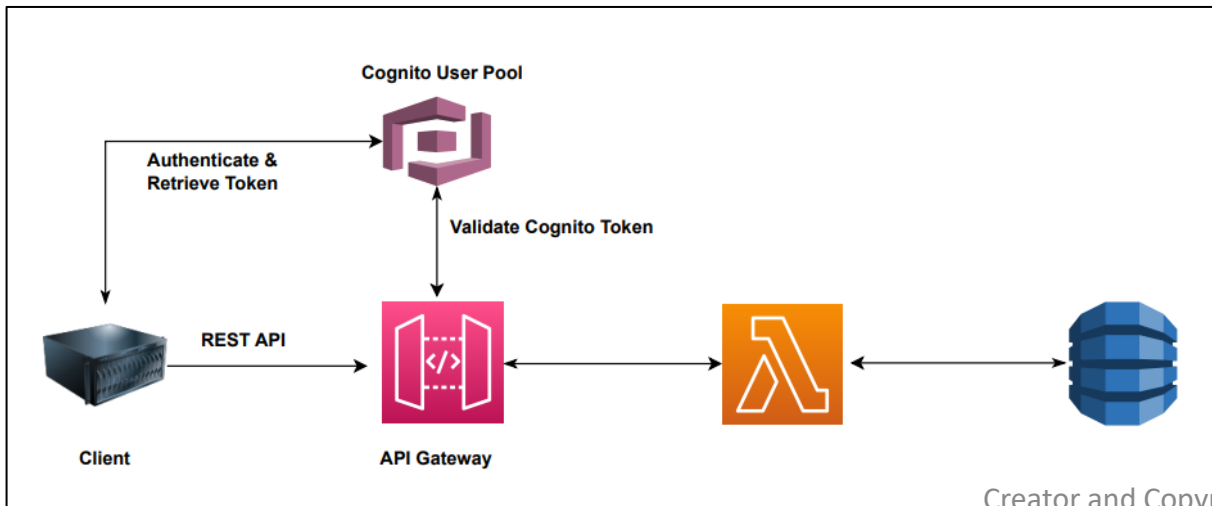
7. AWS API Gateway - Authentication and Authorization



1. API Security – IAM

Authentication – IAM
Authorization – IAM Policy

Use Case : Internal AWS Services, Cross Account Access



2. API Security – Cognito

Authentication – Cognito User Pool
Authorization – API Gateway Methods

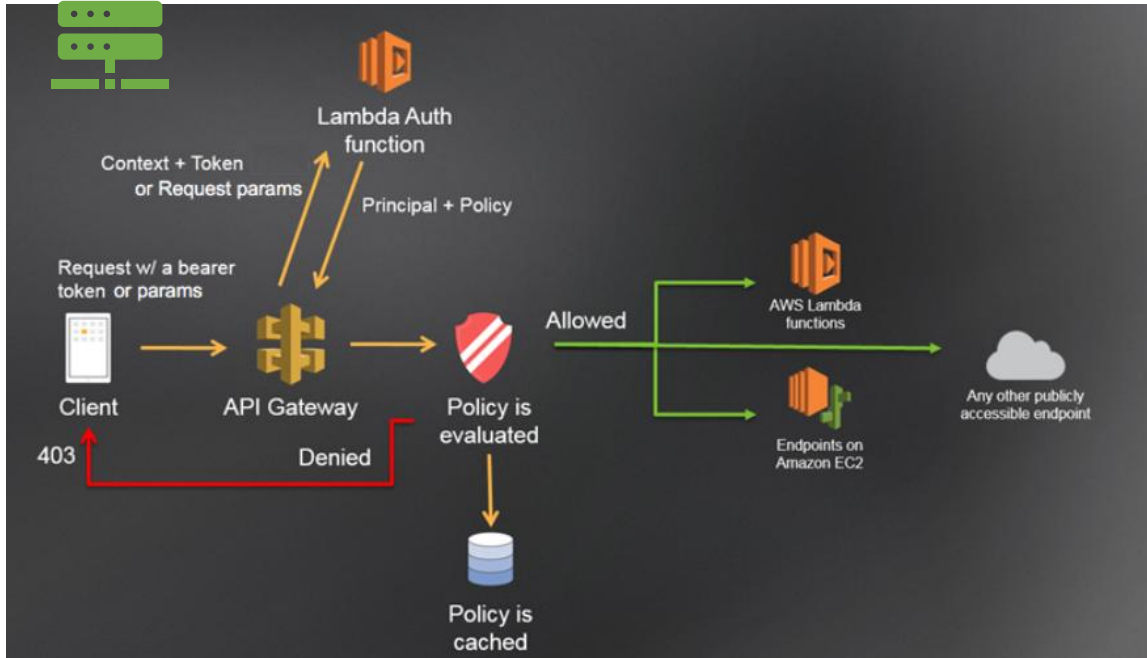
Use Case : External Users for Web/Mobile Apps

API Gateway - Overview

7. AWS API Gateway - Authentication and Authorization



3rd Party IDP



3. API Security – Lambda Authorizer

Authentication – External

Authorization – Lambda Function

Use Case : Third Party Identity Provider such as OAuth 2.0

<https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-use-lambda-authorizer.html>

API Gateway - Overview

8. Usage Plans



Usage Plans

Use Case : Differentiate between Basic and Premium Customers

Sets the **target request rate - Throttling, Burst and Quota Limit for each API Key**

9. API Keys



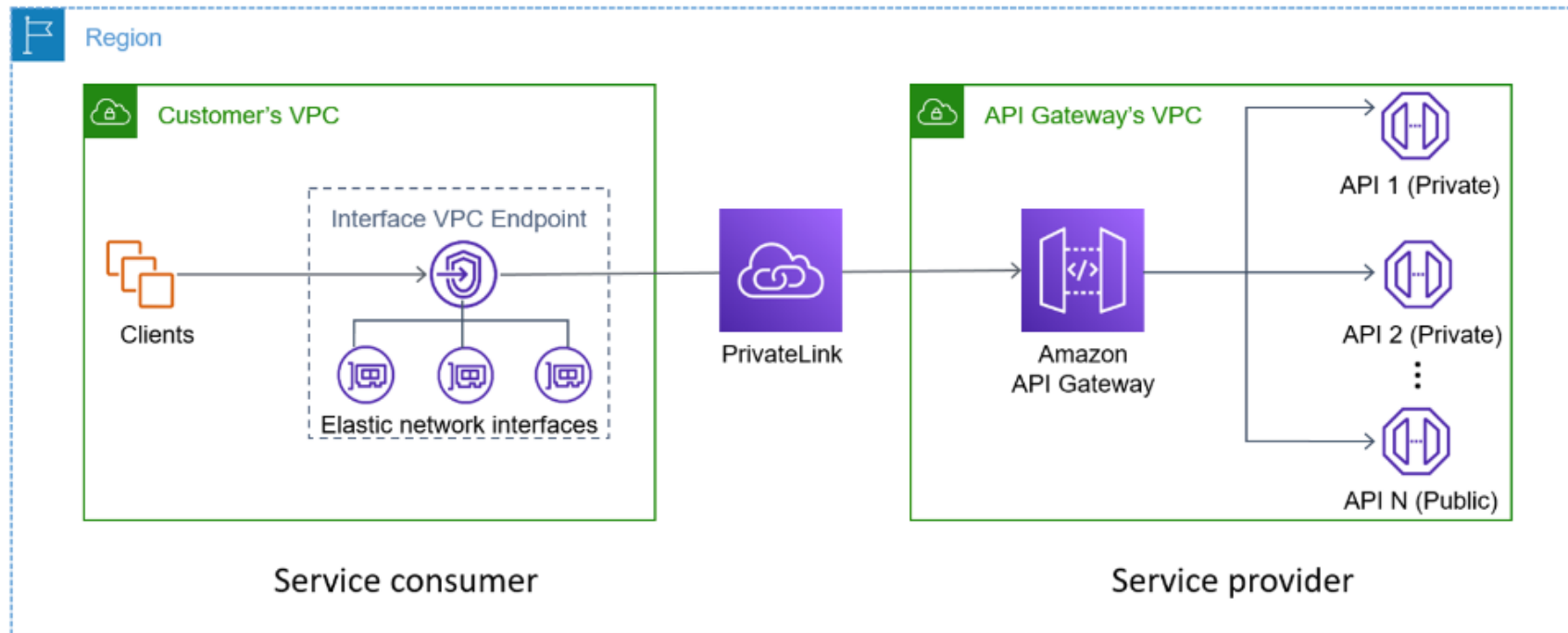
- An **alphanumeric string that API Gateway uses to identify an app developer** who uses your REST or WebSocket API.
- You can use API keys together with [Lambda authorizers](#) or [usage plans](#) to control access to your APIs.

API Gateway - Overview

10. API Integration and Private API's

Private API's

A private API means that the **API endpoint is reachable only through the VPC**. Private APIs are **accessible only** from **clients within the VPC** or from clients that have **network connectivity to the VPC**.

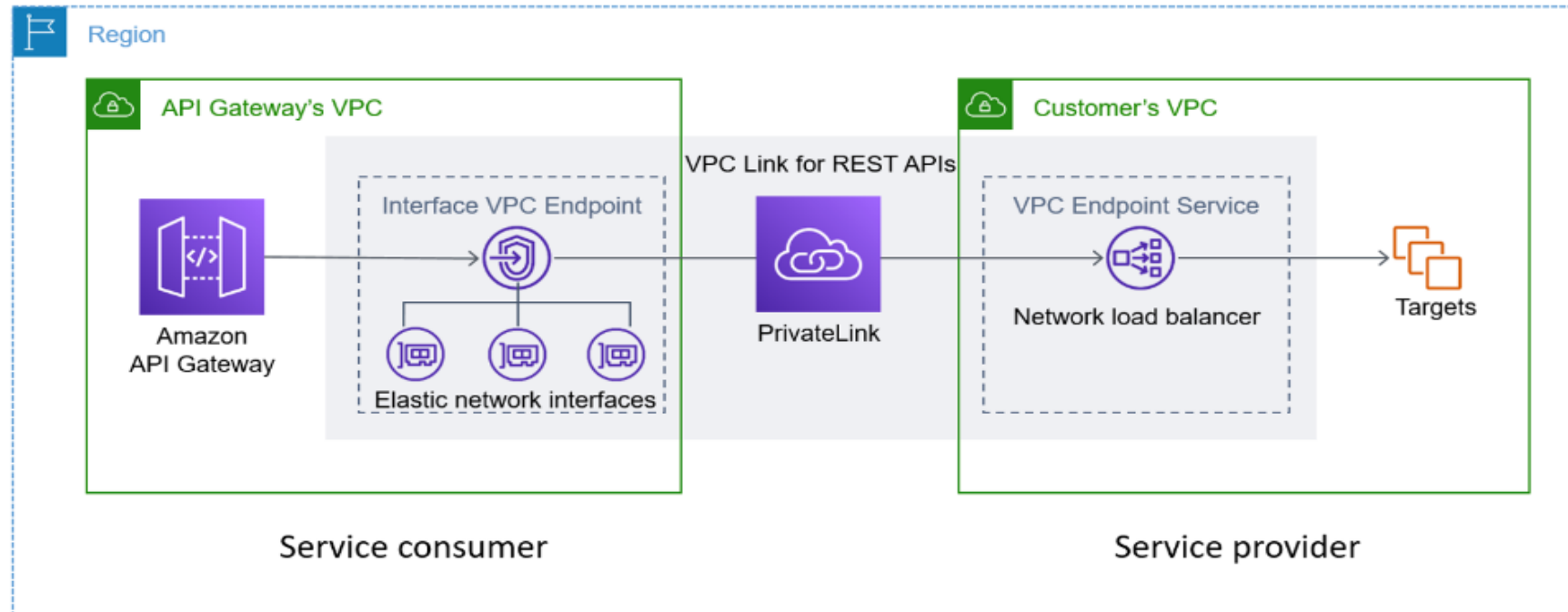


API Gateway - Overview

10. API Integration, Private API's and VPC Link

Private Integration

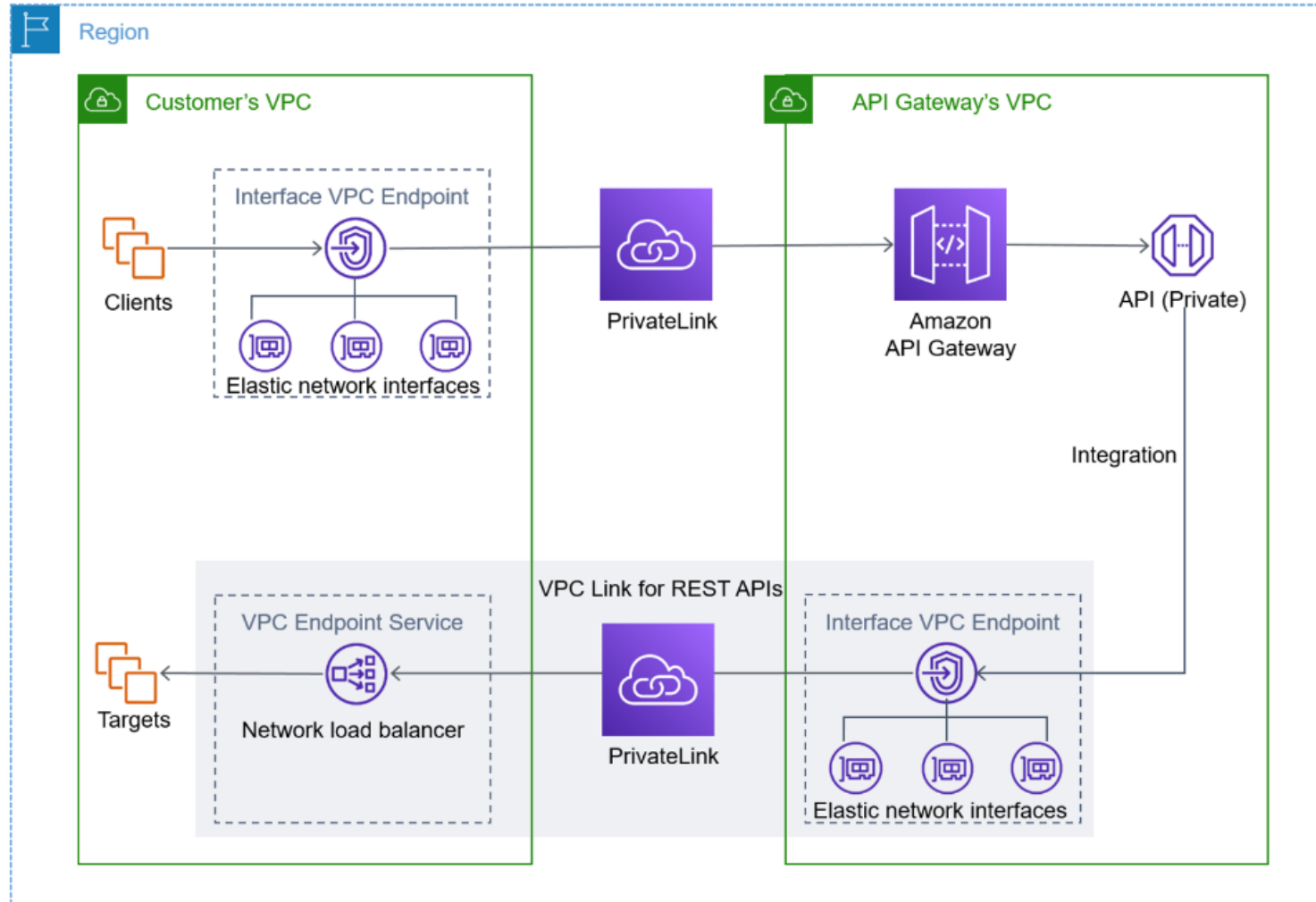
An API Gateway integration type for a client to access resources inside a customer's VPC through a private REST API endpoint without exposing the resources to the public internet.



Source : <https://aws.amazon.com/blogs/compute/understanding-vpc-links-in-amazon-api-gateway-private-integrations/>

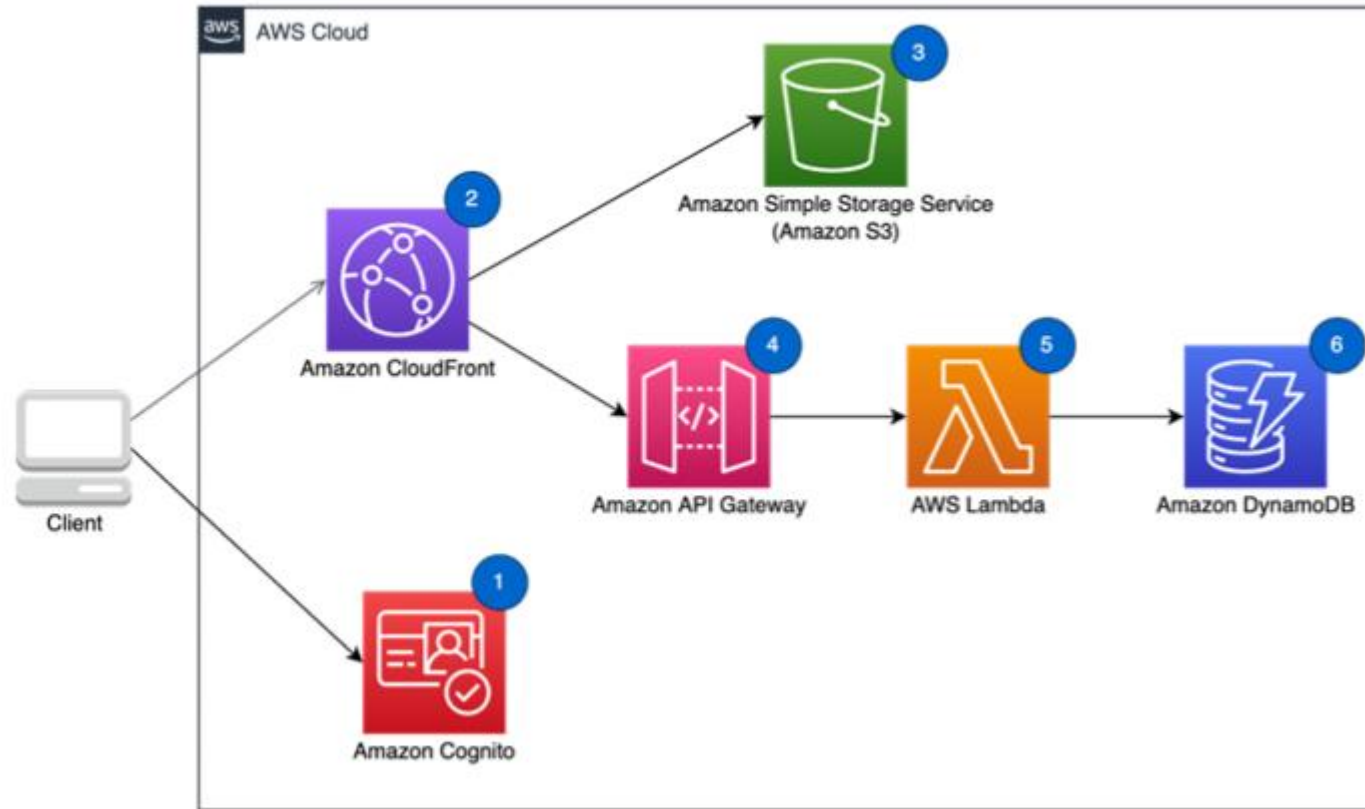
API Gateway - Overview

10. API Integration and Private API's



API Gateway - Overview

Sample Architecture



Thank You